Control Systems

Detailed Solutions

GATE INSTRUMENTATION ENGINEERING

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Previous Year Questions Solutions

Contents

SOLUTIONS

1

Basics of Control Systems

Topics

1. Introduction to Control Systems
2. Laplace Transform and Concept of a System
3. Transfer Function
4. Concept of Feedback, OLTF and Sensitivity
5. Poles, Zeros and Final Value Theorem
6. Dominant Pole, Impulse and Step Response

Q1.24.1 MCQ 2024 1M Ans: B

Valve type	$\frac{df}{dx}$	f(x)
Linear	Constant	x
Equal %	Increasing	α^{x-1}
Square-root	Decreasing	$\sqrt{x}$

Hence, the valve with decreasing sensitivity has $f(x) = \sqrt{x}$

$$\boxed{(B)\ \sqrt{x}}$$

Q1.22.1 MCQ 2022 1M Ans: B

The open-loop transfer function (loop-gain) of the system is: $L(s) = \dfrac{6}{s(s-5)}$

The characteristic equation of the unity-gain negative-feedback system is given by $1 + L(s) = 0$:

$$1 + \frac{6}{s(s-5)} = 0 \implies s^2 - 5s + 6 = 0$$

$$(s-2)(s-3) = 0 \implies s = 2,\ 3$$

- **Causality:** degree of the denominator is greater than the numerator, making it proper, and therefore causal, meaning their impulse response $h(t) = 0$ for $t < 0$. Therefore, the region of convergence (ROC) must extend to the right of the rightmost pole ($s > 3$).
- **Stability:** Both closed-loop poles ($s = 2$ and $s = 3$) lie entirely in the right-half of the s-plane ($\text{Re}\{s\} > 0$). Since there are poles in the right-half plane, the causal system is **unstable**.

$$\boxed{\text{(B) Causal and unstable}}$$

Key Points

Any closed-loop pole located in the right-half plane causes the bounded-input bounded-output (BIBO) response to grow exponentially over time making the system unstable.

Q1.19.1 MCQ 2019 1M Ans: A

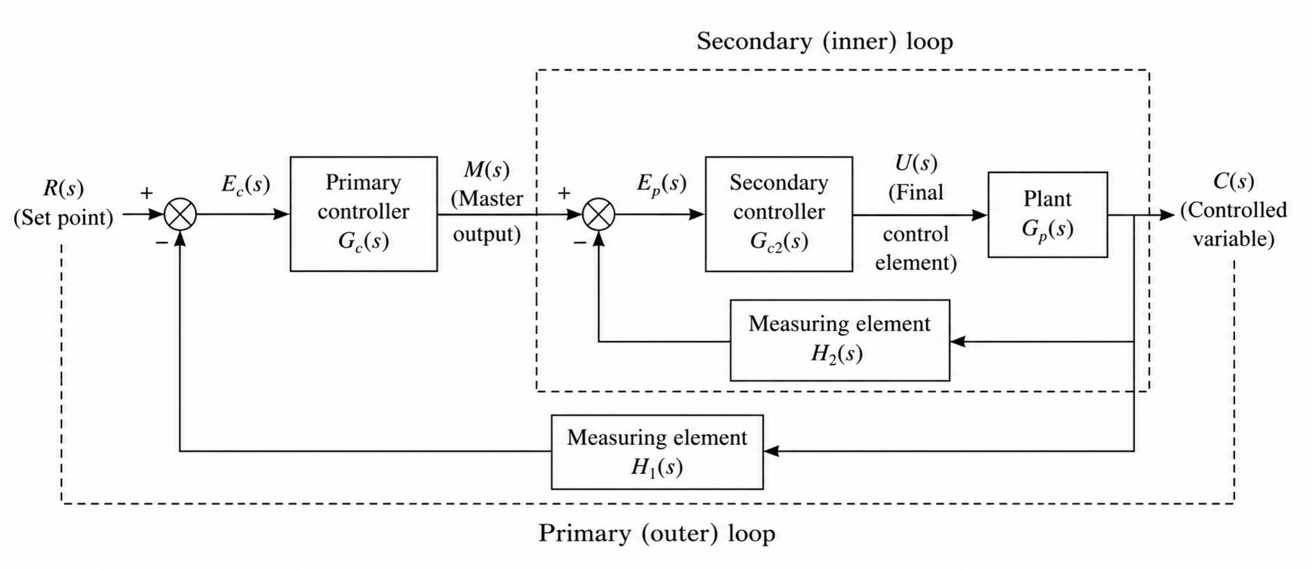


Fig. Solution Q1.19.1

In cascade control, the inner loop (secondary loop) handles high-frequency disturbances and must react significantly faster than the outer loop (primary loop). In fact, it is the basic characteristic of a cascade control strategy.

Generally, the speed of response of a loop is inversely proportional to the time constant of the transfer function.

So, the time constant of the inner loop τ_1 should be less compared to the time constant of the outer loop τ_2.

$$\tau_1 \ll \tau_2$$

$$\boxed{\text{(A) } \tau_1 \text{ is much less than } \tau_2}$$

Q1.17.1 MCQ 2017 1M Ans: A ● ○ ○

An ON-OFF controller is designed to tolerate a range of error to avoid self-sustained oscillations; this range or dead-band is known as the hysteresis characteristics.

Without hysteresis, the final control element (like a valve or switch) would constantly cycle or "chatter" rapidly around the setpoint, leading to excessive mechanical wear and tear.

Therefore, the term hysteresis is directly associated with ON-OFF control.

$$\boxed{\text{(A) ON-OFF control}}$$

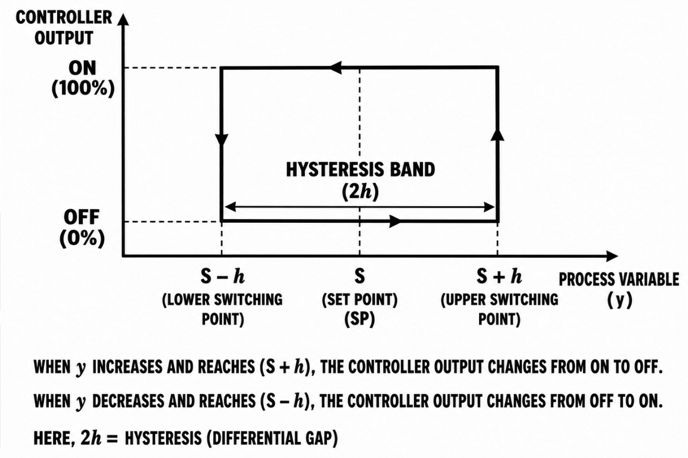


Fig. Solution Q1.17.1

Q1.17.2 MCQ 2017 1M Ans: C ● ● ○

In the process control industry, the standard for long-distance analog signal transmission is a $4 - 20\,\text{mA}$ loop.

To differentiate between a system fault (such as a broken wire or open circuit) and a true zero input condition, the lower baseline value of the signal is intentionally kept at a non-zero value of $4\,\text{mA}$ (known as a live zero). If a wire breaks, the current drops to $0\,\text{mA}$, which immediately alerts the controller to a fault condition.

$$\boxed{\text{(C) } 4 - 20\,\text{mA}}$$

Q1.16.1 MCQ 2016 2M Ans: A ● ○ ○

The given differential equation :

$$M\frac{d^2x(t)}{dt^2} + D\frac{dx(t)}{dt} + Kx(t) = f(t)$$

Taking the Laplace transform on both sides under zero initial conditions:

$$Ms^2X(s) + DsX(s) + KX(s) = F(s)$$

$$\left(Ms^2 + Ds + K\right)X(s) = F(s)$$

Thus, the transfer function $G(s)$ is:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Ds + K}$$

Substituting the given parameters $M = 0.1$, $D = 2$, and $K = 10$:

$$G(s) = \frac{1}{0.1s^2 + 2s + 10} = \frac{10}{s^2 + 20s + 100}$$

$$\boxed{(A)\ \frac{10}{s^2 + 20s + 100}}$$

Key Points

The transfer function is always defined assuming all initial conditions are zero.

Q1.10.1 MCQ 2010 2M Ans: B

Given that the maximum voltage of the control transformer rotor is $V_m = 1.0\,\text{V}$ and the transmitter rotor is rotated by an angle of $\phi = 30°$.

The induced rotor voltage V_r in a synchro control transformer pair varies as a cosine function of the angular displacement from its null position:

$$V_r = V_m \cos\phi$$

$$V_r = 1.0 \cdot \cos 30°$$

$$V_r = 0.866\,\text{V}$$

$$\boxed{\text{(B) } 0.866\text{ V}}$$

Q1.07.1 MCQ 2007 1M Ans: C

The closed-loop transfer function of the given control system is:

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$

Sensitivity with respect to forward path $G(s)$:

$$S_G^T = \frac{\partial T}{\partial G} \cdot \frac{G}{T}$$

Differentiating $T(s)$ with respect to $G(s)$:

$$\frac{\partial T}{\partial G} = \frac{[1 + KG(s)H(s)]K - KG(s)[KH(s)]}{[1 + KG(s)H(s)]^2}$$

$$S_G^T = \frac{G}{T} \times \frac{K}{[1 + KG(s)H(s)]^2} = \frac{1}{1 + KG(s)H(s)}$$

Given that the forward path gain K is very high ($K \to \infty$):

$$S_G^T \to 0 \quad \text{(System becomes less sensitive to } G(s))$$

Sensitivity with respect to feedback path $H(s)$:

$$S_H^T = \frac{\partial T}{\partial H} \cdot \frac{H}{T}$$

Differentiating $T(s)$ with respect to $H(s)$:

$$\frac{\partial T}{\partial H} = \frac{-[KG(s)] \cdot [KG(s)]}{[1 + KG(s)H(s)]^2}$$

$$S_H^T = -\frac{KG(s)}{[1 + KG(s)H(s)]^2} \cdot \frac{[1 + KG(s)H(s)]H(s)}{KG(s)}$$

$$S_H^T = -\frac{KG(s)H(s)}{1 + KG(s)H(s)}$$

As the forward path gain $K \to \infty$:

$$S_H^T \to -1 \implies |S_H^T| \to 1 \quad \text{(Highly sensitive to } H(s))$$

$$\boxed{\text{(C) sensitive to perturbations in } H(s) \text{ but not to perturbations in } G(s)}$$

Key Points

- Sensitivity function $S_X^T = \dfrac{\%\text{ change in } T}{\%\text{ change in } X} = \dfrac{\partial T}{\partial X} \cdot \dfrac{X}{T}$.
- Increasing the loop gain decreases the system sensitivity to variations in the forward path elements but increases dependency on the accuracy of feedback path elements.

Q1.06.1 MCQ 2006 1M Ans: D ● ● ○

A two-phase servomotor is specifically designed for fast transient response and stable control operation, differing significantly from a conventional two-phase power induction motor.

- **High Resistance (R):** Designed with thin, high-resistivity rotor conductors to reduce the torque-speed slope and eliminate oscillations.
- **Low Reactance (X):** Achieved via a light, low-inertia rotor structure (often drag-cup or thin squirrel-cage type).

Ratio and Inertia Comparison: As commonly compared in specific machines literature for this specialized configuration, these parameters yield:

- A characteristic **larger R/X ratio**.
- A physically **lighter rotor** allowing the motor to accelerate and decelerate quickly. The rotor utilizes lightweight materials or drag-cup geometries to facilitate rapid acceleration and braking.

$\boxed{\text{(D) a larger } R/X \text{ ratio and a lighter rotor}}$

Q1.05.1 MCQ 2005 1M Ans: D ● ○ ○

In the given control loop configuration:

- The primary objective is to maintain the temperature inside the jacketed CSTR, monitored by the Temperature Transmitter (TT) and regulated by the Temperature Controller (TC).
- The output of the Temperature Controller (TC) acts directly as the dynamic **set point** for the inner loop's Flow Controller (FC).
- The inner loop consists of the Flow Transmitter (FT) and Flow Controller (FC) regulating the cooling water valve to rapidly suppress flow disturbances before they affect the process temperature.

Since the output of one controller determines the set point of another controller operating in a nested configuration, this scheme is a classic implementation of a **Cascade Control** system.

$\boxed{\text{(D) Cascade Control}}$

Key Points

Design Objective: The inner loop isolates fast-acting disturbances (like cooling fluid line pressure variations) before they can propagate into the primary process variable.

Q1.05.2 MCQ 2005 2M Ans: A ● ● ○

Given data for the single stack stepper motor:

- Number of stator slots (phases/stator poles group), $m = 4$
- Number of rotor poles, $N_r = 18$

The full-step size (β) for a stepper motor is given by:

$$\beta = \frac{360^\circ}{m \times N_r} = \frac{360^\circ}{4 \times 18} = \frac{360^\circ}{72} = 5^\circ$$

For a half-stepping scheme, the step angle is halved:

$$\beta_{\text{half}} = \frac{\beta}{2} = \frac{5^\circ}{2} = 2.5^\circ$$

2.5° is not listed in the options.

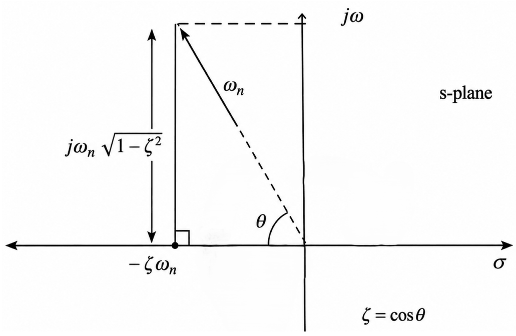
$\boxed{\text{MTA}}$

Q1.04.1 MCQ 2004 1M Ans: C

Given the complex conjugate dominant poles located at $s = -2 \pm j2$.

Comparing this with the standard second-order location $s = -\zeta\omega_n \pm j\omega_d$, we identify:

$$\zeta\omega_n = 2 \quad \text{and} \quad \omega_d = \omega_n\sqrt{1-\zeta^2} = 2$$



Method 1

The natural frequency ω_n is the distance from the origin to the pole location:

$$\omega_n = \sqrt{(\zeta\omega_n)^2 + \omega_d^2} = \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2}$$

Now, solving directly for the damping ratio ζ:

$$\zeta = \frac{2}{\omega_n} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707$$

Method 2

The damping ratio can also be expressed as $\zeta = \cos\theta$, where θ is the angle made by the pole phasor line with the negative real axis:

$$\theta = \tan^{-1}\left(\frac{\omega_d}{\zeta\omega_n}\right) = \tan^{-1}\left(\frac{2}{2}\right) = 45^\circ$$

$$\zeta = \cos(45^\circ) = \frac{1}{\sqrt{2}} \approx 0.707$$

$$\boxed{(C)\ \ 0.707}$$

Q1.04.2 MCQ 2004 2M Ans: C

From the given linear torque-speed curve:

- At zero speed ($\omega = 0$), the stall torque is $T_{\text{stall}} = 0.5$ Nm.
- At zero torque ($T_m = 0$), the no-load speed is $N_0 = 6000$ rpm.

Converting the no-load speed to radians per second: $\omega_0 = \dfrac{2\pi \times 6000}{60} = 200\pi$ rad/s

For an armature-controlled DC servomotor, the torque equation is:

$$T_m = \frac{K_t}{R_a}e_a - \frac{K_bK_t}{R_a}\omega$$

At stall condition ($\omega = 0$): $T_{\text{stall}} = \dfrac{K_t}{R_a}e_a \implies 0.5 = \dfrac{K_t}{R_a}(4) \implies \dfrac{K_t}{R_a} = 0.125$

Testing option (C) where $R_a = 2.00\,\Omega$:

$$K_t = 0.125 \times 2.00 = 0.25\ \text{Nm/A}$$

$$\boxed{\text{(C) (2.00, 0.25)}}$$

Q1.03.1 MCQ 2003 1M Ans: A

The step angle or resolution (α) of a multi-stack variable reluctance stepper motor is defined as the angular rotation per input pulse. The step angle or resolution (α) for an n-stack stepper motor with T teeth per stack is given by:

$$\alpha = \frac{360^\circ}{n \times T}$$

$$\text{Number of stacks (n)} = 3 \quad \textit{and} \quad \text{Number of teeth (T)} = 12$$

$$\alpha = \frac{360^\circ}{3 \times 12} = \frac{360^\circ}{36} = 10^\circ$$

$$\boxed{(A)\ \ 10^\circ}$$

Q1.03.2 MCQ 2003 1M Ans: A ● ● ○

In a synchro system, the control transmitter (CX) acts as a balanced three-phase system on its stator side.

When the rotor winding of the control transmitter is excited by an AC voltage, it induces voltages in the three stator windings (S_1, S_2, S_3) through electromagnetic induction. Because the three stator windings are physically distributed 120° apart in space, the induced voltages will always have **equal maximum magnitudes but will be displaced by a successive phase difference of** 120°.

$$\boxed{\text{(A) equal magnitudes but different phases}}$$

Q1.98.1 MCQ 1998 4M Ans: a-f, b-e, c-g, d-h ● ○ ○

The matching between the given control system components and their corresponding transfer functions is evaluated as follows:

- **(a) All pass filter:** An all-pass filter passes all frequencies with equal gain but introduces a frequency-dependent phase shift. Its transfer function has zeros in the right half of the s-plane that are mirror images of its left-half plane poles. Hence, **(a)** → **(f)** $\left(\frac{1-s}{1+s}\right)$.
- **(b) Transport delay:** A pure time delay or dead-time element introduces a phase lag proportional to frequency without changing the magnitude. Its Laplace domain representation is e^{-as}. Hence, **(b)** → **(e)**.
- **(c) Lag network:** A phase-lag network has a transfer function of the form $\dfrac{1+as}{1+bs}$ where the pole is closer to the origin than the zero, which means the pole time constant is greater than the zero time constant ($b > a$ or $a < b$). Hence, **(c)** → **(g)**.
- **(d) Servomotor:** A standard armature-controlled DC servomotor driving an inertial load acts as an integrator with a field/armature time constant lag. Its transfer function is typically represented as $\dfrac{K}{s(1+as)}$. Hence, **(d)** → **(h)**.

$$\boxed{\text{(a)-(f), (b)-(e), (c)-(g), (d)-(h)}}$$

Q1.96.1 MCQ 1996 2M Ans: a-f, b-e, c-i, d-g ● ○ ○

The characteristics match with the system transfer functions as follows:

- **(a) All pass** $\implies$ **(f)** $\dfrac{1-s}{1+s}$. An all-pass filter satisfies $|H(j\omega)| = 1$ for all frequencies, with its poles and zeros lying symmetrically across the imaginary ($j\omega$) axis.
- **(b) Transport delay** $\implies$ **(e)** $e^{-0.5s}$. A pure time-delay system with a delay of 0.5 sec has the Laplace domain representation of $e^{-0.5s}$.
- **(c) Lead compensator** $\implies$ **(i)** $\dfrac{10(1+0.3s)}{(1+0.1s)}$. The zero is at $s = -3.33$ and the pole is at $s = -10$. Since the zero is closer to the origin than the pole, it provides a positive phase lead:

$$\phi = \tan^{-1}(0.3\omega) - \tan^{-1}(0.1\omega) > 0$$

- **(d) Unstable** $\implies$ **(g)** $\dfrac{1}{s^2+2s-3}$. The characteristic equation is $(s+3)(s-1) = 0$, giving a pole at $s = 1$ in the right half of the s-plane.
- Additionally, **(h)** $\dfrac{s}{s^2+s+100}$ represents a Band Pass Filter (B.P.F.).

$$\boxed{\text{a-f, b-e, c-i, d-g}}$$

Q1.94.1 MCQ 1994 1M Ans: D

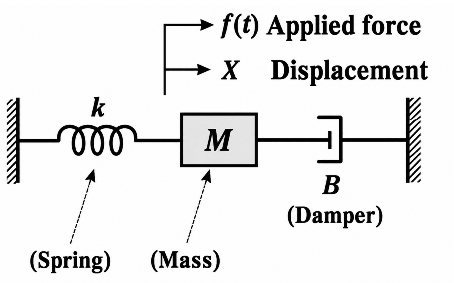


Fig. Solution Q1.94.1

The differential equation of motion for a second-order mechanical system by applying Newton's second law is:

$$M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx = f(t)$$

Taking the Laplace transform on both sides assuming zero initial conditions:

$$\left(Ms^2 + Bs + K\right) X(s) = F(s) \implies \frac{X(s)}{F(s)} = \frac{1/M}{s^2 + \frac{B}{M}s + \frac{K}{M}}$$

Comparing this with the standard second-order characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$:

$$\omega_n = \sqrt{\frac{K}{M}} \quad \text{and} \quad 2\zeta\omega_n = \frac{B}{M} \implies \zeta = \frac{B}{2\sqrt{KM}}$$

To just prevent any overshoot (i.e., achieving a critically damped response), the damping ratio must be exactly equal to 1:

$$\zeta = 1 \implies \frac{B}{2\sqrt{KM}} = 1 \implies B = 2\sqrt{KM}$$

$$\boxed{\text{(D) equal to } 2\sqrt{KM}}$$

Key Points

- Underdamped ($\zeta < 1$): System exhibits oscillations and overshoot.
- Critically damped ($\zeta = 1$): System returns to steady state quickest without overshoot ($B = 2\sqrt{KM}$).
- Overdamped ($\zeta > 1$): Sluggish response, no overshoot ($B > 2\sqrt{KM}$).

Q1.94.2 MCQ 1994 1M Ans: c

The standard transfer function of a first-order system with a static gain K, time constant τ, and a transport delay t_d is represented by:

$$T(s) = \frac{K \cdot e^{-t_d s}}{1 + \tau s}$$

Given parameters from the question:

- Static gain, $K = 1$
- Time constant, $\tau = 1$ sec
- Transport delay, $t_d = 0.1$ sec

Substituting these given parameters into the standard equation:

$$T(s) = \frac{1 \cdot e^{-0.1s}}{1 + 1 \cdot s} = \frac{e^{-0.1s}}{1 + s}$$

$$\boxed{(C) \quad \frac{e^{-0.1s}}{1+s}}$$

Q1.94.3 MCQ 1994 4M Ans: a-f, b-e, c-h, d-g ● ● ○

To find the correct match between control components and their mathematical transfer functions:

- **(a) Pneumatic actuator:** A standard pneumatic actuator typically a constat gain properties, matching with **(f)** k.
- **(b) A transport delay:** A dead-time parameter is modeled exactly by an exponential shift operator. Hence, **(b)** → **(e)** $\left(e^{-sT}\right)$.
- **(c) A servomotor:** An electrical DC/AC servomotor features inherent speed-to-position integration combined with mechanical field/armature time constants. Hence, **(c)** → **(h)** $\left(\frac{K}{s(1+as)}\right)$.
- **(d) A nozzle-flapper:** A mechanical nozzle-flapper structure possesses generic second-order dynamic response behaviors under full loading. Hence, **(d)** → **(g)** $\left(\frac{K}{as^2+bs+c}\right)$.

Combining these matches gives the correct choice sequence:

$$\boxed{\text{(a)-(f), (b)-(e), (c)-(h), (d)-(g)}}$$

Q1.93.1 NAT 1993 2M Ans: amplification, error detection ● ○ ○

A synchro system consists of various configurations designed for specific functional roles in control systems:

- **Synchro Transmitter-Repeater Pair:** This configuration functions as a torque transmission system. Since the repeater reproduces the angular position of the transmitter with an increased driving torque capability, it is primarily used for torque transmission and mechanical positioning amplification.
- **Synchro Transmitter-Control Transformer Pair:** This configuration works as an error detector. The output from the control transformer's rotor winding is an electrical voltage proportional to the sine of the angular misalignment between the transmitter and transformer shafts, making it ideal for feedback loops.

$$\boxed{\text{amplification, error detection}}$$

Q1.92.1 MCQ 1992 1M Ans: A ● ○ ○

The open-loop transfer function is given as $G(s)H(s) = \frac{K}{(s+1)^3}$. For a negative feedback control system, the characteristic equation is defined by $1 + G(s)H(s) = 0$:

$$(s+1)^3 + K = 0$$

Method 1

Using the magnitude criteria, any point lying on the root locus must satisfy $|G(s)H(s)| = 1$.

$$\left|\frac{K}{(s+1)^3}\right| = 1 \implies K = |s+1|^3$$

Substitute the value of the pole $s = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$:

$$s + 1 = \frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$|s+1| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$$

Therefore, the required gain value is:

$$K = (1)^3 = 1$$

Method 2

Substitute the value of s directly into the characteristic equation:

$$\left[\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) + 1\right]^3 + K = 0 \implies \left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)^3 + K = 0$$

Expressing $\frac{1}{2} + j\frac{\sqrt{3}}{2}$ in polar form: $1 \cdot e^{j60^\circ}$.

$$\left(1 \cdot e^{j60^\circ}\right)^3 + K = 0 \implies 1^3 \cdot e^{j180^\circ} + K = 0$$

Since $e^{j180^\circ} = -1$:

$$-1 + K = 0 \implies K = 1$$

$\boxed{\text{(A) } 1}$

Key Points

- **Magnitude Condition:** For any valid pole on the root locus, $|G(s)H(s)| = 1$.
- Convert complex terms into phasor/polar form ($re^{j\theta}$) to simplify higher power calculations easily.

Q1.92.2 NAT 1992 2M Ans: *

Given a complex pole pair at $s = -1 \pm j2$ and a real zero at $s = -3$. The general form of the transfer function $G(s)$ is:

$$G(s) = \frac{k(s+3)}{(s+1+j2)(s+1-j2)} = \frac{k(s+3)}{(s+1)^2 - (j2)^2} = \frac{k(s+3)}{s^2+2s+5}$$

For a unit step input $R(s) = \frac{1}{s}$, the steady-state output y_{ss} can be found using the Final Value Theorem:

$$y_{ss} = \lim_{s\to 0} s \cdot Y(s) = \lim_{s\to 0} s \cdot G(s) \cdot \frac{1}{s} = G(0)$$

Given that the steady state output is 2:

$$G(0) = \frac{k(0+3)}{0^2 + 2(0) + 5} = 2 \implies \frac{3k}{5} = 2 \implies k = \frac{10}{3}$$

Substituting k back into the expression for $G(s)$:

$$\boxed{G(s) = \frac{10(s+3)}{3(s^2+2s+5)}}$$

DEBUG INFO

Chapter I

Basics of Control Systems

Total Questions : 23

MCQ : 21

MSQ : 0

NAT : 2

SOLUTIONS

11

Block Diagram and Signal Flow Graphs

Topics

1. Block Diagram Reduction Techniques
2. Signal Flow Graphs and Mason's Gain Formula
3. Limitations of Mason's Gain Formula

Q2.22.1 MCQ 2022 2M Ans: A

By applying Mason's Gain Formula, the overall transfer function is given by:

$$\frac{Y(s)}{X(s)} = \frac{\sum P_k \Delta_k}{\Delta}$$

- **Forward Paths (P_k):** $P_1 = 2G_1(s)G_2(s)$ and $P_2 = 2G_1(s)G_3(s)$
- **Individual Loops (L_m):** $L_1 = -G_2(s)$ and $L_2 = -G_3(s)$
- **Path Determinants (Δ_k):** Since both paths touch all loops, $\Delta_1 = 1$ and $\Delta_2 = 1$
- **System Determinant (Δ):** Since the loops touch each other, $\Delta = 1 - (L_1 + L_2) = 1 + G_2(s) + G_3(s)$

$$\frac{Y(s)}{X(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{2G_1(s)G_2(s) + 2G_1(s)G_3(s)}{1 + G_2(s) + G_3(s)}$$

$$\boxed{\text{(A) } \frac{2G_1(s)G_2(s) + 2G_1(s)G_3(s)}{1 + G_2(s) + G_3(s)}}$$

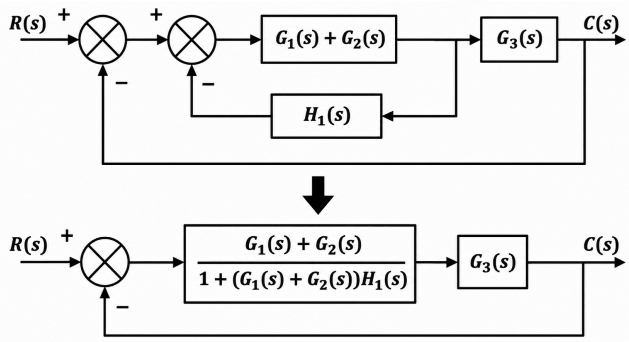
Key Points

- Mason's Gain Formula: $\frac{C(s)}{R(s)} = \frac{\sum P_k \Delta_k}{\Delta}$
  - P_k: Gain of the k^{th} forward path
  - $\Delta = 1 -$ (sum of individual loop gains) $+$ (sum of gain products of two non-touching loops) $- \ldots$
  - Δ_k: The cofactor of the k^{th} forward path, calculated by removing the loops touching the k^{th} path.

Q2.17.1 MCQ 2017 2M Ans: A

Method 1

Block Diagram Reduction:



Simplifying the unity negative feedback ($H(s) = 1$) yields:

$$\frac{C(s)}{R(s)} = \frac{\dfrac{(G_1(s) + G_2(s))G_3(s)}{1 + (G_1(s) + G_2(s))(H_1(s))}}{1 + \dfrac{(G_1(s) + G_2(s))G_3(s)}{1 + (G_1(s) + G_2(s))(H_1(s))}}$$

$$\frac{C(s)}{R(s)} = \frac{(G_1(s) + G_2(s))G_3(s)}{1 + (G_1(s) + G_2(s))(H_1(s) + G_3(s))}$$

Method 2

Mason's Gain Formula: $T = \frac{\sum P_k \Delta_k}{\Delta}$

- **Forward Paths:**

$$P_1 = G_1(s)G_3(s),\ \Delta_1 = 1; \quad P_2 = G_2(s)G_3(s),\ \Delta_2 = 1$$

- **Individual Loops:**

$$L_1 = -G_1(s)H_1(s) \quad L_2 = -G_2(s)H_1(s)$$

$$L_3 = -G_1(s)G_3(s) \quad L_4 = -G_2(s)G_3(s)$$

- **Determinant (Δ):**

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4)$$

$$\Delta = 1 + (G_1(s) + G_2(s))(H_1(s) + G_3(s))$$

Thus, the overall transfer function matches:

$$\frac{C(s)}{R(s)} = \frac{(G_1(s) + G_2(s))G_3(s)}{1 + (G_1(s) + G_2(s))(H_1(s) + G_3(s))}$$

$$\boxed{(A)\ \frac{(G_1(s) + G_2(s))G_3(s)}{1 + (G_1(s) + G_2(s))(H_1(s) + G_3(s))}}$$

Q2.14.1 MCQ 2014 2M Ans: D

Method 1

Using block diagram algebraic reduction equations directly at the summing nodes:

$$Y(s) = [-50Y(s) + KD(s)]\frac{1}{s+2} + D(s)$$

Grouping $Y(s)$ and $D(s)$ terms:

$$Y(s)\left[1 + \frac{50}{s+2}\right] = \left[\frac{K}{s+2} + 1\right]D(s)$$

$$Y(s)\left[\frac{s+52}{s+2}\right] = \left[\frac{K+s+2}{s+2}\right]D(s)$$

$$Y(s) = \frac{K+s+2}{s+52}D(s)$$

Converting to the frequency domain ($s = j\omega$):

$$Y(j\omega) = \frac{K+j\omega+2}{j\omega+52}D(j\omega)$$

For the output to have a zero mean value, the response at $\omega = 0$ must satisfy $Y(j0) = 0$:

$$\frac{K+j0+2}{j0+52}D(j0) = 0$$

$$\frac{K+2}{52} = 0 \implies K = -2$$

Method 2

Using **Mason's Gain Formula** for the transfer function $T(s) = \frac{Y(s)}{D(s)}$ with $r(t) = 0$:

$$T(s) = \frac{\sum P_i \Delta_i}{\Delta}$$

Forward Paths from $D(s)$ to $Y(s)$:

$$P_1 = 1, \quad \Delta_1 = 1; \; and \quad P_2 = \frac{K}{s+2}, \quad \Delta_2 = 1$$

Feedback Loops: $L_1 = -50 \cdot \left(\frac{1}{s+2}\right) = -\frac{50}{s+2}$

Determinant: $\Delta = 1 - L_1 = 1 + \frac{50}{s+2} = \frac{s+52}{s+2}$

Evaluating the transfer function:

$$T(s) = \frac{1 + \frac{K}{s+2}}{\frac{s+52}{s+2}} = \frac{s+2+K}{s+52}$$

For a zero-mean steady-state output, the DC gain must be zero ($s = 0$):

$$T(0) = 0$$

$$\frac{2+K}{52} = 0 \implies K = -2$$

$$\boxed{(D) \quad -2}$$

Key Points

Complete disturbance rejection or zero-mean output tracking under constant inputs is achieved when the closed-loop system transmission zeros neutralize the input poles at $\omega = 0$.

Q2.13.1 MCQ 2013 2M Ans: A

- Forward paths and their determinants (Δ_k): $P_1 = \frac{1}{s^2} \implies \Delta_1 = 1; \quad P_2 = \frac{1}{s} \implies \Delta_2 = 1$
- Individual loop gains from the signal flow graph:

$$L_1 = -2s^{-1} = -\frac{2}{s}, \quad L_2 = -2s^{-2} = -\frac{2}{s^2}, \quad L_3 = -4, \quad L_4 = -4s^{-1} = -\frac{4}{s}$$

- Since all loops touch each other, the graph determinant Δ is:

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) = 1 + \frac{2}{s} + \frac{2}{s^2} + 4 + \frac{4}{s} = \frac{5s^2+6s+2}{s^2}$$

Substituting these into Mason's gain formula:

$$\frac{Y(s)}{U(s)} = \frac{\sum P_k \Delta_k}{\Delta} = \frac{\frac{1}{s^2}\times 1 + \frac{1}{s}\times 1}{\frac{5s^2+6s+2}{s^2}} = \frac{s+1}{5s^2+6s+2}$$

$$\boxed{(A)\ \frac{s+1}{5s^2+6s+2}}$$

Q2.11.1 MCQ 2011 2M Ans: C

- **Forward Paths (P_k):** $P_1 = G_1G_2$ and $P_2 = G_1G_3$
- **Individual Loops (L_m):** $L_1 = G_1H_1$, $L_2 = -G_1G_3H_2$, and $L_3 = -G_1G_2H_2$
- **Path Determinants (Δ_k):** Since both paths touch all loops, $\Delta_1 = 1$ and $\Delta_2 = 1$
- **System Determinant (Δ):** $\Delta = 1 - (L_1 + L_2 + L_3) = 1 - G_1H_1 + G_1G_2H_2 + G_1G_3H_2$

According to Mason's gain formula, the overall transfer function $T(s)$ is given by:

$$T(s) = \frac{C}{R} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1G_2 + G_1G_3}{1 - G_1H_1 + G_1G_2H_2 + G_1G_3H_2}$$

$$\boxed{(C)\ \frac{(G_1G_2 + G_1G_3)}{(1 - G_1H_1 + G_1G_2H_2 + G_1G_3H_2)}}$$

Q2.09.1 MCQ 2009 2M Ans: A

Using Mason's Gain Formula, we analyze the signal flow graph:

- **Forward paths:** $P_1 = 1 \cdot 1 \cdot \left(-\frac{1}{s}\right) \cdot 1 = -\frac{1}{s}, \quad P_2 = 1 \cdot k \cdot 1 \cdot (-1) = -k$
- **Individual loop:** $L_1 = 1 \cdot k \cdot 1 \cdot \left(-\frac{1}{s}\right) = -\frac{k}{s}$
- **Path factors:** Since the individual loop touches both forward paths, $\Delta_1 = 1$ and $\Delta_2 = 1$.

The system determinant is $\Delta = 1 - L_1 = 1 - \left(-\frac{k}{s}\right) = 1 + \frac{k}{s}$. The overall transfer function is:

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{-\frac{1}{s} + (-k)}{1 + \frac{k}{s}} = \frac{\frac{-(1+sk)}{s}}{\frac{s+k}{s}} = \frac{-(1+sk)}{s+k}$$

$$\boxed{(A)\ \frac{-(1+ks)}{s+k}}$$

Q2.06.1 MCQ 2006 2M Ans: A

Applying **Mason's Gain Formula**,

- **Forward Paths:** $P_1 = \frac{1}{s}$, $P_2 = \frac{1}{s^3}$, with $\Delta_1 = \Delta_2 = 1$
- **Individual Loops:** $L_1 = -\frac{1}{s^2}$, $L_2 = -\frac{1}{s^2}$, $L_3 = \frac{1}{s^2}$

$$\Delta = 1 - (L_1 + L_2 + L_3) = 1 - \left(-\frac{1}{s^2} - \frac{1}{s^2} + \frac{1}{s^2}\right) = 1 + \frac{1}{s^2}$$

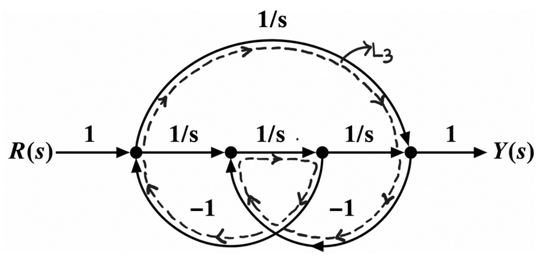


Fig. Solution Q2.06.1

Substituting these into the transfer function formula:

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{\frac{1}{s} + \frac{1}{s^3}}{1 + \frac{1}{s^2}} = \frac{\frac{s^2+1}{s^3}}{\frac{s^2+1}{s^2}} = \frac{1}{s}$$

$$\boxed{(A)\ \frac{1}{s}}$$

Mistakes to be avoided

Ensure all individual feedback loops are identified properly by carefully tracing the direction of arrows (especially the hidden loop 3)

Q2.05.1 MCQ 2005 2M Ans: A ● ○ ○

Applying **Mason's Gain Formula**,

- **Forward Path:** $P_1 = G_1G_2$ with $\Delta_1 = 1$
- **Individual Loops:** $L_1 = -G_1H_1,\ L_2 = -G_2H_2$

$$\Delta = 1 - (L_1 + L_2) = 1 + G_1H_1 + G_2H_2$$

According to Mason's gain formula, the overall transfer function is:

$$T(s) = \frac{C(s)}{R(s)} = \frac{P_1\Delta_1}{\Delta} = \frac{G_1G_2}{1 + G_1H_1 + G_2H_2}$$

$$\boxed{(A)\ \frac{G_1G_2}{1 + G_1H_1 + G_2H_2}}$$

Q2.00.1 NAT 2000 5M Ans: * ● ○ ○

With $u(s) = 0$, considering $w(s)$ as the input and $y(s)$ as the output:

- **Forward Paths:** $P_1 = G_p(s)$ and $P_2 = G_f(s)G_c(s)G_p(s)$
- **Individual Loops:** $L_1 = -G_c(s)G_p(s)H(s)$
- **Path Determinants:** Since the loop touches both forward paths, $\Delta_1 = 1$ and $\Delta_2 = 1$.
- **System Determinant:** $\Delta = 1 - L_1 = 1 + G_c(s)G_p(s)H(s)$.

Applying Mason's Gain Formula: $\dfrac{Y(s)}{W(s)} = \dfrac{\sum P_i\Delta_i}{\Delta}$:

$$\frac{Y(s)}{W(s)} = \frac{G_p(s) + G_f(s)G_c(s)G_p(s)}{1 + G_c(s)G_p(s)H(s)}$$

$$\boxed{\frac{Y(s)}{W(s)} = \frac{G_p(s)\left[1 + G_c(s)G_f(s)\right]}{1 + G_c(s)G_p(s)H(s)}}$$

Q2.00.2 NAT 2000 5M Ans: *

To achieve complete isolation or absolute tracking rejection against incoming load fluctuations, the output $y(s)$ must remain invariant to $w(s)$:

$$\frac{Y(s)}{W(s)} = 0$$

Since the characteristic polynomial denominator governs absolute loop stability metrics, it cannot be rendered infinite. Thus, set the open-loop numerator term identically to zero:

$$1 + G_c(s)G_f(s) = 0$$

$$\boxed{G_f(s) = -\frac{1}{G_c(s)}}$$

Key Points

- **Feedforward Inversion:** The ideal feedforward controller acts as the structural inverse of the feedback control block to cancel paths before errors propagate through the plant.
- **Superposition Principle:** Linear systems allow individual inputs to be analyzed independently by nulling all secondary independent sources.

Q2.99.1 MCQ 1999 1M Ans: C

To find the transfer function between y_1 and y_2, we apply Mason's Gain Formula:

- Forward paths from y_1 to y_2: $P_1 = a$, $P_2 = b$
- Individual loops: $L_1 = c$
- Since the loop L_1 touches both forward paths P_1 and P_2, the path factors are $\Delta_1 = 1 - 0 = 1$ and $\Delta_2 = 1 - 0 = 1$.
- The system determinant is $\Delta = 1 - L_1 = 1 - c$.

$$\text{T.F.} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{a(1-0) + b(1-0)}{1-c} = \frac{a+b}{1-c}$$

$$\boxed{(C)\ \frac{a+b}{1-c}}$$

Q2.95.1 NAT 1995 2M Ans: *

To find the overall closed-loop transfer function of the given signal flow graph, we apply Mason's Gain Formula:

- Forward path: $P_1 = G_1G_2G_3$
- Individual loops: $L_1 = -G_1G_2H_1, \quad L_2 = -G_2G_3H_2$

Since both loops touch the single forward path, the path factor is $\Delta_1 = 1 - 0 = 1$. There are no non-touching loops, so the system determinant Δ is given by:

$$\Delta = 1 - (L_1 + L_2) = 1 + G_1G_2H_1 + G_2G_3H_2$$

Applying Mason's Gain Formula:

$$\text{T.F.} = \frac{P_1\Delta_1}{\Delta} = \frac{G_1G_2G_3(1-0)}{1 + G_1G_2H_1 + G_2G_3H_2}$$

$$\boxed{\frac{G_1G_2G_3}{1 + G_1G_2H_1 + G_2G_3H_2}}$$

DEBUG INFO

Chapter II

Block Diagram and Signal Flow Graphs

Total Questions : 12

MCQ : 9

MSQ : 0

NAT : 3

SOLUTIONS

III

Time Response Analysis

Topics

1. Impulse, Step, Ramp and Parabolic Response
2. Concept of Damping in 2nd Order Systems
3. Complex Frequency and Damping in RLC Circuits
4. Step Response of 2nd Order Systems
5. Time Domain Performance Parameters and their variation
6. Steady State Error - Unity and Non-Unity Feedback
7. Error Constants and System Types

Q3.26.1 NAT | 2026 | 2M | Ans: 0.49 to 0.51 | ● ○ ○

$$G(s) = \frac{10(s+1)}{s(s^2+2s+5)}$$

Since the system has a single pole at the origin (s^1 in the denominator), it is a **Type-1 system**. For a Type-1 system, the steady-state error due to a unit ramp input is determined using the velocity error coefficient (K_v).

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s \cdot 10(s+1)}{s(s^2+2s+5)} = \frac{10(0+1)}{0^2+2(0)+5} = \frac{10}{5} = 2$$

The steady-state error (e_{ss}) : $e_{ss} = \dfrac{1}{K_v} = \dfrac{5}{10} = 0.5$

$$\boxed{0.5}$$

Key Points

Condition	e_{ss}
Type = Input	Constant
Type > Input	0
Type < Input	∞

- **Type-0 & Step:** $e_{ss} = \dfrac{A}{1+K_p}$
- **Type-1 & Ramp:** $e_{ss} = \dfrac{A}{K_v}$
- **Type-2 & Parabolic:** $e_{ss} = \dfrac{A}{K_a}$

Shortcut Method:
For any Type-1 system expressed in polynomial form, K_v can be directly computed as:

$$K_v = \frac{\text{Numerator constant}}{\text{Denominator constant}} = \frac{10}{5} = 2$$

Q3.25.1 NAT | 2025 | 2M | Ans: MTA | ● ○ ○ | CLICK FOR VIDEO SOLUTION

From the given block diagram, the closed-loop output equation at the summing junction is:

$$X(s) = D(s)\cdot\frac{1}{s^n} + Y(s)\cdot\left(-\frac{1}{s^2}\right)$$

$$Y(s) = X(s)\cdot\frac{1}{s+1} = \left[\frac{D(s)}{s^n} - \frac{Y(s)}{s^2}\right]\frac{1}{s+1}$$

$$Y(s)\left[1 + \frac{1}{s^2(s+1)}\right] = \frac{D(s)}{s^n(s+1)}$$

$$\frac{Y(s)}{D(s)} = \frac{s^2}{s^n(s^3+s^2+1)}$$

The characteristic equation of this closed-loop control system is given by the denominator polynomial:

$$q(s) = s^3 + s^2 + 1 = 0$$

This polynomial has poles lying in the right half of the s-plane.

Because the system is inherently unstable, its time-domain response $y(t)$ grows unboundedly as $t \to \infty$. Consequently, the **Final Value Theorem is not applicable**, and a steady-state value cannot be determined.

Key Points

Any third-order system $as^3 + bs^2 + cs + d = 0$, is stable if $bc > ad$. When $bc = ad$, the system becomes marginally stable with poles on the $j\omega$-axis.

Mistakes to be avoided

Final Value Theorem Limit Condition: The relation $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ holds valid **if and only if** all poles of $sF(s)$ are strictly in the left half of the s-plane (LHP) or at the origin (with at most a single pole at the origin).

Q3.24.1 MCQ 2024 2M Ans: A

The transfer function for the error $E(s) = R(s) - Y(s)$ is given by:

$$E(s) = R(s) - T(s)R(s) = R(s)\left[1 - \frac{G(s)}{1+G(s)H(s)}\right]$$

$$E(s) = R(s)\left[1 - \frac{\frac{1}{s(s+1)}}{1+\frac{K}{s(s+1)}}\right] = R(s)\left[1 - \frac{1}{s^2+s+K}\right] = R(s)\left[\frac{s^2+s+K-1}{s^2+s+K}\right]$$

For a unit step input reference, $r(t) = u(t) \implies R(s) = \frac{1}{s}$.

$$E(s) = \frac{1}{s}\left[\frac{s^2+s+K-1}{s^2+s+K}\right]$$

Using the Final Value Theorem, the steady-state error e_{ss} is:

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\cdot\frac{1}{s}\left[\frac{s^2+s+K-1}{s^2+s+K}\right] = \frac{0+0+K-1}{0+0+K} = \frac{K-1}{K}$$

$$\boxed{(A)\ \frac{K-1}{K}}$$

Mistakes to be avoided

Do not use the standard unity feedback steady-state error formula $e_{ss} = \frac{1}{1+K_p}$ directly, as this system has a non-unity feedback factor K.

Q3.24.2 MCQ 2024 2M Ans: A CLICK FOR VIDEO SOLUTION

The characteristic equation of a standard second-order system is $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$, with roots (poles) given by $s = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$.

- **Damping Ratio (ξ):**
  $\theta = 180^\circ - 120^\circ = 60^\circ$
  The shaded region lies to the left of this angular line, which corresponds to an increased damping ratio ($\theta \le 60^\circ$):
  $$\xi = \cos\theta \ge \cos(60^\circ) = \frac{1}{2}$$
- **Natural Frequency (ω_n):**
  The shaded region lies outside the semi-circular arc of radius 2. Since the distance from the origin represents ω_n, we have:
  $$\omega_n \ge 2$$

$$\boxed{(A)\ \xi \ge \frac{1}{2} \text{ and } \omega_n \ge 2}$$

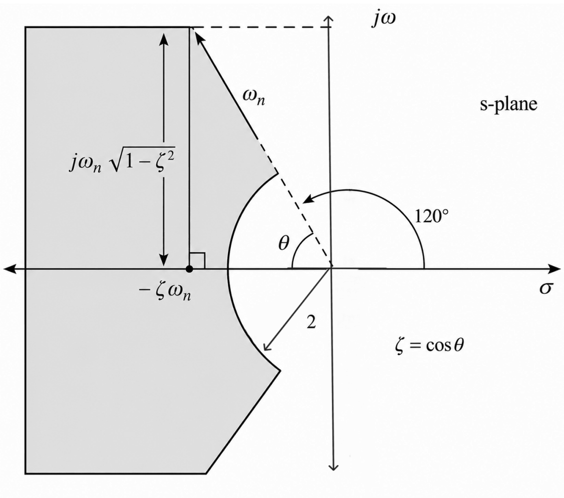


Fig. Solution Q3.24.2

Q3.21.1 MSQ | 2021 | 1M | Ans: A,B | ● ● ○

An oscillatory step response with oscillations dying down over time indicates that the system is stable and underdamped in nature.

- **Order of the System :** A first-order system has a characteristic equation of the form $\tau s + 1 = 0$, giving a single real pole which produces a purely exponential response without any oscillation. For a real system to exhibit oscillatory behavior, its least possible order must be 2 (requiring at least a complex conjugate pole-pair). Thus, it is of **atleast second order**.
- **Damping Behavior :** The decaying oscillations imply that the system contains at least one dominant complex-conjugate pole-pair with a damping ratio in the range $0 < \xi < 1$ (underdamped condition).
- **Presence of Real Poles :** Higher-order systems (such as a 3^{rd}-order system) can possess both real poles and an underdamped complex pole-pair while still maintaining an overall underdamped, oscillatory transient response. Therefore, it is incorrect to infer that the system cannot have a real pole.

$\boxed{\text{(A) that it is of at least second order.}}$ $\boxed{\text{(B) that it has at least one pole-pair that is underdamped.}}$

Q3.21.2 NAT | 2021 | 2M | Ans: 1 | ● ○ ○

Given the system transfer function and input signal:

$$G(s) = \frac{Y(s)}{X(s)} = \frac{2}{s+1}, \quad X(s) = \mathcal{L}\{\mu(t)\} = \frac{1}{s}$$

$$Y(s) = G(s)X(s) = \frac{2}{s(s+1)}$$

Given : $e(t) = y(t) - \mu(t)$
Taking the Laplace transform on both sides:

$$E(s) = Y(s) - \frac{1}{s} = \frac{2}{s(s+1)} - \frac{1}{s} = \frac{2-(s+1)}{s(s+1)} = \frac{1-s}{s(s+1)}$$

Applying the Final Value Theorem to determine the steady-state limit as $t \to \infty$:

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\left[\frac{1-s}{s(s+1)}\right] = \lim_{s\to 0}\frac{1-s}{s+1} = \frac{1}{1} = 1$$

$$\boxed{1}$$

Mistakes to be avoided

Do not blindly apply the standard steady-state error formula $e_{ss} = \frac{1}{1+K_p}$ without verifying the error definition sign convention. Doing so yields -1 instead of $+1$.

Q3.20.1 NAT | 2020 | 1M | Ans: 0.78 to 0.79 | ● ○ ○

$$s = -3 \pm j4$$

Comparing this with the characteristic pole position expression for an underdamped second-order system:

$$s = -\xi\omega_n \pm j\omega_d = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$\omega_d = \omega_n\sqrt{1-\xi^2} = 4\,\text{rad/s}$$

The maximum value of a step response occurs at the peak time t_p. This peak time corresponds to the first overshoot $(n = 1)$:

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{4} \approx 0.7853\ \text{seconds}$$

$$\boxed{0.79}$$

Key Points

The transient response peak time formula for a standard second-order control system is $t_p = \dfrac{n\pi}{\omega_d}$.

Q3.20.2 NAT 2020 2M Ans: 4

From the given block diagram in Fig. (a), the closed-loop transfer function (CLTF) is found as:

$$\text{CLTF} = \frac{Y(s)}{R(s)} = \frac{\frac{K}{\tau s+1}}{1+\frac{K}{\tau s+1}} = \frac{K}{\tau s + 1 + K}$$

The steady-state output value is found via the Final Value Theorem for the input $r(t) = 10\,u(t) \implies R(s) = \dfrac{10}{s}$:

$$y(\infty) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} s\left[\frac{K}{\tau s + 1 + K}\cdot\frac{10}{s}\right]$$

$$y(\infty) = 10\cdot\left(\frac{K}{1+K}\right)$$

From the time response curve in Fig. (b), the steady-state value of the output is:

$$y(\infty) = 8$$

$$\therefore\ 10\cdot\left(\frac{K}{1+K}\right) = 8 \implies 10K = 8 + 8K \implies K = 4$$

$$\boxed{4}$$

Key Points

- The DC gain of a stable system is obtained by substituting $s = 0$ into its closed-loop transfer function.
- For any step input of magnitude A, the final steady-state output value is given by $y(\infty) = A \times (\text{DC Gain})$.

Q3.19.1 NAT 2019 2M Ans: 0.46 to 0.48

$$G(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+3} \implies sY(s) + 3Y(s) = X(s)$$

Taking the inverse Laplace transform, the differential equation governing the system is:

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

Applying a unit-step input $x(t) = u(t)$ for $t \geq 0$, we have $x(t) = 1$. The complete time response $y(t)$ consists of both the forced response and natural response:

$$y(t) = y(\infty) + \left[y(0) - y(\infty)\right]e^{-3t}$$

Given the initial condition $y(0) = 3$, we determine the steady-state value $y(\infty)$ by setting $\dfrac{dy(t)}{dt} = 0$:

$$3y(\infty) = 1 \implies y(\infty) = \frac{1}{3}$$

Substituting these values back into the response equation:

$$y(t) = \frac{1}{3} + \left(3 - \frac{1}{3}\right)e^{-3t} = \frac{1}{3} + \frac{8}{3}e^{-3t}$$

Now, calculating the value of $y(t)$ at time $t = 1$ s:

$$y(1) = \frac{1}{3} + \frac{8}{3}e^{-3} = 0.4661 \approx 0.47$$

$$\boxed{0.47}$$

Q3.17.1 | NAT | 2017 | 2M | Ans: 0.32 to 0.4 | ● ○ ○

The characteristic equation of the closed-loop system is given by $1 + G(s)H(s) = 0$:

$$1 + \frac{k}{s(s+1)} \cdot (1 + k_p s) = 0$$

$$s^2 + s(1 + kk_p) + k = 0$$

Comparing this to the standard form of a second-order characteristic equation ($s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$):

$$\omega_n^2 = k \implies \omega_n = \sqrt{k} \;\; and \quad 2\zeta\omega_n = 1 + kk_p$$

Given that the undamped natural frequency $\omega_n = 4\,\text{rad/s}$:

$$\sqrt{k} = 4 \implies k = 16$$

Substituting the given damping ratio $\zeta = 0.8$, $\omega_n = 4$, and $k = 16$:

$$2(0.8)(4) = 1 + 16k_p$$

$$k_p = \frac{5.4}{16} = 0.3375$$

$$\boxed{0.337}$$

Q3.16.1 | MCQ | 2016 | 2M | Ans: MTA | ● ○ ○

The input is a unit-step signal, so $U(s) = \frac{1}{s}$. The steady-state value $c(\infty)$ is given by:

$$c(\infty) = \lim_{s\to 0} s \cdot C(s) = \lim_{s\to 0} s \cdot H(s) \cdot U(s) = \lim_{s\to 0} H(s)$$

For $P(s)$:

$$P(s) = \frac{-1}{s+1}$$

$$c(\infty) = \lim_{s\to 0} \frac{-1}{s+1} = -1$$

This matches curve **(1)**.

For $Q(s)$:

$$Q(s) = \frac{2(s-1)}{(s+10)(s+2)}$$

$$c(\infty) = \lim_{s\to 0} \frac{2(s-1)}{(s+10)(s+2)} = -0.5$$

Doesn't match any option

For $R(s)$:

$$R(s) = \frac{1}{(s+1)^2}$$

$$c(\infty) = \lim_{s\to 0} \frac{1}{(s+1)^2} = 1$$

This matches curve **(3)**.

Key Points

For a unit-step input $U(s) = \frac{1}{s}$, the steady-state output value equals the DC gain of the system, i.e., $c(\infty) = H(0)$.

Q3.15.1 | MCQ | 2015 | 1M | Ans: C | ● ○ ○

The given sinusoidal force input is:

$$F(t) = 10\sin(70t)\,\text{N}$$

$$\omega = 70\,\text{rad/s}$$

Fig. Solution Q3.15.1

For a stable linear time-invariant (LTI) system operating in the steady state, the output response changes only in magnitude and phase, while its frequency remains identical to the input frequency.

Therefore, the steady-state vibration frequency of the mass is: $f = \frac{\omega}{2\pi} = \frac{70}{2\pi} = \frac{35}{\pi}$ Hz

An object illuminated by a stroboscope appears completely stationary when the strobe flashing frequency matches the fundamental frequency of the object's periodic motion. Thus, the strobe frequency must be $\frac{35}{\pi}$ Hz.

$$\boxed{\text{(C) } \frac{35}{\pi}}$$

Q3.15.2 NAT 2015 2M Ans: 100

The given system transfer function is: $G(s) = \dfrac{1}{s^2 + (0 \times s) + 1}$
Comparing this with the standard second-order characteristic equation, $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$, we get:

$$\omega_n = 1 \text{ rad/s}, \quad \zeta = 0$$

Since the damping ratio $\zeta = 0$, the system is completely **undamped**. Percentage overshoot ($\%M_p$) is:

$$\%M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\%$$

Substituting $\zeta = 0$: $$\%M_p = e^{-0} \times 100\% = 1 \times 100\% = 100\%$$

$$\boxed{100}$$

Q3.14.1 MCQ 2014 1M Ans: A

At low frequencies ($\omega \to 0$), the Laplace variable $s = j\omega \to 0$. The low-frequency response is dominated by the pole closest to the origin.

$$G_p(s) = \frac{20}{(s+0.1)(s+2)(s+100)}$$

The poles of the system are located at $s = -0.1$, $s = -2$, and $s = -100$. The dominant pole nearest to the origin is at $s = -0.1$.

To approximate the model by retaining only this dominant pole while keeping the low-frequency steady-state gain unchanged, we approximate the insignificant high-frequency poles by setting $s \to 0$ in those terms:

$$G_p(s) \approx \frac{20}{(s+0.1) \times 2 \times 100} = \frac{20}{200(s+0.1)} = \frac{0.1}{s+0.1}$$

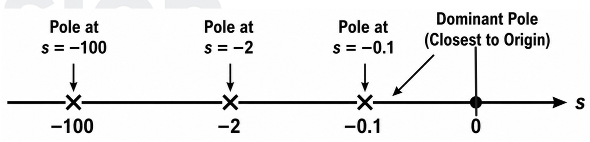


Fig. Solution Q3.14.1

$$\boxed{(A) \ \frac{0.1}{s+0.1}}$$

Key Points

- **Dominant Poles:** Poles closer to the imaginary axis dominate the transient response as they have larger time constants.
- **Insignificant Poles:** Poles located far to the left (typically > 5 times the real part of dominant poles) can be neglected.

Mistakes to be avoided

Do not simply remove the other terms without normalizing the numerator; failing to account for the gain will result in an incorrect steady-state value.

Q3.13.1 MCQ | 2013 | 2M | Ans: C | ● ○ ○

Given the open-loop transfer function of a dc motor:

$$\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s} = \frac{10}{1+\tau s}$$

$$\tau = 10$$

The closed-loop transfer function is given by:

$$\frac{\omega(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\dfrac{10K_a}{1+10s}}{1+\dfrac{10K_a}{1+10s}} = \frac{10K_a}{1+10K_a+10s}$$

To find the closed-loop time constant, we rearrange the denominator into the standard form $(1+\tau_{cl}s)$:

$$\frac{\omega(s)}{R(s)} = \frac{\dfrac{10K_a}{1+10K_a}}{1+\left(\dfrac{10}{1+10K_a}\right)s} \implies \tau_{cl} = \frac{10}{1+10K_a}$$

According to the problem statement, the closed-loop time constant is reduced by one hundred times compared to the open-loop time constant:

$$\tau_{cl} = \frac{\tau}{100}$$

$$\frac{10}{1+10K_a} = \frac{1}{100}\times 10$$

$$K_a = 9.9 \approx 10$$

$$\boxed{(C)\ \ 10}$$

Key Points

- The standard form of a first-order system transfer function is $\dfrac{K}{1+\tau s}$, where τ represents the time constant.
- Negative feedback reduces the time constant of a system by a factor of $(1+GH)$, thereby increasing the speed of response (bandwidth).

Mistakes to be avoided

Incorrect Time Constant Extraction: Do not mistake the coefficient of s as the time constant before ensuring the constant term in the denominator is normalized to 1. Always bring the expression to the form $1+\tau s$.

Q3.11.1 MCQ | 2011 | 2M | Ans: D | ● ○ ○

Closed-loop Transfer Function: $T(s) = \dfrac{C(s)}{R(s)} = \dfrac{G(s)}{1+G(s)H(s)} = \dfrac{\dfrac{6}{s+5}}{1+\dfrac{6}{s+5}} = \dfrac{6}{s+11}$

Unit-step Input: $R(s) = \dfrac{1}{s}$

$$C(s) = R(s)\cdot T(s) = \frac{1}{s}\cdot\frac{6}{s+11}$$

Partial Fraction Decomposition:

$$C(s) = \frac{6}{11}\cdot\frac{1}{s} - \frac{6}{11}\cdot\frac{1}{s+11}$$

Taking the inverse Laplace transform to find the time-domain response $c(t)$:

$$c(t) = \left(\frac{6}{11} - \frac{6}{11}e^{-11t}\right)u(t)$$

$$\boxed{(D)\ \frac{6}{11} - \frac{6}{11}e^{-11t}}$$

Key Points

The Laplace transform pairs utilized here are $\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$ and $\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$.

Q3.11.2 MCQ 2011 2M Ans: B

The closed-loop characteristic equation is

$$1 + \frac{K}{(s+5)^3} = 0$$

$$(s+5)^3 + K = 0$$

$$s^3 + 15s^2 + 75s + (125+K) = 0$$

Since the dominant closed-loop poles have damping ratio $\zeta = 0.5$, the characteristic equation can be written as

$$(s+p)(s^2 + \omega_n s + \omega_n^2) = 0$$

$$s^3 + (p+\omega_n)s^2 + (\omega_n^2 + p\omega_n)s + p\omega_n^2 = 0$$

Comparing coefficients,

$$\omega_n^2 + p\omega_n = 75$$
$$\omega_n(\omega_n + p) = 75$$
$$\omega_n(15) = 75$$
$$\omega_n = 5$$

$$p + \omega_n = 15$$
$$p + 5 = 15$$
$$p = 10$$

$$p\omega_n^2 = 125 + K$$
$$(10)(5)^2 = 125 + K$$
$$K = 125$$

$$\boxed{(B)\ 125}$$

Key Points

Since the characteristic equation is of third order, it is factorized as $(s+p)(s^2 + 2\zeta\omega_n s + \omega_n^2) = 0$, where the quadratic factor represents the dominant complex conjugate pole pair and the remaining pole is real.

Q3.10.1 MCQ 2010 2M Ans: B

$$G(s) = \frac{2}{s(s+1)}$$

Since it is a unity feedback system ($H(s) = 1$), we analyze the static error constants for a unit ramp input:

- **Velocity Error Constant (K_v):** $K_v = \lim_{s\to 0} sG(s)H(s) = \lim_{s\to 0} s \cdot \dfrac{2}{s(s+1)} = \dfrac{2}{0+1} = 2$
- **Steady-State Error (e_{ss}):** For a standard unit ramp input, the steady-state error is given by:

$$e_{ss} = \frac{1}{K_v} = \frac{1}{2} = 0.5$$

$$\boxed{(B)\ 0.5}$$

Q3.10.2 MCQ 2010 2M Ans: D ● ○ ○

Given the open-loop transfer function $G(s) = \dfrac{k}{s(s+3)}$ with a unity feedback system ($H(s) = 1$).

The closed-loop transfer function is expressed as: $T(s) = \dfrac{G(s)}{1 + G(s)H(s)} = \dfrac{k}{s^2 + 3s + k}$

Comparing $s^2 + 3s + k = 0$ with the standard second-order characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$:

$$\omega_n = \sqrt{k} \quad \text{and} \quad 2\zeta\omega_n = 3$$

Substituting the given damping ratio $\zeta = 0.5$:

$$2 \times 0.5 \times \sqrt{k} = 3 \implies \sqrt{k} = 3 \implies k = 9$$

$$\boxed{(D)\ 9}$$

Q3.09.1 MCQ 2009 2M Ans: B ● ○ ○

The characteristic equation for the given unity feedback system ($H(s) = 1$) is: $1 + G(s)H(s) = 0$

$$1 + \frac{K(s+b)}{s^2(s+20)} = 0 \implies K = -\frac{s^3 + 20s^2}{s+b}$$

To find the breakaway or break-in points where the roots meet, we set $\dfrac{dK}{ds} = 0$:

$$\frac{(s+b)(3s^2+40s) - (s^3+20s^2)(1)}{(s+b)^2} = 0$$

$$s\left[2s^2 + (3b+20)s + 40b\right] = 0$$

Thus, $s = 0$ is one intersection point. For all three roots to meet at a single point, the remaining two roots from the quadratic equation must also coincide at $s = 0$. For a single point of convergence, the discriminant of the quadratic factor $2s^2 + (3b+20)s + 40b = 0$ must be zero ($B^2 - 4AC = 0$):

$$(3b+20)^2 - 4(2)(40b) = 0$$

$$9b^2 - 200b + 400 = 0$$

$$(b-20)\left(b - \frac{20}{9}\right) = 0 \implies b = 20 \text{ or } b = \frac{20}{9}$$

If $b = 20$, the open-loop zero at $s = -20$ cancels the open-loop pole at $s = -20$, reducing the system to a second-order system (only 2 roots total), which contradicts having three roots.

$$\boxed{(B)\ \frac{20}{9}}$$

Q3.08.1 MCQ 2008 2M Ans: B ● ○ ○

The closed-loop transfer function is expressed as:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{100}{s^2 + ps + 100}$$

Comparing this with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = 100 \implies \omega_n = 10\,\text{rad/s}$$

The peak time t_p for a unit step input is given by:

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} = \frac{\pi}{8}\text{ s}$$

$$\frac{\pi}{10\sqrt{1-\xi^2}} = \frac{\pi}{8}$$

$$1-\xi^2 = 0.64 \implies \xi = 0.6$$

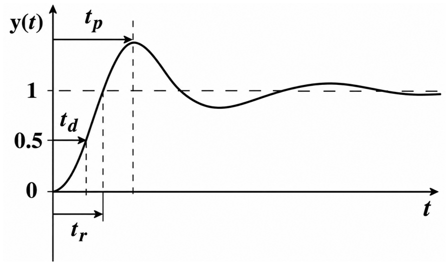


Fig. Solution Q3.08.1

$$\boxed{(B)\ \ 0.6}$$

Q3.08.2 MCQ 2008 2M Ans: B

From the comparison of the characteristic equation in the previous question:

$$s^2 + ps + 100 = s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

Equating the coefficients of s:

$$p = 2\xi\omega_n$$

Substituting the calculated values $\omega_n = 10\,\text{rad/s}$ and $\xi = 0.6$:

$$p = 2\times 0.6\times 10 \implies p = 12$$

$$\boxed{(B)\ \ 12}$$

Q3.05.1 MCQ 2005 2M Ans: C

Method 1

$$G(s) = \frac{K}{s^2}$$

The error signal in the Laplace domain is given by:

$$E(s) = \frac{R(s)}{1+G(s)H(s)}$$

For a unit step input, $R(s) = \frac{1}{s}$. Using the Final Value Theorem, the steady-state error e_{ss} is calculated as:

$$e_{ss} = \lim_{s\to 0}\frac{sR(s)}{1+G(s)}$$

$$e_{ss} = \lim_{s\to 0}\frac{s\cdot\frac{1}{s}}{1+\frac{K}{s^2}} = \lim_{s\to 0}\frac{1}{1+\frac{K}{s^2}}$$

$$e_{ss} = \frac{1}{1+\infty} = 0$$

Method 2

Alternatively, we can evaluate the steady-state error using the static error coefficients method :

$$K_p = \lim_{s\to 0} G(s)H(s)$$

Substituting $G(s) = \frac{K}{s^2}$ and $H(s) = 1$:

$$K_p = \lim_{s\to 0}\frac{K}{s^2} = \infty$$

The steady-state error e_{ss} for a standard unit step input is given by the formula:

$$e_{ss} = \frac{1}{1+K_p}$$

Substituting the value of $K_p = \infty$:

$$e_{ss} = \frac{1}{1+\infty} = 0$$

$$\boxed{(C)\ \ 0}$$

Q3.05.2 MCQ 2005 2M Ans: A

The controller block transfer function $G_c(s)$ can be found using the inverting op-amp configuration:

$$G_c(s) = \frac{V_o}{V_{in}} = -A \cdot \frac{Z_f}{Z_1}$$

where the pre-amplifier has a gain $A = -1$.

Thus, $G_c(s) = -\left(-\frac{Z_f}{Z_1}\right) = \frac{Z_f}{Z_1}$.

$$Z_f = 100\,\text{k}\Omega + \frac{1}{10^{-6}s} = 10^5 + \frac{1}{10^{-6}s}$$

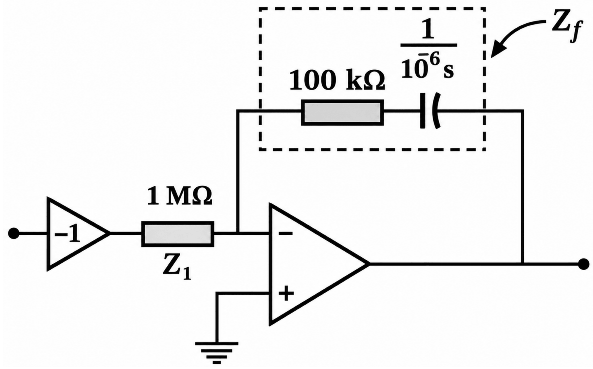


Fig. Solution Q3.05.2

Substituting the values of Z_f and Z_1:

$$G_c(s) = \frac{10^5 + \frac{10^6}{s}}{10^6} = \frac{s+10}{10s} = \frac{0.1s+1}{s}$$

$$\boxed{(A)\ \frac{1+0.1s}{s}}$$

Key Points

For an inverting amplifier configuration, the ideal voltage transfer function is given by $-\frac{Z_f}{Z_1}$.

Q3.05.3 MCQ 2005 2M Ans: C

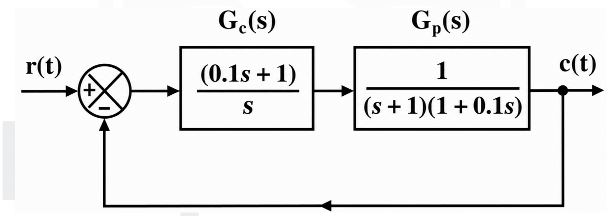


Fig. Solution Q3.05.3

The simplified forward path transfer function $G(s)$ is:

$$G(s) = G_c(s)G_p(s) = \left(\frac{0.1s+1}{s}\right)\left(\frac{1}{(s+1)(1+0.1s)}\right)$$

$$G(s) = \frac{1}{s(s+1)}$$

The closed-loop transfer function with unity feedback ($H(s) = 1$) is:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{1}{s(s+1)+1} = \frac{1}{s^2+s+1}$$

Comparing this with the standard system equation: $s^2 + 2\xi\omega_n s + \omega_n^2$

$$\omega_n^2 = 1 \implies \omega_n = 1 \text{ rad/sec}$$

$$2\xi\omega_n = 1 \implies \xi = 0.5$$

The percentage peak overshoot ($\%M_p$) is:

$$\%M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \times 100\%$$

$$\%M_p = e^{-\frac{0.5\pi}{\sqrt{1-0.5^2}}} \times 100\% = 16.3\%$$

$$\boxed{(C)\ 16.3\%}$$

Q3.04.1 MCQ 2004 1M Ans: A

To find the steady-state error e_{ss} due to the disturbance input $D(s)$, we set the reference input $R(s) = 0$. The reduced block diagram with respect to the disturbance is shown below:

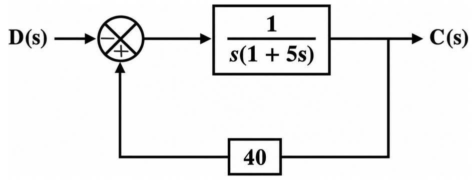


Fig. Solution Q3.04.1

The standard form of steady-state error is given by: $e_{ss} = \lim_{s\to 0} \dfrac{s \cdot R(s)}{1 + G(s)H(s)}$
Here, the effective input to the forward path block is :

$$\text{Input} = -D(s) = -\frac{1}{s}$$

Substituting the input and the loop blocks into the standard formula:

$$e_{ss} = \lim_{s\to 0} \frac{s \cdot \left(-\frac{1}{s}\right)}{1 + \frac{40}{s(1+5s)}} == \frac{-1}{\frac{40}{0} + 1} = \frac{-1}{\infty} = 0$$

$$\boxed{(A)\ 0}$$

Key Points

- When finding steady-state error due to a disturbance input $D(s)$, set $R(s) = 0$ and trace the loop from the disturbance injection node.
- Error for non-unity feedback: $E(s) = \dfrac{R(s)}{1 + G(s)H(s)}$.

Mistakes to be avoided

- $e_{ss} = \lim_{s\to 0} s[R(s) - C(s)]$ is strictly for unity feedback systems ($H(s) = 1$). Do not use it for non-unity feedback systems.
- **Conversion to Unity Feedback:** Converting a non-unity system to an equivalent unity feedback system results in a wrong answer for error because the "Error Signal" is redefined.
  - The Core Conflict: While the conversion preserves the overall Transfer Function $\dfrac{C(s)}{R(s)}$, it changes the physical location and mathematical definition of the error signal.
  - Hence, for steady-state error analysis, always solve the system in its original non-unity form.

Q3.04.2 MCQ 2004 1M Ans: A

The step response of a first-order system is governed by the expression:

$$y(t) = Y_{ss}(1 - e^{-t/\tau})$$

where $\tau = RC$ represents the time constant of the system and Y_{ss} is the steady-state final value.
The settling time t_s for a specified error band Δ is the time required for the output to stay within $(1 - \Delta)$ of its final value:

$$1 - e^{-t_s/\tau} = 1 - \Delta \implies e^{-t_s/\tau} = \Delta$$

$$t_s = -\tau \ln(\Delta)$$

- For a 5% error band ($\Delta = 0.05$): $t_s = -\tau \ln(0.05) \approx 2.996\,\tau \approx 3\,RC$
- For a 2% error band ($\Delta = 0.02$): $t_s = -\tau \ln(0.02) \approx 3.912\,\tau \approx 4\,RC$

Thus, a 5% settling time is equal to three times the time constant.

$\boxed{\text{(A) three times the time constant}}$

Key Points

- Time constant $\tau = RC$ is the time taken to reach 63.2% of the final steady-state value.
- $t_s = 3\,\tau$ corresponds to 95% response completion (5% tolerance band).
- $t_s = 4\,\tau$ corresponds to 98% response completion (2% tolerance band).

Q3.04.3 | MCQ | 2004 | 2M | Ans: B

Given the maximum percentage overshoot is $M_p = 16\% = 0.16$.
The maximum overshoot formula is:

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\ln(0.16) = -\frac{\pi\zeta}{\sqrt{1-\zeta^2}}$$

$$\frac{\zeta^2}{1-\zeta^2} = \left(\frac{1.8326}{\pi}\right)^2$$

$$\zeta^2 = 0.3403 - 0.3403\zeta^2 \implies \zeta \approx 0.5$$

The **decay ratio (DR)** is defined as the ratio of the 2nd peak overshoot to the 1st peak overshoot:

$$\text{DR} = \frac{2^{\text{nd}}\text{ peak overshoot}}{1^{\text{nd}}\text{ peak overshoot}} = \frac{AK \cdot e^{-\frac{3\pi\zeta}{\sqrt{1-\zeta^2}}}}{AK \cdot e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}}$$

$$\text{DR} = e^{-\left(\frac{3\pi\zeta}{\sqrt{1-\zeta^2}} - \frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)} = e^{-\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}}$$

Since $M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.16$:

$$\text{DR} = \left(e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right)^2 = (0.16)^2 = 0.0256$$

$\boxed{(B)\ \ (0.5, 0.0256)}$

Key Points

Decay Ratio: It represents how quickly oscillations die out and is mathematically equal to the square of the maximum overshoot: Decay Ratio $= e^{-\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}} = (M_p)^2$.

Q3.03.1 | MCQ | 2003 | 1M | Ans: D

Given the forward path transfer function $G(s) = \dfrac{A}{1+s\tau}$ and feedback path gain $H(s) = \beta$. The closed-loop transfer function $T(s)$ for the negative feedback system is:

$$T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{A}{1+s\tau}}{1+\frac{A\beta}{1+s\tau}} = \frac{A}{1+s\tau+A\beta}$$

To find the new time constant, rewrite the denominator in standard standard first-order form $(1+s\tau')$:

$$T(s) = \frac{A}{(1+A\beta)+s\tau} = \frac{\frac{A}{1+A\beta}}{1+s\left(\frac{\tau}{1+A\beta}\right)}$$

Comparing this with the standard form $T(s) = \dfrac{A_{cl}}{1 + s\tau'}$, the closed-loop time constant τ' is found to be:

$$\tau' = \frac{\tau}{1 + A\beta}$$

$$\boxed{(D) \quad \frac{\tau}{(1 + A\beta)}}$$

Q3.03.2 MCQ 2003 1M Ans: B

Method 1

Using the transfer function approach:

$$\frac{C(s)}{R(s)} = \frac{K_p K}{1 + s\tau + K_p K}$$

The error signal transfer function is:

$$\frac{E(s)}{R(s)} = 1 - \frac{C(s)}{R(s)} = \frac{1 + s\tau}{1 + s\tau + K_p K}$$

For a unit step input $R(s) = \dfrac{1}{s}$, the steady-state error e_{ss} via the Final Value Theorem is:

$$e_{ss} = \lim_{s \to 0} s \cdot E(s)$$

$$e_{ss} = \lim_{s \to 0} s \cdot \left[\frac{1}{s} \cdot \frac{1 + s\tau}{1 + s\tau + K_p K} \right] = \frac{1}{1 + KK_p}$$

Method 2

Using the error constant method:
The open-loop transfer function $G(s)H(s)$ for this unity feedback system is:

$$G(s)H(s) = K_p \cdot \frac{K}{1 + s\tau} = \frac{KK_p}{1 + s\tau}$$

Since there are no open-loop poles at the origin, this is a **Type 0** system.
The positional error constant K'_p is defined as:

$$K'_p = \lim_{s \to 0} G(s)H(s) = \lim_{s \to 0} \frac{KK_p}{1 + s\tau} = KK_p$$

The steady-state error for a unit step input is:

$$e_{ss} = \frac{1}{1 + K'_p} = \frac{1}{1 + KK_p}$$

$$\boxed{\frac{1}{1 + KK_p}}$$

Key Points

Shortcut for Type-0 System Constant (K'_p): For standard form open-loop transfer functions $\dfrac{N(s)}{D(s)}$ where there are no poles at the origin, K'_p simplifies directly to the ratio of the constant terms: $\dfrac{\text{Numerator constant}}{\text{Denominator constant}}$.

Q3.02.1 MCQ 2002 1M Ans: B

For a unity feedback system subjected to a unit ramp input of magnitude k, the steady-state tracking error is given by a constant value:

$$e_{ss} = \lim_{t \to \infty} e(t) = k\tau$$

The Integral of Squared Error (ISE) performance index is defined as:

$$\text{ISE} = \int_0^\infty e^2(t)\, dt$$

Since the error approaches a non-zero constant value $k\tau$ as $t \to \infty$, the integrand does not vanish at infinity. Thus, evaluating the integral yields:

$$\text{ISE} = \int_0^\infty (k\tau)^2\, dt = k^2\tau^2 \int_0^\infty dt = k^2\tau^2\, [t]_0^\infty = \infty$$

Because the steady-state error is sustained indefinitely, the cumulative squared error grows boundlessly.

$$\boxed{(B)\ \infty}$$

Key Points

- For any system where the steady-state error e_{ss} is a non-zero finite constant or grows with time, the integral of the squared error will always diverge to ∞.
- ISE converges to a finite value only when the steady-state error to the given input is exactly zero.

Q3.02.2

MCQ | 2002 | 2M | Ans: D

Given the forward path transfer function of a unity feedback system: $G(s) = \dfrac{(1+5s)(1+10s)(1+2s)}{(1+s)(1+8s)(1+20s)}$

The error signal $E(s)$ in the Laplace domain for a unity feedback system is given by: $E(s) = \dfrac{R(s)}{1+G(s)}$

For a unit impulse input $r(t) = \delta(t)$, its Laplace transform is: $R(s) = 1$
The given performance index is: $J = \int_0^\infty e(t)\,dt$
From the definition of the Laplace transform, we know that:

$$E(s) = \int_0^\infty e(t)e^{-st}\,dt$$

Taking the limit as $s \to 0$:

$$\lim_{s\to 0} E(s) = \int_0^\infty e(t)\,dt = J$$

$$J = E(0) = \frac{R(0)}{1+G(0)}$$

Now, evaluate $G(s)$ at $s = 0$:

$$G(0) = \frac{(1+0)(1+0)(1+0)}{(1+0)(1+0)(1+0)} = 1$$

Substituting $R(0) = 1$ and $G(0) = 1$ into the expression for J:

$$J = \frac{1}{1+1} = 0.5$$

$$\boxed{(D)\ 0.5}$$

Key Points

The integral of a time-domain signal from 0 to ∞ corresponds to its Laplace transform evaluated at $s = 0$, provided the integral converges (i.e., $\int_0^\infty f(t)\,dt = F(0)$).

Q3.02.3

MCQ | 2002 | 2M | Ans: C

Given the loop transfer function of a unity feedback system: $G(s)H(s) = \dfrac{10e^{-0.1s}}{s(s+5)(s^2+4s+16)}$

The system is of **Type-1** because there is a single pole at the origin.

Method 1

For a unit parabolic input $r(t) = \frac{t^2}{2}u(t)$, the Laplace transform is $R(s) = \frac{1}{s^3}$. The steady-state error e_{ss} using the final value theorem is:

$$e_{ss} = \lim_{s\to 0} \frac{s\cdot R(s)}{1+G(s)H(s)} = \lim_{s\to 0} \frac{s\cdot \dfrac{1}{s^3}}{1+\dfrac{10e^{-0.1s}}{s(s+5)(s^2+4s+16)}}$$

$$e_{ss} = \lim_{s \to 0} \frac{1}{s^2 \left[1 + \frac{10e^{-0.1s}}{s(s+5)(s^2+4s+16)}\right]} = \frac{1}{0} = \infty$$

Method 2

Using the static error coefficients method: For a standard parabolic input $r(t) = \frac{t^2}{2}$, the steady-state error is given by:

$$e_{ss} = \frac{1}{K_a}$$

where K_a is the parabolic (acceleration) error constant defined as:

$$K_a = \lim_{s \to 0} s^2 G(s)H(s)$$

$$K_a = \lim_{s \to 0} s^2 \left[\frac{10e^{-0.1s}}{s(s+5)(s^2+4s+16)}\right] = \lim_{s \to 0} \frac{10s \cdot e^{-0.1s}}{(s+5)(s^2+4s+16)} = 0$$

Now, calculating the steady-state error:

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \infty \text{ (infinity)}$$

$$\boxed{(C) \quad \infty}$$

Mistakes to be avoided

Handling the Time Delay Term ($e^{-0.1s}$): Do not get confused by the exponential term. Since $\lim_{s \to 0} e^{-0.1s} = 1$, it does not affect the system type or the final value computation.

Q3.02.4 NAT | 2002 | 5M | **Ans:** $50, 3.08 \times 10^{-3}$

1. Characteristic Equation & Parameters

From the block diagram, the characteristic equation is:

$$1 + KG(s) = 1 + \frac{Kn}{s(Js+B)} = 0$$

$$Js^2 + Bs + nK = 0 \implies s^2 + \frac{B}{J}s + \frac{nK}{J} = 0$$

Comparing with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$:

$$2\zeta\omega_n = \frac{B}{J}, \quad \omega_n^2 = \frac{nK}{J}$$

2. Damping Ratio from Overshoot

Given $M_p = 16\% = 0.16$:

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\zeta = \frac{-\ln(0.16)}{\sqrt{\pi^2 + \ln^2(0.16)}} \approx 0.504$$

3. Disturbance Analysis for K

With $R(s) = 0$ and step disturbance $D(s) = \frac{10}{s}$:

$$E(s) = \frac{n}{Js^2 + Bs + nK} \cdot \frac{10}{s}$$

Applying the Final Value Theorem for $e_{ss} = 0.2$ rad:

$$e_{ss} = \lim_{s \to 0} sE(s) = \frac{10n}{nK} = \frac{10}{K}$$

$$0.2 = \frac{10}{K} \implies K = 50$$

4. Calculation of Value J

Squaring and rearranging the parameter relations:

$$4\zeta^2 \left(\frac{nK}{J}\right) = \frac{B^2}{J^2} \implies J = \frac{B^2}{4\zeta^2 nK}$$

Given $B = 0.125$ Nm/rad/s and $n = 0.1$:

$$J = \frac{0.125^2}{4 \cdot 0.504^2 \cdot 0.1 \cdot 50} \approx 3.08 \times 10^{-3}\,\text{kg} \cdot \text{m}^2$$

$$\boxed{K = 50, \quad J = 3.08 \times 10^{-3}\,\text{kg} \cdot \text{m}^2}$$

Q3.00.1 MCQ | 2000 | 1M | Ans: A

The closed-loop transfer function (T.F.) is :

$$T(s) = \frac{\frac{9}{s(s+2)}}{1 + \frac{9}{s(s+2)}} = \frac{9}{s(s+2)+9} = \frac{9}{s^2+2s+9}$$

Comparing the characteristic equation $s^2 + 2s + 9 = 0$ with the standard second-order characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = 9 \implies \omega_n = 3$$

$$2\zeta\omega_n = 2 \implies 2\zeta \times 3 = 2 \implies \zeta = \frac{1}{3}$$

$$\boxed{(A)\ \frac{1}{3}}$$

Q3.99.1 MCQ | 1999 | 1M | Ans: C

The response of a first-order system to a unit step input is given by:

$$y(t) = y_{ss}(1 - e^{-t/\tau})$$

Given that at $t = 10$ s, the response reaches 50% of its final value ($y(t) = 0.5\, y_{ss}$):

$$0.5\, y_{ss} = y_{ss}(1 - e^{-10/\tau})$$

$$0.5 = 1 - e^{-10/\tau}$$

$$-\frac{10}{\tau} = \ln(0.5) \implies \tau = \frac{10}{0.693} \approx 14.43\,\text{s}$$

$$\boxed{(C)\ 14.4\,\text{s}}$$

Key Points

- The time taken to reach 50% of the final value is known as the delay time t_d for some conventions, where $t_{0.5} = \tau \ln(2) \approx 0.693\,\tau$.
- Similarly, the time to reach 63.2% is exactly $1\,\tau$.

Q3.97.1 MCQ | 1997 | 1M | Ans: B

Given transfer function:

$$\frac{C(s)}{R(s)} = \frac{25}{s^2+6s+25}$$

Comparing the denominator with $s^2 + 2\zeta\omega_n s + \omega_n^2$:

$$\omega_n^2 = 25 \implies \omega_n = 5\,\text{rad/s}$$

$$2\zeta\omega_n = 6 \implies 2\zeta(5) = 6 \implies \zeta = 0.6$$

The first maximum value of the step response occurs at the peak time t_p (where $n = 1$):

$$t_{\max} = t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$t_{\max} = \frac{\pi}{5\sqrt{1-(0.6)^2}} = \frac{\pi}{5 \times 0.8} = \frac{\pi}{4}$$

$$\boxed{(B)\ \frac{\pi}{4}}$$

Key Points

- Peak time formula: $t_p = \dfrac{n\pi}{\omega_d} = \dfrac{n\pi}{\omega_n\sqrt{1-\zeta^2}}$.
- Overshoot maxima occur at odd values of n ($n = 1, 3, 5, \ldots$), whereas undershoot minima occur at even values of n ($n = 2, 4, 6, \ldots$).

Q3.96.1 MCQ | 1996 | 2M | Ans: A-(ii), B-(iii), C-(i), D-(iv)

The correct matching between roots in the s-plane and their impulse response is determined below:

- **(a) Two imaginary roots:** Roots lie at $s = \pm j\omega$. The response is purely sinusoidal with constant amplitude $\implies$ (ii).
- **(b) Two complex roots in the left half plane:** Roots lie at $s = -\sigma \pm j\omega_d$. The response is an exponentially decaying sinusoid $\implies$ (iii).
- **(c) A single root on the negative real axis:** Root lies at $s = -\alpha$. The response is a purely decaying exponential curve $\implies$ (i).
- **(d) A single root at the origin:** Root lies at $s = 0$. The transfer function is $1/s$, and the impulse response is a step function (constant magnitude) $\implies$ (iv).

$$\boxed{\text{(a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)}}$$

Key Points

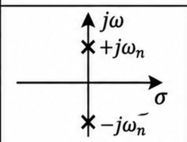
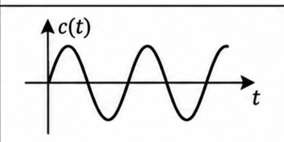
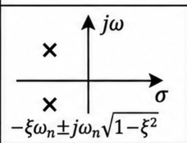
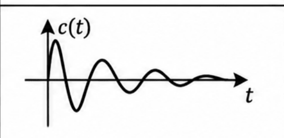
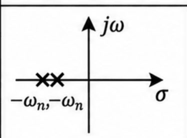
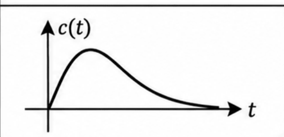
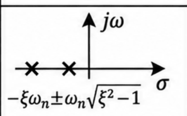
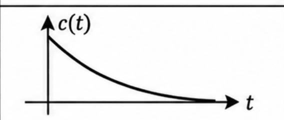
ξ	Damping	s-plane	Impulse Response	Poles	Stability
$\xi = 0$	Undamped			Imaginary Conjugate	Marginally stable
$0 < \xi < 1$	Underdamped			Complex conjugate roots on the LHS of s-plane	Stable
$\xi = 1$	Critically damped			Negative Equal Real Roots	Stable
$\xi > 1$	Overdamped			Negative Unequal Real Roots	Stable

Fig. Solution Q3.96.1

Q3.94.1 MCQ 1994 1M Ans: D

The closed-loop transfer function is given as:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{8}{s^2 + 0.6s + 9}$$

Since it is a unity feedback system, $H(s) = 1$.

Method 1

Using the open-loop conversion:

$$\frac{G(s)}{1 + G(s)} = \frac{8}{s^2 + 0.6s + 9}$$

$$G(s)[s^2 + 0.6s + 9] = 8[1 + G(s)]$$

$$G(s) = \frac{8}{s^2 + 0.6s + 1}$$

This is a **Type 0** system. The position error coefficient K_p is:

$$K_p = \lim_{s\to 0} G(s) = \frac{8}{1} = 8$$

The steady-state error e_{ss} is:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + 8} = \frac{1}{9} \approx 0.1$$

Method 2

Using the direct closed-loop formula for a unit step input $R(s) = \frac{1}{s}$:

$$e_{ss} = \lim_{s\to 0} s \cdot R(s) \cdot [1 - T(s)]$$

Substituting the values:

$$\begin{aligned} e_{ss} &= \lim_{s\to 0} s \cdot \frac{1}{s} \cdot \left[1 - \frac{8}{s^2 + 0.6s + 9}\right] \\ &= 1 - \frac{8}{9} \\ &= \frac{1}{9} \approx 0.1 \end{aligned}$$

$$\boxed{(D)\ 0.1}$$

Q3.93.1 NAT 1993 2M Ans: *

The general transfer function of a first-order system is: $\dfrac{Y(s)}{R(s)} = \dfrac{K}{\tau s + 1}$

Given a unit step input $R(s) = \frac{1}{s}$:

$$Y(s) = \frac{K}{s(\tau s + 1)} \implies y(t) = K(1 - e^{-t/\tau})$$

- At steady state $(t \to \infty)$, $y(\infty) = 2$ units $\implies K = 2$.
- At $t = 10$ mins, $y(10) = 1.264$ units:

$$1.264 = 2(1 - e^{-10/\tau}) \implies 0.632 = 1 - e^{-10/\tau}$$

$$e^{-10/\tau} = 0.368 \approx e^{-1} \implies \frac{10}{\tau} = 1 \implies \tau = 10\,\text{mins}$$

Substituting $K = 2$ and $\tau = 10$ mins back into the transfer function expression:

$$\boxed{\frac{Y(s)}{R(s)} = \frac{2}{10s + 1}}$$

DEBUG INFO

Chapter III

Time Response Analysis

Total Questions : 40

MCQ : 29

MSQ : 1

NAT : 10

SOLUTIONS

IV

Routh's Stability Criterion

Topics

1. Hurwitz Polynomial and Routh Array
2. Special Cases in Routh Stability
3. Mapping Theorem in Routh-Hurwitz

Q4.25.1 MCQ | 2025 | 1M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

To find the zeros of the transfer function $G(s)$, we determine the roots of its numerator polynomial:

$$G(s) = 1 + \frac{2s - 1}{s^3 + 5s^2 + 3s + 22} = \frac{s^3 + 5s^2 + 5s + 21}{s^3 + 5s^2 + 3s + 22}$$

The numerator polynomial representing the zeros is:

$$N(s) = s^3 + 5s^2 + 5s + 21 = 0$$

Method 1

Routh-Hurwitz Tabulation:

$$\begin{array}{c|cc} s^3 & 1 & 5 \\ s^2 & 5 & 21 \\ s^1 & \frac{5\times5 - 1\times21}{5} = \frac{4}{5} & 0 \\ s^0 & 21 & \end{array}$$

- All elements in the first column are strictly positive, meaning there are **no sign changes**.
- Thus, there are no roots in the right-half of the s-plane or the $j\omega$-axis.

Method 2

Shortcut Rule:

For a standard 3rd-order characteristic equation: $as^3 + bs^2 + cs + d = 0$

The condition for stability is:

$$\text{Inner Product} > \text{Outer Product}$$

$$b \times c > a \times d$$

$$a = 1, \quad b = 5, \quad c = 5, \quad d = 21$$

$$5 \times 5 > 1 \times 21$$

Thus, there are no roots in the right-half of the s-plane or the $j\omega$-axis.

Since $N(s)$ is a 3rd-order polynomial, all 3 roots must lie in the left-half of the s-plane.

$$\boxed{(D)\ 3}$$

Key Points

The number of sign changes in the first column of the Routh array equals the number of roots lying in the right-half of the s-plane (unstable roots).

Mistakes to be avoided

Wrong Polynomial Selection: Do not accidentally perform Routh analysis on the characteristic equation (denominator).

Q4.23.1 NAT | 2023 | 1M | Ans: 2 | ● ○ ○

The given characteristic polynomial is: $P(s) = s^3 + 2s^2 + 5s + 80$

Routh Array:

$$\begin{array}{c|cc} s^3 & 1 & 5 \\ s^2 & 2 & 80 \\ s^1 & \frac{(2\times5) - (1\times80)}{2} = -35 & \\ s^0 & 80 & \end{array}$$

Since there are 2 sign changes in the first column, the polynomial has exactly 2 zeros in the right-half plane.

$$\boxed{2}$$

Q4.18.1 NAT | 2018 | 2M | Ans: 8.9 to 9.1 | ● ○ ○

The characteristic equation of the closed-loop system is given by $1 + G(s)H(s) = 0$.

$$1 + \frac{1}{(s+1)(s+2)} \cdot \frac{s+\alpha}{s} = 0 \implies s^3 + 3s^2 + 3s + \alpha = 0$$

Method 1

Using the Routh-Hurwitz criterion:

$$\begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 3 & \alpha \\ s^1 & \frac{9-\alpha}{3} & \\ s^0 & \alpha & \end{array}$$

For poles to lie on the imaginary axis (marginally stable condition), the s^1 row must be zero:

$$\frac{9-\alpha}{3} = 0 \implies \alpha = 9$$

Method 2

Shortcut:
For a standard third-order system:

$$a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0$$

For marginal stability (poles on the imaginary axis), the product of the inner coefficients equals the product of the outer coefficients:

Inner Product = Outer Product

$$3 \times 3 = 1 \times \alpha \implies \alpha = 9$$

$$\boxed{9}$$

Key Points

- For a third-order characteristic equation $s^3 + a_2 s^2 + a_1 s + a_0 = 0$, the system is marginally stable with poles on the $j\omega$-axis if $a_2 a_1 = a_0$.
- **Shortcut:** The frequency of sustained oscillations is given by $\omega_n = \sqrt{a_1} = \sqrt{\frac{a_0}{a_2}} = \sqrt{\frac{\text{independent term}}{\text{coefficient of } s^2}}$.

Q4.17.1 MCQ | 2017 | 2M | Ans: D | ● ○ ○

The characteristic equation of the given closed-loop system is $1 + G(s)H(s) = 0$, where $H(s) = 1$:

$$1 + \frac{s+\alpha}{s^3 + 2\alpha s^2 + \alpha s + 1} = 0 \implies s^3 + 2\alpha s^2 + (\alpha+1)s + (\alpha+1) = 0$$

Method 1

Using Routh-Hurwitz table:

$$\begin{array}{c|cc} s^3 & 1 & \alpha+1 \\ s^2 & 2\alpha & \alpha+1 \\ s^1 & \frac{2\alpha(\alpha+1)-(\alpha+1)}{2\alpha} & \\ s^0 & \alpha+1 & \end{array}$$

For stability, there must be no sign changes in 1st column:

- From the s^2 row: $2\alpha > 0 \implies \alpha > 0$
- From the s^1 row:

$$2\alpha(\alpha+1) - (\alpha+1) > 0 \implies (\alpha+1)(2\alpha-1) > 0$$

Since $\alpha > 0$, $(\alpha + 1)$ is positive,

$$so \;\; 2\alpha - 1 > 0 \implies \alpha > 0.5$$

Method 2

Shortcut: For a standard third-order system $a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0$ with positive coefficients, the condition for stability is:

Inner Product > Outer Product

$$a_2 a_1 > a_3 a_0$$

Substituting the coefficients:

$$2\alpha(\alpha+1) > 1 \cdot (\alpha+1)$$

$$2\alpha > 1$$

$$\alpha > 0.5$$

$$\boxed{(D) \;\; \alpha > 0.5}$$

Q4.16.1 MCQ 2016 1M Ans: D

For roots to lie strictly in the left half of the s-plane, all coefficients must be positive, meaning $a_0 > 0$.

Method 1

Using the Routh-Hurwitz criterion:

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 3 & a_0 \\ s^1 & \dfrac{6-a_0}{3} & 0 \\ s^0 & a_0 & \end{array}$$

For stability, the s^1 row element must be positive:

$$\frac{6-a_0}{3} > 0 \implies a_0 < 6$$

Combining conditions: $0 < a_0 < 6$.

Method 2

For a stable third-order polynomial $s^3+a_2s^2+a_1s+a_0 = 0$, the necessary and sufficient condition is:

$$\text{Inner Product} > \text{Outer Product}$$

$$a_2a_1 > a_3a_0$$

Substituting the given parameters:

$$3 \times 2 > 1 \times a_0 \implies 6 > a_0$$

Thus, for stability, we must have $a_0 < 6$.

$$\boxed{(D)\ 5}$$

Q4.16.2 MCQ 2016 2M Ans: C

The closed-loop characteristic equation is given by $1 + C(s)G(s) = 0$,

$$where,\ \ G(s) = \frac{1}{(s+1)^2} = \frac{1}{s^2+2s+1}$$

For a standard third-order characteristic equation $s^3 + a_2s^2 + a_1s + a_0 = 0$, the system is stable if all coefficients are positive and satisfies the following stability condition:

$$\text{Inner Product} > \text{Outer Product} \implies a_2a_1 > a_0$$

Option (A):
For $C(s) = \frac{s+3}{s}$:

$$1 + \frac{s+3}{s(s^2+2s+1)} = 0$$

$$s^3 + 2s^2 + 2s + 3 = 0$$

Checking stability:

$$a_2a_1 = 2 \times 2 = 4$$

$$a_0 = 3$$

Since $4 > 3$, **stable**.

Option (B):
For $C(s) = \frac{3s+7}{s}$:

$$1 + \frac{3s+7}{s(s^2+2s+1)} = 0$$

$$s^3 + 2s^2 + 4s + 7 = 0$$

Checking stability:

$$a_2a_1 = 2 \times 4 = 8$$

$$a_0 = 7$$

Since $8 > 7$,**stable**.

Option (C):
For $C(s) = \frac{3s+9}{s}$:

$$1 + \frac{3s+9}{s(s^2+2s+1)} = 0$$

$$s^3 + 2s^2 + 4s + 9 = 0$$

Checking stability:

$$a_2a_1 = 2 \times 4 = 8$$

$$a_0 = 9$$

Since $8 \ngtr 9$, **unstable**.

Option (D):
For $C(s) = \frac{1}{s}$:

$$1 + \frac{1}{s(s^2+2s+1)} = 0$$

$$s^3 + 2s^2 + s + 1 = 0$$

Checking stability:

$$a_2a_1 = 2 \times 1 = 2$$

$$a_0 = 1$$

Since $2 > 1$, **stable**.

$$\boxed{(C)\ \ C(s) = 3 + \frac{9}{s}}$$

Q4.12.1 MCQ 2012 2M Ans: A

The characteristic equation of the unity negative feedback system is:

$$1 + G(s)H(s) = 0 \implies s^3 + as^2 + (K+2)s + (K+1) = 0$$

RH Table:

$$\begin{array}{c|cc} s^3 & 1 & K+2 \\ s^2 & a & K+1 \\ s^1 & \dfrac{a(K+2)-(K+1)}{a} & \\ s^0 & K+1 & \end{array}$$

For sustained oscillations, the s^1 row must be a row of zeros:

$$a(K+2) - (K+1) = 0$$

$$a = \frac{K+1}{K+2} \quad \text{— (i)}$$

The auxiliary equation from the s^2 row is:

$$A(s) = as^2 + (K+1) = 0$$

Given the oscillation frequency is $\omega = 2$ rad/s, we substitute $s = j\omega = j2$ into $A(s) = 0$:

$$a(j2)^2 + (K+1) = 0 \implies -4a + K + 1 = 0 \implies K = 4a - 1 \quad \text{— (ii)}$$

Substituting equation (i) into equation (ii):

$$K = 4\left(\frac{K+1}{K+2}\right) - 1 \implies K^2 + 2K = 3K + 2 \implies (K-2)(K+1) = 0$$

For system stability configurations, the open-loop gain must be positive ($K > 0$), which yields:

$$K = 2$$

Substituting $K = 2$ back into equation (i) determines the parameter a:

$$a = \frac{2+1}{2+2} = \frac{3}{4} = 0.75$$

$$\boxed{(A) \ \ K = 2 \text{ and } a = 0.75}$$

Key Points

- A sustained oscillation condition implies a system is marginally stable.
- The roots of the auxiliary equation $A(s) = 0$ specify the exact frequency of these sustained oscillations on the imaginary axis.

Q4.11.1 MCQ 2011 1M Ans: D

The completed Routh's array is:

$$\begin{array}{c|cc} s^3 & 3 & 3 \\ s^2 & 4 & 4 \\ s^1 & 0(8) & 0 \\ s^0 & 4 & \end{array}$$

The s^1 is a special case where a row of zeros appears prematurely. We form the auxiliary equation $A(s)$ using the coefficients of the s^2 row:

$$A(s) = 4s^2 + 4 = 0$$

Differentiating $A(s)$ with respect to s:

$$\frac{dA(s)}{ds} = 8s = 0$$

There are no sign changes in the first column, which means there are no roots in the right-half of the s-plane.

Solving the auxiliary equation $A(s) = 0$:

$$4s^2 + 4 = 0 \implies s^2 = -1 \implies s = \pm j$$

Therefore, the system has a pair of complex conjugate poles on the imaginary axis of the s-plane making it marginally stable.

$$\boxed{\text{(D) a pair of complex conjugate poles on the imaginary axis of the } s\text{-plane}}$$

Key Points

- A premature row of zeros in a Routh array indicates the presence of roots symmetric about the origin.
- The auxiliary equation $A(s)$ is always formed from the row directly above the row of zeros and is always of even degree.
- Roots of $A(s) = 0$ are also roots of the main characteristic equation.

Q4.10.1 MCQ 2010 1M Ans: D

The characteristic equation for the unity feedback system is:

$$1 + G(s) = 1 + \frac{k(s+3)}{(s+1)(s+2)} = 0$$

$$(s+1)(s+2) + k(s+3) = 0$$

$$s^2 + 3s + 2 + ks + 3k = 0$$

$$s^2 + (3+k)s + (2+3k) = 0$$

According to R-H array:

$$\begin{array}{c|cc} s^2 & 1 & 2+3k \\ s^1 & 3+k & 0 \\ s^0 & 2+3k & 0 \end{array}$$

For stability, all coefficients in the first column must be greater than zero:

$$3 + k > 0 \implies k > -3$$

$$2 + 3k > 0 \implies k > -\frac{2}{3}$$

Taking the intersection of both conditions, the system is stable for:

$$k > -\frac{2}{3}$$

Since the question asks for the range of **positive** values of k, we look at the range where $k > 0$:

$$0 < k < \infty$$

$$\boxed{(D)\ \ 0 < k < \infty}$$

Q4.08.1 MCQ 2008 2M Ans: C

The characteristic equation of the closed-loop system is: $1 + G(s)H(s) = 0$

$$1 + \frac{K_c}{s(s+3)} = 0 \implies s^2 + 3s + K_c = 0$$

Method 1

The closed-loop poles are given by the roots:

$$s_1, s_2 = \frac{-3 \pm \sqrt{9 - 4K_c}}{2}$$

For the real parts of all poles to be more negative than -1, we analyze the condition:

$$\text{Re}(s_{1,2}) < -1$$

Case 1: If the roots are complex conjugate ($9 - 4K_c < 0$), the real part is constant:

$$\text{Re}(s_{1,2}) = -\frac{3}{2} = -1.5 < -1$$

This holds true for any $K_c > \frac{9}{4} = 2.25$.

Case 2: If the roots are real ($9 - 4K_c \geq 0$), the dominant pole must satisfy:

$$\frac{-3 + \sqrt{9 - 4K_c}}{2} < -1$$

$$\sqrt{9 - 4K_c} < 1 \implies K_c > 2$$

Combining with the condition $K_c \leq 2.25$ for real roots, we get $2 < K_c \leq 2.25$.

Thus, for all cases, the condition $K_c > 2$ must be satisfied.

Method 2

Alternatively, we can use the Routh-Hurwitz criterion by shifting the axis. To ensure all poles have real parts more negative than -1, we substitute $s = z - 1$:

$$(z-1)^2 + 3(z-1) + K_c = 0$$

$$z^2 + z + (K_c - 2) = 0$$

Constructing the Routh-Hurwitz table for this z-domain polynomial:

$$\begin{array}{c|cc} z^2 & 1 & K_c - 2 \\ z^1 & 1 & 0 \\ z^0 & K_c - 2 & \end{array}$$

For the shifted system to be stable (all roots of z having negative real parts), there must be no sign changes in the first column of the Routh array. Therefore, all elements in the first column must be strictly positive:

$$K_c - 2 > 0 \implies K_c > 2$$

$$\boxed{(C)\;\; K_C > 2}$$

Key Points

To find the parameter range for poles located to the left of $s = -\sigma$, shift the s-plane using the transformation $s = z - \sigma$ and apply the Routh-Hurwitz criterion on the z-polynomial.

Q4.04.1 MCQ 2004 2M Ans: B

The characteristic equation of the closed-loop system is given by: $1 + G(s)H(s) = 0$

$$1 + \frac{K}{s(s+1)(s+5)} = 0 \implies s^3 + 6s^2 + 5s + K = 0$$

Constructing the Routh-Hurwitz array:

$$\begin{array}{c|cc} s^3 & 1 & 5 \\ s^2 & 6 & K \\ s^1 & \frac{30-K}{6} & 0 \\ s^0 & K & 0 \end{array}$$

For the system to sustain continuous oscillations, the row of s^1 must be zero (marginal stability condition):

$$\frac{30-K}{6} = 0 \implies K_{\text{mar}} = 30$$

The auxiliary equation $(A(s))$ is formed from the s^2 row:

$$6s^2 + K = 0$$

Substituting $K = 30$:

$$6s^2 + 30 = 0 \implies s^2 + 5 = 0$$

$$s = \pm j\sqrt{5}$$

Angular frequency of oscillation, $\omega = \sqrt{5}$ rad/sec

$$\boxed{(B)\;\; \sqrt{5}\text{ rad/sec}}$$

Q4.03.1 MCQ | 2003 | 2M | Ans: D

From the given block diagram, the open-loop transfer function is:

$$G(s)H(s) = 2\left(1 + \frac{1}{\tau_1 s}\right) \cdot \frac{5}{s(s+1)} = \frac{10(\tau_1 s + 1)}{\tau_1 s^2 (s+1)}$$

The characteristic equation is given by $1 + G(s)H(s) = 0$:

$$1 + \frac{10(\tau_1 s + 1)}{\tau_1 s^2 (s+1)} = 0 \implies \tau_1 s^3 + \tau_1 s^2 + 10\tau_1 s + 10 = 0$$

Method 1

Constructing the Routh array:

$$\begin{array}{c|cc} s^3 & \tau_1 & 10\tau_1 \\ s^2 & \tau_1 & 10 \\ s^1 & \dfrac{10\tau_1^2 - 10\tau_1}{\tau_1} & 0 \\ s^0 & 10 & 0 \end{array}$$

For stability, all coefficients in the first column must be positive ($\tau_1 > 0$):

$$\frac{10\tau_1(\tau_1 - 1)}{\tau_1} > 0$$

$$10(\tau_1 - 1) > 0 \implies \tau_1 > 1.0$$

Method 2

For a standard third-order characteristic equation $a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0$ with all positive coefficients, the system is stable if the product of inner coefficients is greater than the product of outer coefficients:

$$\text{Inner Product} > \text{Outer Product}$$

$$a_2 \cdot a_1 > a_3 \cdot a_0$$

Substituting the coefficients ($\tau_1 \cdot 10\tau_1 > \tau_1 \cdot 10$):

$$10\tau_1^2 > 10\tau_1 \xrightarrow{\text{Since } \tau_1 > 0} \tau_1 > 1.0$$

$$\boxed{(D)\ \ \tau_1 > 1.0}$$

Q4.02.1 MCQ | 2002 | 2M | Ans: A

Given partial Routh array:

$$\begin{array}{c|ccc} s^4 & 1 & a & 8 \\ s^3 & 3 & 12 & \end{array}$$

For the system to sustain continuous oscillations at an angular frequency $\omega = 2$ rad/s, a row of zeros must occur at the s^1 row.

Consequently, the row directly above it (s^2 row) forms the auxiliary equation $A(s) = 0$, which contains only even powers of s. Since the s^4 row represents the next higher even-powered row, the roots lying on the imaginary axis must also satisfy the auxiliary equation formed by the coefficients of the s^4 row if the intermediate s^2 row coefficients maintain a proportional relationship:

$$s^4 + as^2 + 8 = 0$$

Substituting $s = j\omega = j2$ into this equation:

$$(j2)^4 + a(j2)^2 + 8 = 0$$

$$16 - 4a + 8 = 0$$

$$4a = 24 \implies a = 6$$

$$\boxed{(A)\ \ 6}$$

Q4.93.1 NAT | 1993 | 2M | Ans: 0.25

Given characteristic equation:

$$s^2 + Ks + (4K - 1) = 0$$

By Routh-Hurwitz stability criterion:

$$\begin{array}{c|cc} s^2 & 1 & 4K-1 \\ s^1 & K & 0 \\ s^0 & 4K-1 & 0 \end{array}$$

For the system to be stable, all elements in the first column of the Routh array must be strictly positive:

1. $K > 0$
2. $4K - 1 > 0 \implies K > \frac{1}{4}$

Taking the intersection of both conditions: $K > 0.25$

$\boxed{0.25}$

DEBUG INFO

Chapter IV

Routh's Stability Criterion

Total Questions : 14

MCQ : 11

MSQ : 0

NAT : 3

SOLUTIONS

V

Root Locus

Topics

- Basics of Root Locus
- Rules for Sketching Root Locus
- Angle of Departure and Arrival
- Complementary Root Locus

Q5.24.1 NAT 2024 2M Ans: 2 CLICK FOR VIDEO SOLUTION

The system has three open-loop poles located at $s = 0$, $s = -1$, and $s = -4$.

$$G(s)H(s) = \frac{K}{s(s+1)(s+4)}$$

The characteristic equation is given by $1 + G(s)H(s) = 0$:

$$s(s+1)(s+4) + K = 0 \implies s^3 + 5s^2 + 4s + K = 0$$

RH Array:

$$\begin{array}{c|cc} s^3 & 1 & 4 \\ s^2 & 5 & K \\ s^1 & \dfrac{20-K}{5} & 0 \\ s^0 & K & \end{array}$$

For the root locus to cross the imaginary axis, the system becomes marginally stable at $K = K_{\text{mar}}$:

$$\frac{20 - K_{\text{mar}}}{5} = 0 \implies K_{\text{mar}} = 20$$

Shortcut: For a third-order system $as^3 + bs^2 + cs + d = 0$, the system becomes marginally stable when the inner product equals the outer product ($bc = ad$), provided a,b,c,d are positive : $5 \times 4 = 1 \times K \implies K_{\text{mar}} = 20$.

Forming the auxiliary equation from the s^2 row:

$$A(s) = 5s^2 + K_{\text{mar}} = 0$$

$$5s^2 + 20 = 0 \implies s^2 = -4$$

$$s = \pm j2$$

$$\boxed{\alpha = 2}$$

Q5.20.1 NAT 2020 2M Ans: 5.0 to 5.1

The characteristic equation of the negative feedback control system is $1 + G(s)H(s) = 0$:

$$1 + \frac{K}{s(s+2)(s+6)} = 0$$

$$K = -s(s^2 + 8s + 12) = -(s^3 + 8s^2 + 12s)$$

To find the breakaway points, we set $\frac{dK}{ds} = 0$:

$$\frac{dK}{ds} = -(3s^2 + 16s + 12) = 0 \implies 3s^2 + 16s + 12 = 0$$

Solving this quadratic equation using the root formula:

$$s = \frac{-16 \pm \sqrt{16^2 - 4(3)(12)}}{2(3)} = \frac{-16 \pm \sqrt{112}}{6}$$

$$s_1 \approx -0.9, \quad s_2 \approx -4.43$$

$s = -0.9$ is the valid breakaway point as it lies on the root locus segment.

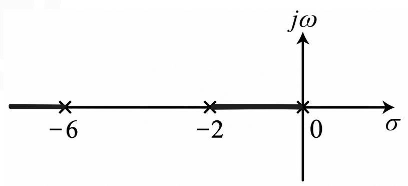


Fig. Solution Q5.20.1

Evaluating K at $s = -0.9$ via the magnitude criterion:

$$\frac{K}{|s(s+2)(s+6)|_{s=-0.9}} = 1$$

$$K = |-0.9(-0.9+2)(-0.9+6)|$$

$$K = |-0.9(1.1)(5.1)|$$

$$K = 5.049$$

$$\boxed{5.0}$$

Key Points

- Breakaway points always occur where $\frac{dK}{ds} = 0$ and the corresponding gain value satisfies $K > 0$.
- A segment on the real axis is part of the root locus if the total number of real poles and zeros to its right is odd.

- At any valid point on the root locus, the absolute parameter value is determined via the standard magnitude criterion: $|G(s)H(s)| = 1$.

Q5.19.1

MCQ | 2019 | 2M | Ans: C

The open-loop transfer function of the unity feedback control system is:

$$G(s)H(s) = C(s)P(s) = \frac{K}{s+p}\cdot\frac{3}{s} = \frac{3K}{s(s+p)}$$

Method 1

Since the points $s = -3 \pm j3\sqrt{3}$ lie on the root-locus, they must satisfy the angle condition:

$$\angle G(s_1)H(s_1) = \pm 180^\circ$$

Evaluating the angle at $s_1 = -3 + j3\sqrt{3}$:

$$\angle 3K - \angle s_1 - \angle(s_1 + p) = \pm 180^\circ$$

$$0^\circ - \angle(-3 + j3\sqrt{3}) - \angle((p-3) + j3\sqrt{3}) = -180^\circ$$

$$-(180^\circ - \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right)) - \tan^{-1}\left(\frac{3\sqrt{3}}{p-3}\right) = -180^\circ$$

$$-120^\circ - \tan^{-1}\left(\frac{3\sqrt{3}}{p-3}\right) = -180^\circ$$

$$\frac{3\sqrt{3}}{p-3} = \sqrt{3} \implies p = 6$$

Method 2

Alternatively, using the characteristic equation $1 + G(s)H(s) = 0$:

$$s^2 + ps + 3K = 0$$

Comparing with $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$, the roots are,

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

Given the roots are $-3 \pm j3\sqrt{3}$, we can directly equate the real part:

$$\zeta\omega_n = 3$$

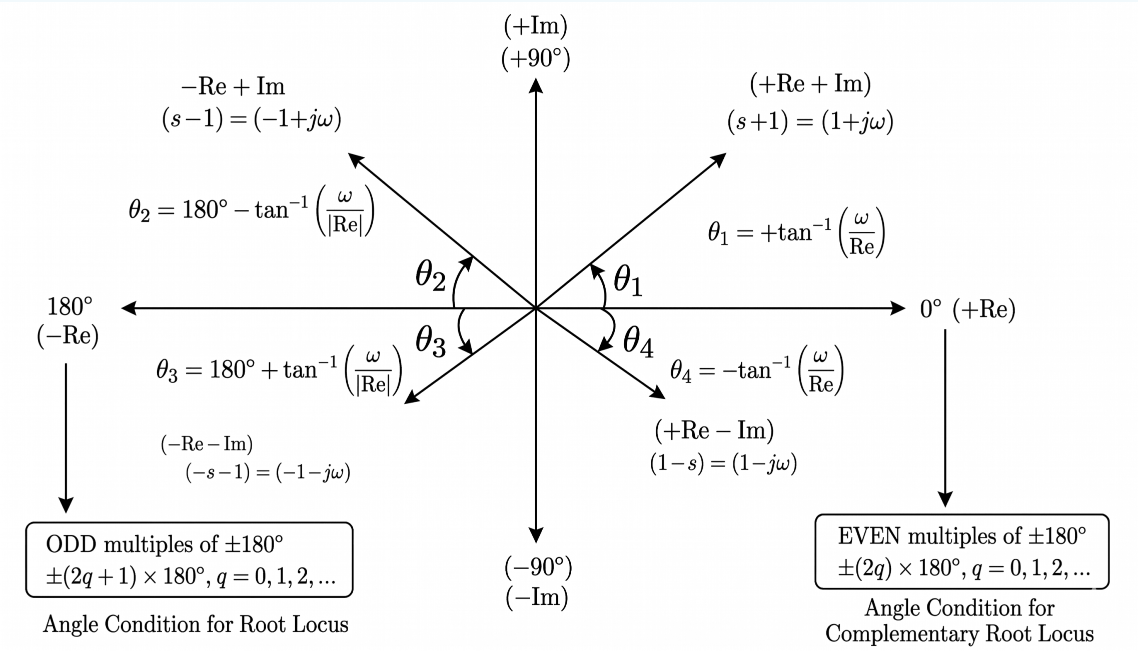
From the polynomial coefficients, we know that $p = 2\zeta\omega_n$. Substituting the value:

$$p = 2 \times 3 = 6$$

$$\boxed{(C)\ 6}$$

Key Points

Angle Criterion Rule:



Q5.17.1 NAT 2017 2M **Ans: -1.2 to -1.0**

The characteristic equation of the system is given by $1 + G(s)H(s) = 0$:

$$1 + \frac{K(s+6)}{s(s+2)} = 0 \implies K = -\frac{s^2+2s}{s+6}$$

For finding the breakaway and break-in points, we set $\frac{dK}{ds} = 0$:

$$\frac{dK}{ds} = -\frac{(2s+2)(s+6) - (s^2+2s)(1)}{(s+6)^2} = 0 \implies s^2 + 12s + 12 = 0$$

Using the quadratic formula to solve for s:

$$s = \frac{-12 \pm \sqrt{12^2 - 4(1)(12)}}{2}$$

$$s = -6 \pm 2\sqrt{6} = -1.101, -10.899$$

Since $s = -1.10$ lies between 2 poles, it is the valid breakaway point whereas -10.899 is the breakin oint.

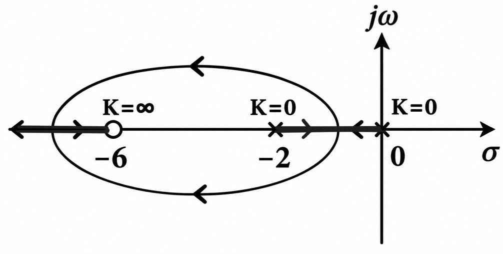


Fig. Solution Q5.17.1

$$\boxed{-1.1}$$

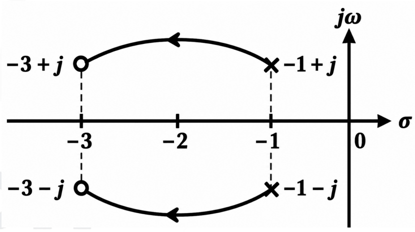
Key Points

Breakaway points occur between two adjacent poles on the real axis if that segment is part of the root locus. Break-in points occur between two adjacent zeros on the real axis if that segment is part of the root locus.

Q5.15.1 MCQ 2015 2M **Ans: A**

Given open-loop transfer function: $G(s) = \frac{s^2+6s+10}{s^2+2s+2}$

$$z_1, z_2 = -3 \pm j1$$
$$p_1, p_2 = -1 \pm j1$$



Method 1

Using phasor phase angle method :

$$\phi_A = 180^\circ - \angle G(s)H(s)\Big|_{s=-3+j1}$$

Evaluating the total transfer function angle directly at the upper complex zero $s_1 = -3 + j1$:

$$G(-3+j1) = \frac{(-3+j1+3+j1)(-3+j1+3-j1)}{(-3+j1+1+j1)(-3+j1+1-j1)}$$

$$G(-3+j1) = \frac{(2j)(0)}{(-2+2j)(-2)}$$

$$\angle G(-3+j1) = \angle(2j) - \angle(-2+2j) - \angle(-2)$$

$$\angle G(-3+j1) = 90^\circ - 135^\circ - 180^\circ = -225^\circ \equiv 135^\circ$$

Applying the arrival criterion:

$$\phi_A = 180^\circ - 135^\circ = 45^\circ = \frac{\pi}{4}$$

Method 2

Using explicit summation method :

$$\phi_A = 180^\circ - \phi \quad \left[\text{where } \phi = \sum \phi_{z'} - \sum \phi_p\right]$$

Calculating geometric vector angles from remaining components to $z_1 = -3 + j1$:

- From $z_2 = -3 - j1$: $\phi_{z2} = \tan^{-1}\left(\frac{1-(-1)}{-3-(-3)}\right) = 90^\circ$
- From $p_1 = -1 + j1$: $\phi_{p1} = \tan^{-1}\left(\frac{1-1}{-3-(-1)}\right) = 180^\circ$
- From $p_2 = -1 - j1$: $\phi_{p2} = \tan^{-1}\left(\frac{1-(-1)}{-3-(-1)}\right) = 135^\circ$

$$\phi_a = 180^\circ - 90^\circ + (180^\circ + 135^\circ) = 405^\circ \equiv 45^\circ = \frac{\pi}{4}$$

$$\boxed{(A)\ \pm\frac{\pi}{4}}$$

Key Points

Angle of departure applies to complex poles, whereas the angle of arrival applies to complex zeros where root locus branches terminate.

Q5.14.1 NAT 2014 1M Ans: -4

Given the loop transfer function: $G(s)H(s) = \dfrac{K(s+2)}{s^2(s+10)}$

- Poles (P): $s = 0, 0, -10 \implies$ Number of poles $(n) = 3$
- Zeros (Z): $s = -2 \implies$ Number of zeros $(m) = 1$

The centroid (σ) is the point on the real axis where all the root locus asymptotes intersect.

$$\sigma = \frac{\sum(\text{real parts of poles}) - \sum(\text{real parts of zeros})}{\text{poles} - \text{zeros}}$$

$$\sigma = \frac{(0+0-10)-(-2)}{3-1} = \frac{-10+2}{2} = -4$$

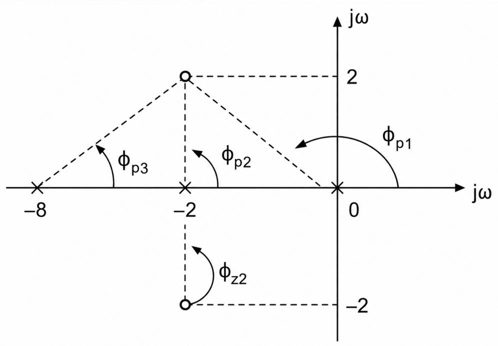
$$\boxed{-4}$$

Q5.12.1 MCQ 2012 2M Ans: D

Given the open loop transfer function:

$$G(s) = \frac{s^2+4s+8}{s(s+2)(s+8)}$$

- Zeros (Z): $s^2+4s+8 = 0 \implies s = -2 \pm j2$
- Poles (P): $s = 0, -2, -8$



Method 1

Using phasor phase angle method: $\phi_A = 180° + \angle G(s)H(s)\Big|_{s=-2+j2}$

$$\angle G(s)H(s) = \frac{\angle(s-(-2-j2))\angle(s-(-2+j2))}{\angle s\angle(s+2)\angle(s+8)}$$

$$\angle G(-2+j2) = \frac{\angle(-2+j2-(-2-j2))\angle(0)}{\angle(-2+j2)\angle(-2+j2+2)\angle(-2+j2+8)} = \frac{\angle 4j}{\angle(-2+j2)\angle(j2)\angle(6+j2)}$$

Calculating individual angles:

$$\angle(4j) = \tan^{-1}\left(\frac{4}{0}\right) = \tan^{-1}(\infty) = 90°; \quad \angle(-2+j2) = \tan^{-1}\left(\frac{2}{-2}\right) = \tan^{-1}(-1) = 135°$$

$$\angle(j2) = \tan^{-1}\left(\frac{2}{0}\right) = \tan^{-1}(\infty) = 90°; \quad \angle(6+j2) = \tan^{-1}\left(\frac{2}{6}\right) = \tan^{-1}\left(\frac{1}{3}\right)$$

Thus, the net phase contribution is:

$$\angle G(-2+j2) = 90° - \left(135° + 90° + \tan^{-1}\left(\frac{1}{3}\right)\right) = -135° - \tan^{-1}\left(\frac{1}{3}\right)$$

Applying the arrival criterion: $\phi_A = 180° - 135° - \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)$

Method 2

Using explicit summation method:

$$\phi_A = 180° - \phi \quad \left[\text{where } \phi = \sum \phi_{z'} - \sum \phi_p\right]$$

Calculating geometric vector angles from remaining components to $z_1 = -2 + j2$:

- From $z_2 = -2 - j2$:

$$\phi_{z2} = \tan^{-1}\left(\frac{2-(-2)}{-2-(-2)}\right) = 90°$$

- From $p_1 = 0$:

$$\phi_{p1} = \tan^{-1}\left(\frac{2-0}{-2-0}\right) = 135°$$

- From $p_2 = -2$:

$$\phi_{p2} = \tan^{-1}\left(\frac{2-0}{-2-(-2)}\right) = 90°$$

- From $p_3 = -8$:

$$\phi_{p3} = \tan^{-1}\left(\frac{2-0}{-2-(-8)}\right) = \tan^{-1}\left(\frac{1}{3}\right)$$

Substituting into the arrival formula:

$$\phi_A = 180° - \left(90° - 135° - 90° - \tan^{-1}\left(\frac{1}{3}\right)\right)$$

$$\phi_A = 315° + \tan^{-1}\left(\frac{1}{3}\right) \equiv -45° + \tan^{-1}\left(\frac{1}{3}\right)$$

Taking the absolute magnitude:

$$|\phi_A| = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)$$

$$\boxed{(D)\ \ |\theta| = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)}$$

Q5.11.1 MCQ 2011 2M Ans: C

Given the characteristic equation:

$$s(s+3) + K(s+5) = 0$$

Expressing gain K as a function of s:

$$K = -\frac{s(s+3)}{s+5} = -\frac{s^2+3s}{s+5}$$

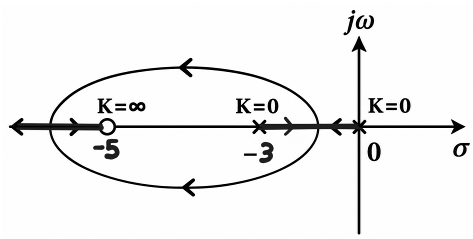


Fig. Solution Q5.11.1

For breakaway or break-in points, set $\frac{dK}{ds} = 0$:

$$\frac{dK}{ds} = -\frac{(2s+3)(s+5) - (s^2+3s)(1)}{(s+5)^2} = 0$$

$$s^2 + 10s + 15 = 0$$

Solving the quadratic equation for s:

$$s = \frac{-10 \pm \sqrt{10^2 - 4\times 1\times 15}}{2} = \frac{-10 \pm 2\sqrt{10}}{2} = -5 \pm \sqrt{10}$$

Both values lie on sections of the real axis where root locus exists, representing valid breakaway and break-in points.

$$\boxed{(C)\ \ -5 \pm \sqrt{10}}$$

Q5.08.1 MCQ 2008 2M Ans: C

The open loop transfer function is given by:

$$G(s)H(s) = \frac{K(s+2)}{(s+1+j1)(s+1-j1)} = \frac{K(s+2)}{s^2+2s+2}$$

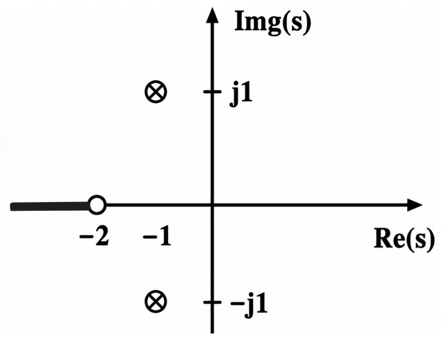


Fig. Solution Q5.08.1

For breakaway/break-in points, we set $\frac{dK}{ds} = 0$:

$$K = -\frac{s^2+2s+2}{s+2}$$

Differentiating with respect to s:

$$\frac{dK}{ds} = -\frac{(2s+2)(s+2)-(s^2+2s+2)(1)}{(s+2)^2} = 0$$

$$s^2+4s+2 = 0$$

Solving the quadratic equation:

$$s = \frac{-4 \pm \sqrt{16-8}}{2} = -2 \pm \sqrt{2}; \quad s_1 = -0.59 \quad \text{and} \quad s_2 = -3.41$$

$s = -3.41$ is a valid break-in point as it lies on the root locus.

$$\boxed{\text{(C) one breakaway point located at } s = -3.41}$$

Q5.07.1 MCQ 2007 2M Ans: B

Given the plant transfer function has three poles at -1, -2, and -3 with no finite zeros, it can be represented as:

$$G(s) = \frac{k}{(s+1)(s+2)(s+3)}$$

For unity DC gain, we substitute $s = 0$:

$$G(0) = \frac{k}{(0+1)(0+2)(0+3)} = 1 \implies k = 6$$

Thus, the plant transfer function is:

$$\boxed{(B)\ \ G(s) = \frac{6}{(s+1)(s+2)(s+3)}}$$

Q5.07.2 MCQ 2007 2M Ans: C

The open-loop transfer function with a proportional controller K is:

$$G(s)H(s) = \frac{6K}{(s+1)(s+2)(s+3)}$$

The characteristic equation of the closed-loop system is $1 + G(s)H(s) = 0$:

$$(s+1)(s+2)(s+3) + 6K = 0 \implies s^3 + 6s^2 + 11s + (6+6K) = 0$$

Method 1

Using the Routh-Hurwitz array:

$$\begin{array}{c|cc} s^3 & 1 & 11 \\ s^2 & 6 & 6(1+K) \\ s^1 & \dfrac{66-6(1+K)}{6} & 0 \\ s^0 & 6(1+K) & \end{array}$$

For marginal stability, the s^1 row becomes zero, giving the auxiliary equation from the s^2 row:

$$6s^2 + 6(1+K) = 0$$

Substituting $s = j\omega = j\sqrt{11}$ into the auxiliary equation:

$$6(j\sqrt{11})^2 + 6(1+K) = 0$$

$$-66 + 6 + 6K = 0 \implies K = 10$$

Method 2

Shortcut Method (Inner-Outer Product):
For a standard third-order characteristic equation $a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0$:

$$s^3 + 6s^2 + 11s + 6(1+K) = 0$$

At the verge of instability (marginal stability), the product of the inner coefficients equals the product of the outer coefficients:

$$\text{Inner Product} = \text{Outer Product}$$

$$a_2 \cdot a_1 = a_3 \cdot a_0$$

$$6 \times 11 = 1 \times 6(1+K)$$

$$66 = 6 + 6K$$

$$K = 10$$

$$\boxed{\text{(C) } 10}$$

Key Points

Inner-Outer Product Rule: For a stable third-order system $s^3 + as^2 + bs + c = 0$, marginal stability occurs when $ab = c$, and the frequency of sustained oscillations is $\omega = \sqrt{b}$.

Q5.06.1

MCQ | 2006 | 2M | Ans: C | ● ● ●

From the given root locus, the open-loop poles are located at $s = 0$, $s = -4$, and $s = -8$. There are no finite zeros. The open-loop gain in pole-zero form is expressed as:

$$k = K' \cdot |s(s+4)(s+8)|$$

Given that at the imaginary axis crossing, $\omega = 4\sqrt{2}$ rad/sec (i.e., $s = j4\sqrt{2}$) and the gain $k = 384$:

$$384 = K' \cdot \left|(j4\sqrt{2})(j4\sqrt{2}+4)(j4\sqrt{2}+8)\right|$$

$$384 = K' \cdot (4\sqrt{2}) \cdot \sqrt{16+32} \cdot \sqrt{64+32}$$

$$384 = K' \cdot (4\sqrt{2}) \cdot (4\sqrt{3}) \cdot (4\sqrt{6})$$

$$K' \cdot 384 \implies K' = 1$$

Now, we compute the gain k at the point $s = -1.5 + j1.5$:

$$k = |(-1.5 + j1.5)(-1.5 + j1.5 + 4)(-1.5 + j1.5 + 8)|$$

$$k = |(-1.5 + j1.5)(2.5 + j1.5)(6.5 + j1.5)|$$

$$k = \sqrt{1.5^2 + 1.5^2} \cdot \sqrt{2.5^2 + 1.5^2} \cdot \sqrt{6.5^2 + 1.5^2}$$

$$k = \sqrt{4.5} \cdot \sqrt{8.5} \cdot \sqrt{44.5} = 41.25$$

$$\boxed{(C)\ 41.25}$$

Key Points

Shortcut: The system gain k at any point on the root locus equals the product of lengths from the poles to that point divided by the product of lengths from the zeros to that point.

Q5.05.1

MCQ | 2005 | 2M | Ans: A

From the given Routh-Hurwitz array:

$$\begin{array}{c|cc} s^3 & 1 & b \\ s^2 & a & c \\ s^1 & \dfrac{6-k}{3} & 0 \\ s^0 & k & \end{array}$$

For critical stability, the row of s^1 must be zero:

$$\frac{6-k}{3} = 0 \implies k = 6$$

The auxiliary equation is formed from the s^2 row:

$$A(s) = as^2 + c = 0$$

Given $a = 3$:

$$3s^2 + c = 0 \implies s = \pm j\sqrt{\frac{c}{3}}$$

From the root locus diagram, the system crosses the imaginary axis at $j\omega = \pm j1.414 = \pm j\sqrt{2}$. Comparing the roots:

$$\sqrt{\frac{c}{3}} = \sqrt{2} \implies c = 6$$

$$\boxed{\text{(A) 6 and 6}}$$

Q5.99.1

MCQ | 1999 | 1M | Ans: B

Let N be the degree of the numerator polynomial and M be the degree of the denominator polynomial of a transfer function $G(s) = \dfrac{P(s)}{Q(s)}$.

The total number of poles of the system is M, and the total number of zeros is M.

By definition, zeros at infinity occur when the magnitude of s approaches infinity, causing $G(s) \to 0$. The number of zeros located at infinity is given by the difference between the denominator degree and the numerator degree:

$$\text{Number of zeros at infinity} = M - N$$

Given that the transfer function has two zeros at infinity:

$$M - N = 2 \implies N = M - 2$$

$$\boxed{\text{(B) } N = M - 2}$$

Q5.95.1

MCQ | 1995 | 1M | Ans: B

Given open-loop transfer function:

$$G(s)H(s) = \frac{K(s+10)}{(s+2)(s+5)}$$

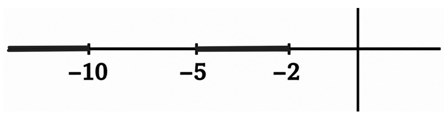


Fig. Solution Q5.95.1

The open-loop poles are at $s = -2, -5$ and the open-loop zero is at $s = -10$. Since the segment between the two open-loop poles $s = -2$ and $s = -5$ lies entirely on the root locus, the branches must break away from each other within this interval.

Therefore, without calculating the exact derivative, the breakaway point must lie **between** -2 **and** -5

$$\boxed{\text{(B) between } -2 \text{ and } -5}$$

Q5.94.1 MCQ 1994 1M Ans: B ● ○ ○

From the given plot:

- The root locus starts at $s = 0$ and moves left to form a break-away point, indicating a pole at $s = 0$ *and* -4. This **eliminates option (d) and (c).**
- The root locus branches move towards $s = -8$, implying a zero at $s = -8$. This **eliminates option (a).**

Testing Option (b): The characteristic equation is:

$$1 + G(s)H(s) = 1 + \frac{K(s+8)}{s(s+4)} = 0 \implies K = \frac{-(s^2+4s)}{s+8}$$

$$\frac{dK}{ds} = -\frac{(2s+4)(s+8) - (s^2+4s)(1)}{(s+8)^2} = 0$$

$$s^2 + 16s + 32 = 0$$

$$s = \frac{-16 \pm \sqrt{16^2 - 4(1)(32)}}{2} = \frac{-16 \pm 11.31}{2}$$

$$s_1 = -2.35 \quad \text{(Breakaway point);} \quad s_2 = -13.65 \quad \text{(Break-in point)}$$

These values perfectly match the provided root locus geometry (a breakaway between 0 and -4, and a circle centered at -8).

$$\boxed{(B) \quad \frac{K(s+8)}{s(s+4)}}$$

Q5.92.1 NAT 1992 2M Ans: 0° ● ○ ○

From the given s-plane diagram, the system has the following poles and zeros:

- Poles at: $P_1 = 0$, $P_2 = -2 + 2j$, and $P_3 = -2 - 2j$
- Zero at: $Z_1 = -4$

The angle of departure ϕ_d at the complex conjugate pole $P_2(-2+2j)$ is given by the angle criterion:

$$\phi_{d(-2+2j)} = 180^\circ - \sum \phi_p + \sum \phi_z$$

$$\phi_{d(-2+2j)} = 180^\circ - (\phi_{p1} + \phi_{p3}) + (\phi_{z1})$$

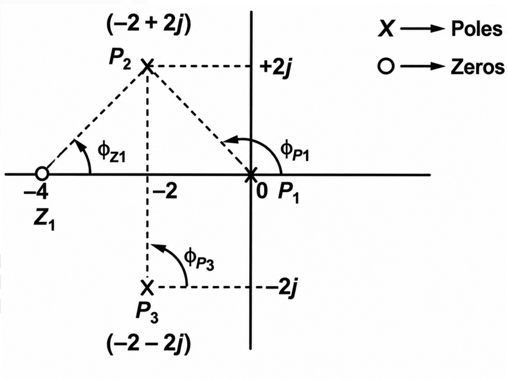


Fig. Solution Q5.92.1

Where the individual angles contributed by the remaining poles and zeros to P_2 are:

$$\phi_{p1} = 180^\circ - \tan^{-1}\left(\frac{2}{2}\right) = 135^\circ$$

$$\phi_{p3} = +90^\circ$$

$$\phi_{z1} = \tan^{-1}\left(\frac{2}{2}\right) = +45^\circ$$

Substituting these values back into the expression:

$$\phi_{d(-2+2j)} = 180^\circ - (135^\circ + 90^\circ) + 45^\circ = 0^\circ$$

Since the root locus is symmetric about the real axis, the angle of departure at $P_3(-2-2j)$ is:

$$\phi_{d(-2-2j)} = -\phi_{d(-2+2j)} = 0^\circ$$

$$\boxed{0^\circ}$$

DEBUG INFO

Chapter V

Root Locus

Total Questions : 17

MCQ : 12

MSQ : 0

NAT : 5

SOLUTIONS

VI

Polar Plot and Frequency Response of Second Order Systems

Topics

1. Basics of Frequency Response Analysis
2. Frequency Response of 2nd Order Systems
3. Polar Plot
4. Gain and Phase Crossover Frequency
5. Gain Margin and Phase Margin

Q6.26.1 NAT 2026 2M Ans: 0.39 to 0.41 ● ● ○

For

$$G(s) = \frac{K}{(s+1)^4}$$

The phase angle is, $\angle G(j\omega) = -4\tan^{-1}\omega$
At the phase crossover frequency, $\angle G(j\omega_{pc}) = -180°$

$$-4\tan^{-1}\omega_{pc} = -180°$$

$$\tan^{-1}\omega_{pc} = 45°$$

$$\omega_{pc} = 1 \text{ rad/s}$$

Magnitude at $\omega_{pc} = 1$ rad/s is

$$|G(j1)| = \frac{K}{|1+j|^4} = \frac{K}{(\sqrt{2})^4} = \frac{K}{4}$$

Gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})|} = \frac{4}{K}$$

Given,

$$GM_{dB} = 20 \text{ dB}$$

$$20\log_{10}\left(\frac{4}{K}\right) = 20$$

$$\frac{4}{K} = 10$$

$$K = 0.4$$

$$\boxed{0.40}$$

Q6.24.1 MCQ 2024 1M Ans: A ● ○ ○

The polar plot starts from $G(j0) = 1$ and moves vertically downward along the line $\Re\{G(j\omega)\} = 1$.
Hence, the real part remains constant and equal to 1, while the imaginary part becomes increasingly negative with increasing ω. Therefore, the sinusoidal transfer function satisfying these conditions is:

$$G(j\omega) = 1 - j\omega T, \qquad T > 0.$$

$$\boxed{(A)\ 1 - j\omega T}$$

Q6.23.1 NAT 2023 2M Ans: 0 ● ● ○

For

$$G(s) = \frac{2(1-s)}{(1+s)^2}$$

The magnitude is

$$|G(j\omega)| = \frac{2|1-j\omega|}{|1+j\omega|^2} = \frac{2\sqrt{1+\omega^2}}{(1+\omega^2)} = \frac{2}{\sqrt{1+\omega^2}}$$

At the gain crossover frequency,

$$|G(j\omega_{gc})| = 1$$

$$\frac{2}{\sqrt{1+\omega_{gc}^2}} = 1$$

$$\sqrt{1+\omega_{gc}^2} = 2$$
$$\omega_{gc} = \sqrt{3}\,\text{rad/s}$$

The phase of the transfer function is

$$\angle G(j\omega) = \angle(1-j\omega) - 2\angle(1+j\omega) = -\tan^{-1}\omega - 2\tan^{-1}\omega = -3\tan^{-1}\omega$$

At $\omega = \omega_{gc} = \sqrt{3}$,

$$\angle G(j\omega_{gc}) = -3\tan^{-1}(\sqrt{3}) = -3(60^\circ) = -180^\circ$$

Therefore,

$$PM = 180^\circ + \angle G(j\omega_{gc}) = 180^\circ - 180^\circ = 0^\circ$$

$$\boxed{0}$$

Q6.21.1 | NAT | 2021 | 2M | Ans: 0.95 to 1.05 | ● ○ ○

The input is $x(t) = \sqrt{2}\sin t\,\mu(t)$
Therefore, the input amplitude is $A_i = \sqrt{2}$
For

$$G(s) = \frac{1}{s+1}$$
$$G(j\omega) = \frac{1}{1+j\omega}$$

At $\omega = 1$ rad/s,

$$|G(j1)| = \frac{1}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}}$$

Hence, the steady-state output amplitude is

$$A_o = A_i|G(j1)| = \sqrt{2}\times\frac{1}{\sqrt{2}} = 1$$

$$\boxed{1}$$

Key Points

- For an LTI system, the steady-state response to a sinusoidal input remains sinusoidal at the same frequency. Only the amplitude and phase change.
- If the input is $x(t) = A\sin(\omega_0 t + \phi)$, then the steady-state output is

$$y(t) = A|G(j\omega_0)|\sin(\omega_0 t + \phi + \angle G(j\omega_0))\,.$$

Q6.20.1 | MCQ | 2020 | 1M | Ans: D | ● ○ ○

Given,

$$\frac{C(s)}{R(s)} = \frac{1}{1+s}$$

For the input $r(t) = \sin t$, the excitation frequency is $\omega = 1$ rad/s. Therefore,

$$T(j) = \frac{1}{1+j} = \frac{1}{\sqrt{2}}\angle -\frac{\pi}{4}$$

Hence, the steady-state response is

$$c(t) = \frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)$$

$$\boxed{\text{(D) } \frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)}$$

Key Points

- If the input to a transfer function $G(s)$ is $x(t) = A\sin(\omega_0 t + \phi)$, then the steady-state output is:

$$y(t) = A|G(j\omega_0)|\sin(\omega_0 t + \phi + \angle G(j\omega_0))$$

Q6.20.2 NAT 2020 2M Ans: 65.4 to 65.6

The open-loop transfer function is

$$G(s)H(s) = \frac{2(s+1)}{s^2}$$

The gain crossover frequency satisfies $|G(j\omega)H(j\omega)| = 1$
Therefore,

$$\frac{2\sqrt{1+\omega^2}}{\omega^2} = 1$$

$$4(1+\omega^2) = \omega^4$$

$$\omega^4 - 4\omega^2 - 4 = 0$$

$$\text{Let} \quad x = \omega^2$$

$$x^2 - 4x - 4 = 0$$

$$x = 2 + 2\sqrt{2}$$

$$\omega_{gc} = \sqrt{2+2\sqrt{2}} = 2.197 \text{ rad/s}$$

The phase of the open-loop transfer function is, $\angle G(j\omega)H(j\omega) = \tan^{-1}(\omega) - 180°$
At the gain crossover frequency, $\angle G(j\omega_{gc})H(j\omega_{gc}) = \tan^{-1}(2.197) - 180° = -114.5°$
Hence, Phase Margin $= 180° - 114.5° = 65.5°$

$$\boxed{65.5°}$$

Q6.19.1 MCQ 2019 1M Ans: D

In a Bode plot, two important frequencies are:

- Gain crossover frequency (ω_{gc}): the frequency where the magnitude curve crosses 0 dB.
- Phase crossover frequency (ω_{pc}): the frequency where the phase curve crosses $-180°$.

It is given that $\omega_{gc} < \omega_{pc}$
This means the magnitude reaches 0 dB before the phase reaches $-180°$.
Therefore, when the magnitude is 0 dB, the phase is still greater than $-180°$.
Hence, the system has a positive phase margin, which indicates that the closed-loop system is stable.

$$\boxed{(D)\text{ stable}}$$

Key Points

- If $\omega_{gc} < \omega_{pc}$, then the phase margin is positive and the closed-loop system is stable.
- If $\omega_{gc} = \omega_{pc}$, then the phase margin is zero and the system is marginally stable.
- If $\omega_{gc} > \omega_{pc}$, then the phase margin is negative and the closed-loop system is unstable.

Q6.19.2 NAT 2019 2M Ans: 37 to 39

The loop transfer function is

$$L(s) = \frac{9e^{-0.1s}}{s}$$

The gain crossover frequency is obtained from $|L(j\omega)| = 1$
Since, $\left|e^{-j0.1\omega}\right| = 1$,

$$\frac{9}{\omega} = 1$$

$$\omega_{gc} = 9 \text{ rad/s}$$

The phase at the gain crossover frequency is,

$$\angle L(j\omega_{gc}) = -90^\circ - 0.1\omega_{gc}\left(\frac{180}{\pi}\right)$$

$$\angle L(j9) = -90^\circ - 9(0.1)\left(\frac{180}{\pi}\right) = -90^\circ - 51.57^\circ = -141.57^\circ$$

Therefore, the phase margin is

$$PM = 180^\circ + \angle L(j\omega_{gc}) = 180^\circ - 141.57^\circ = 38.43^\circ$$

$$\boxed{38.43^\circ}$$

Key Points

- The phase contributed by a time delay is $-\omega\tau_D$ radians $= -\omega\tau_D\left(\frac{180}{\pi}\right)^\circ$

Mistakes to be avoided

- Do not mix radians and degrees. The phase contribution of the exponential term is initially obtained in radians and, if everything is in degrees, then it must be converted to degrees before adding it to the phase.

Q6.18.1 NAT 2018 1M Ans: 90 to 90 or -270 to -270

The input frequency is $\omega = 1$ rad/s. Therefore,

$$G(j1) = \frac{j-1}{j+1} = \frac{\sqrt{1^2+1^2}}{\sqrt{1^2+1^2}}\angle 180^\circ - \tan^{-1}\left(\frac{1}{1}\right) - \tan^{-1}\left(\frac{1}{1}\right) = 1\angle 90^\circ$$

Therefore, the steady-state output is, $y(t) = |G(j1)|\sin(t+90^\circ) = \sin(t+90^\circ)$

$$\boxed{90^\circ}$$

Q6.18.2 NAT 2018 1M Ans: 180 to 180 or -180 to -180

The gain crossover frequency is obtained from $|G(j\omega)| = 1$

$$|G(j\omega)| = \frac{2}{\sqrt{1+\omega^2}\sqrt{4+\omega^2}}$$

Since

$$\sqrt{1+\omega^2} \geq 1, \qquad \sqrt{4+\omega^2} \geq 2,$$

$$|G(j\omega)| \leq 1$$

Equality occurs only at $\omega = 0$. Therefore, $\omega_{gc} = 0$

The phase at the gain crossover frequency is,

$$\angle G(j0) = -\tan^{-1}(0) - \tan^{-1}(0) = 0^\circ$$

Hence,

$$PM = 180^\circ + \angle G(j\omega_{gc}) = 180^\circ + 0^\circ = 180^\circ.$$

$$\boxed{180^\circ}$$

Q6.16.1 NAT 2016 1M Ans: 0.14 to 0.15 ● ○ ○

The input has $\omega = 3$ rad/s, and amplitude $A_i = 2$.
The magnitude of the frequency response at $\omega = 3$ rad/s is

$$|G(j3)| = \frac{8}{|10 + j3|^2} = \frac{8}{10^2 + 3^2} = \frac{8}{109}$$

Therefore, the output amplitude is

$$A_o = A_i|G(j3)| = 2 \times \frac{8}{109} = \frac{16}{109} = 0.147$$

$$\boxed{0.147}$$

Q6.14.1 NAT 2014 2M Ans: 67 to 69 ● ● ○

The input frequency is $\omega = 100\pi$ rad/s.
The frequency response is,

$$G(j\omega) = \frac{e^{-j\omega/500}}{500 + j\omega}$$

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{\omega}{500}\right) - \frac{\omega}{500}\left(\frac{180}{\pi}\right)$$

$$\angle G(j100\pi) = -\tan^{-1}\left(\frac{100\pi}{500}\right) - \frac{100\pi}{500}\left(\frac{180}{\pi}\right) = -\tan^{-1}(0.2\pi) - 36^\circ = -32.14^\circ - 36^\circ = -68.14^\circ$$

Since $y(t) = A\sin(100\pi t - \phi)$,

$$\phi = 68.14^\circ$$

$$\boxed{68.14^\circ}$$

Q6.14.2 NAT 2014 2M Ans: 0.30 to 0.34 ● ● ○

The phase of the loop transfer function is, $\angle G(j\omega)H(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(9\omega)$
At the phase crossover frequency,

$$\angle G(j\omega)H(j\omega) = -180^\circ$$

$$-90^\circ - \tan^{-1}(\omega) - \tan^{-1}(9\omega) = -180^\circ$$

$$\tan^{-1}(\omega) + \tan^{-1}(9\omega) = 90^\circ$$

Using the identity $\tan^{-1}A + \tan^{-1}B = 90^\circ \quad \Rightarrow \quad AB = 1$, we get,

$$9\omega^2 = 1$$

$$\omega = \frac{1}{3} = 0.333 \text{ rad/s}$$

$$\boxed{0.333}$$

Q6.12.1 MCQ | 2012 | 1M | Ans: C

The steady-state output of a sinusoidal input is $y_{ss}(t) = |G(j\omega)| \sin(\omega t + \phi)$.
For the steady-state output to be zero, $|G(j\omega)| = 0$.
Since

$$G(s) = \frac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$$

$$G(j\omega) = \frac{(-\omega^2+9)(j\omega+2)}{(j\omega+1)(j\omega+3)(j\omega+4)}$$

Therefore, $|G(j\omega)| = 0$ when

$$-\omega^2 + 9 = 0$$

$$\omega^2 = 9$$

$$\omega = 3 \text{ rad/s}$$

$$\boxed{(C)\ 3 \text{ rad/s}}$$

Q6.12.2 MCQ | 2012 | 2M | Ans: C

The open-loop transfer function is

$$G(s) = \frac{150}{s(s+9)(s+25)}$$

The phase angle is

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{9}\right) - \tan^{-1}\left(\frac{\omega}{25}\right)$$

At the phase crossover frequency, $\angle G(j\omega_{pc}) = -180^\circ$.

$$\tan^{-1}\left(\frac{\omega_{pc}}{9}\right) + \tan^{-1}\left(\frac{\omega_{pc}}{25}\right) = 90^\circ$$

Using $\tan^{-1} x + \tan^{-1} y = 90^\circ \quad \Rightarrow \quad xy = 1$, we get

$$\frac{\omega_{pc}}{9} \times \frac{\omega_{pc}}{25} = 1$$

$$\omega_{pc} = 15 \text{ rad/s}$$

The magnitude at the phase crossover frequency is

$$|G(j15)| = \frac{150}{15\sqrt{15^2+9^2}\sqrt{15^2+25^2}} = \frac{150}{15\sqrt{306}\sqrt{850}} = 0.01974$$

Hence,

$$\text{GM} = \frac{1}{|G(j15)|} = 50.65$$

$$\text{GM}_{\text{dB}} = 20\log_{10}(50.65) = 34.1 \text{ dB}$$

$$\boxed{(C)\ 34.1 \text{ dB}}$$

Q6.11.1 MCQ 2011 2M Ans: B ● ● ○

The phase of the open-loop transfer function is,

$$\angle G(j\omega) = -3\tan^{-1}\left(\frac{\omega}{5}\right)$$

$$PM = 180^\circ + \angle G(j\omega_{gc})$$

Since the phase margin is 45°,

$$\angle G(j\omega_{gc}) = -180^\circ + 45^\circ = -135^\circ$$

$$3\tan^{-1}\left(\frac{\omega_{gc}}{5}\right) = 135^\circ$$

$$\tan^{-1}\left(\frac{\omega_{gc}}{5}\right) = 45^\circ$$

$$\omega_{gc} = 5 \text{ rad/s}$$

At gain crossover frequency,

$$|G(j\omega_{gc})| = 1$$

$$\frac{K}{|5+j5|^3} = 1$$

$$|5+j5| = 5\sqrt{2}$$

$$K = (5\sqrt{2})^3 = 250\sqrt{2}$$

$$\boxed{(B)\ 250\sqrt{2}}$$

Q6.09.1 MCQ 2009 1M Ans: D ● ○ ○

The input is $x(t) = \sin t$.
Hence, the excitation frequency is $\omega = 1$ rad/s.
The frequency response at $\omega = 1$ is

$$H(j1) = \frac{1}{\sqrt{1+1^2}}\angle -\tan^{-1}(1) = \frac{1}{\sqrt{2}}\angle -\frac{\pi}{4}$$

For an LTI system, the output sinusoid has the same frequency, while its amplitude is multiplied by $|H(j\omega)|$ and its phase is shifted by $\angle H(j\omega)$. Therefore,

$$y(t) = \frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)$$

$$\boxed{(D)\ \frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)}$$

Q6.09.2 MCQ 2009 2M Ans: A ● ● ○

The open-loop transfer function is,

$$T(s) = \frac{K}{s(s+a)}$$

Given, $PM = 60^\circ$ and $\omega_{gc} = 1$ rad/s
As,

$$PM = 180^\circ + \angle T(j\omega)\Big|_{\omega=1}$$

$$60^\circ = 180^\circ + \left(-90^\circ - \tan^{-1}\frac{\omega}{a}\right)_{\omega=1}$$

$$\tan^{-1}\frac{1}{a} = 30^\circ$$

$$\therefore \quad a = \sqrt{3}$$

The inserted element is, $G_c(s) = \dfrac{s-\sqrt{3}}{s+\sqrt{3}}$

At $\omega = 1$ rad/s, $|G_c(j1)| = 1$. Hence, the gain crossover frequency remains unchanged at $\omega_{gc} = 1$ rad/s
When two transfer functions are connected in cascade,

$$|GH| = |G||H|, \qquad \angle(GH) = \angle G + \angle H$$

The phase of the inserted element at $\omega = 1$ is, $\angle G_c(j1) = -2\tan^{-1}\dfrac{1}{\sqrt{3}} = -60°$

The original phase at the gain crossover frequency is, $\angle T(j1) = -180° + 60° = -120°$
Therefore, the new phase at the gain crossover frequency is $\angle T'(j1) = -120° - 60° = -180°$
Hence,

$$PM = 180° + \angle T'(j1) = 180° - 180° = 0°$$

$$\boxed{(A)\ 0°}$$

Q6.08.1 MCQ 2008 2M Ans: C

The phase of the open-loop transfer function is:

$$\angle G(j\omega) = \tan^{-1}(6\omega) - 180° - \tan^{-1}\omega - \tan^{-1}(2\omega)$$

At the phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180°$$

$$\tan^{-1}(6\omega) = \tan^{-1}\omega + \tan^{-1}(2\omega)$$

Using $\tan^{-1}A + \tan^{-1}B = \tan^{-1}\left(\dfrac{A+B}{1-AB}\right)$,

$$6\omega = \frac{\omega + 2\omega}{1 - 2\omega^2}$$

$$6\omega(1 - 2\omega^2) = 3\omega$$

$$\omega^2 = \frac{1}{4}$$

$$\omega_{pc} = 0.5 \text{ rad/s}$$

$$\boxed{(C)\ 0.5}$$

Q6.08.2 MCQ 2008 2M Ans: A

From Q6.08.1, $\omega_{pc} = 0.5$ rad/s.
The gain margin is:

$$GM = \frac{1}{|G(j\omega_{pc})|}$$

At $\omega = 0.5$ rad/s,

$$|G(j\omega)| = \frac{\sqrt{1+(6\omega)^2}}{\omega^2\sqrt{1+\omega^2}\sqrt{1+(2\omega)^2}} = \frac{\sqrt{10}}{(0.5)^2\sqrt{1.25}\sqrt{2}} = 8$$

Therefore,

$$GM = \frac{1}{8} = 0.125$$

$$\boxed{(A)\ 0.125}$$

Q6.08.3 | MCQ | 2008 | 2M | Ans: C

The open-loop transfer function is,

$$G(s) = \frac{e^{-sT_D}}{s}$$

$$G(j\omega) = \frac{1}{\omega} \angle\left(-\frac{\pi}{2} - \omega T_D\right)$$

The gain crossover frequency is obtained from,

$$|G(j\omega_{gc})| = 1$$

$$\frac{1}{\omega_{gc}} = 1$$

$$\omega_{gc} = 1 \text{ rad/s}$$

At the gain crossover frequency,

$$\angle G(j1) = -\frac{\pi}{2} - T_D$$

Hence, the phase margin is:

$$PM = \pi + \angle G(j1) = \pi - \frac{\pi}{2} - T_D = \frac{\pi}{2} - T_D$$

For the closed-loop system to be stable,

$$PM > 0$$

$$\frac{\pi}{2} - T_D > 0$$

$$T_D < \frac{\pi}{2}$$

$$\boxed{(C)\ \frac{\pi}{2}}$$

Mistakes to be avoided

- Do not approximate the exponential term using Padé approximation when the problem can be solved directly using frequency-response concepts.
- Do not mix degrees and radians while evaluating the phase contributed by the delay term.

Q6.06.1 | MCQ | 2006 | 2M | Ans: D

From the given table, $|G(j8)| = 1$. Hence, $\omega_{gc} = 8$ rad/s.
At this frequency, $\angle G(j8) = -170°$. Hence, the phase margin is:

$$PM = 180° + \angle G(j\omega_{gc}) = 180° - 170° = 10°$$

From the table, $\angle G(j10) = -180°$. Hence, $\omega_{pc} = 10$ rad/s.
At this frequency, $|G(j10)| = 0.64$. Hence, the gain margin is:

$$GM = \frac{1}{0.64} = 1.5625$$

In dB,

$$GM_{\text{dB}} = 20\log_{10}(1.5625) \approx 3.88 \text{ dB}$$

$$\boxed{(D)\ 3.88 \text{ dB},\ 10°}$$

Key Points

- Gain crossover frequency is obtained from $|G(j\omega)| = 1$.
- Phase crossover frequency is obtained from $\angle G(j\omega) = -180°$.
- $PM = 180° + \angle G(j\omega_{gc})$ and $GM = \dfrac{1}{|G(j\omega_{pc})|}$

Q6.04.1 MCQ 2004 2M Ans: A

The transfer function is of the form: $G(s) = K(1 + sT)$.
At $\omega = 0$ rad/sec, $G(j0) = K$.
From the polar plot, $G(j0) = 5$. Hence, $K = 5$
The phase angle of $G(j\omega) = 5(1 + j\omega T)$ is, $\phi = \tan^{-1}(\omega T)$
At $\omega = 10$ rad/s, the phase angle is 45°. Hence,

$$\tan^{-1}(10T) = 45°$$
$$10T = 1$$
$$T = 0.1 \text{ s}$$

Therefore,

$$G(s) = 5(1 + 0.1s)$$

$$\boxed{(A)\ 5(1 + 0.1s)}$$

Q6.04.2 MCQ 2004 2M Ans: C

The phase of the loop transfer function is:

$$\angle G(j\omega) = -\frac{\pi}{2} - 0.1\omega$$

At the phase crossover frequency,

$$\angle G(j\omega_{pc}) = -\pi$$
$$-\frac{\pi}{2} - 0.1\omega_{pc} = -\pi$$
$$\omega_{pc} = \frac{\pi}{0.2} \text{ rad/sec}$$

The phase of the loop transfer function is:

$$\angle G(j\omega) = -90° - 0.1\left(\omega_{pc} \times \frac{180°}{\pi}\right)$$

At the phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180°$$
$$-90° - 0.1\left(\omega_{pc} \times \frac{180°}{\pi}\right) = -180°$$
$$\omega_{pc} = 5\pi = \frac{\pi}{0.2} \text{ rad/sec}$$

$$\boxed{(C)\ \frac{\pi}{0.2}}$$

Mistakes to be avoided

- Do not mix degrees and radians. The phase contribution of the exponential term is always in radians.

Q6.03.1 MCQ 2003 2M Ans: C

The input signal is: $x(t) = 0.5 \sin t$.
Therefore, the input frequency is, $\omega = 1$ rad/s.
The frequency response at $\omega = 1$ rad/s is,

$$G(j1) = \frac{e^{-j0.1}}{1+j}$$

$$\angle G(j1) = -0.1 \times \frac{180}{\pi} - \tan^{-1}(1) = -5.73^\circ - 45^\circ = -50.73^\circ$$

$$\boxed{(C)\ -50.73^\circ}$$

Key Points

- If the input to a transfer function $G(s)$ is $x(t) = A\sin(\omega_0 t + \phi)$, then the steady-state output is:

$$y(t) = A|G(j\omega_0)|\sin(\omega_0 t + \phi + \angle G(j\omega_0))$$

Q6.02.1 MCQ 2002 2M Ans: B

The phase of the loop transfer function is:

$$\angle G(j\omega)H(j\omega) = -\frac{\pi}{2} - \omega L$$

The phase cross-over frequency is $\omega_{pc} = 5$ rad/s. Hence,

$$\angle G(j\omega_{pc})H(j\omega_{pc}) = -\pi$$

$$-\frac{\pi}{2} - 5L = -\pi$$

$$L = \frac{\pi}{10}$$

$$\boxed{(B)\ \frac{\pi}{10}}$$

Mistakes to be avoided

- Do not mix degrees and radians. The phase contribution of the exponential term is always in radians.

Q6.98.1 MCQ 1998 1M Ans: B

The gain crossover frequency satisfies:

$$|GH(j\omega_{gc})| = 1$$

$$\frac{1}{(1+\omega_{gc}^2)^{3/2}} = 1$$

$$\omega_{gc} = 0$$

The phase at the gain crossover frequency is, $\angle GH(j0) = -3\tan^{-1}(0) = 0$
Therefore, the phase margin is:

$$\text{PM} = \pi + \angle GH(j\omega_{gc}) = \pi + 0 = \pi$$

$$\boxed{(B)\ \pi}$$

Q6.97.1 MCQ 1997 1M Ans: B

The open-loop transfer function is

$$G(s)H(s) = \frac{1}{(s+1)^3}$$

$$\angle G(j\omega)H(j\omega) = -3\tan^{-1}\omega$$

The phase crossover frequency is obtained from:

$$-3\tan^{-1}\omega_{pc} = -180^\circ$$

$$\omega_{pc} = \tan 60^\circ = \sqrt{3}$$

The magnitude at ω_{pc} is: $|G(j\omega_{pc})H(j\omega_{pc})| = \dfrac{1}{(1+\omega_{pc}^2)^{3/2}} = \dfrac{1}{(1+3)^{3/2}} = \dfrac{1}{8}$

Therefore, the gain margin is:

$$\text{GM} = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} = \frac{1}{1/8} = 8$$

$$\boxed{(B)\ 8}$$

DEBUG INFO

Chapter VI

Polar Plot and Frequency Response of Second Order Systems

Total Questions : 28

MCQ : 18

MSQ : 0

NAT : 10

SOLUTIONS

VII

Nyquist Plot

Topics

1. Nyquist Plot
2. Nyquist Stability Criterion
3. Gain Crossover Frequency (ω_{gc}) from Nyquist Plot
4. Phase Crossover Frequency (ω_{pc}) from Nyquist Plot
5. Gain Margin (GM) using Nyquist Plot
6. Phase Margin (PM) using Nyquist Plot

Q7.25.1 MCQ | 2025 | 1M | Ans: B | ● ● ○

The given contour encloses two poles of $G(s)H(s)$ in the right-half s-plane, $P = 2$
For closed-loop stability, there should be no zeros of $1 + G(s)H(s)$ inside the contour. Hence, $Z = 0$
By the Nyquist Stability Criterion,

$$N = P - Z$$

$$N = 2 - 0 = 2$$

Thus, the locus of $1 + G(s)H(s)$ must encircle the origin two times in the opposite direction of the nyquist contour, that is, in the clockwise direction.

$$\boxed{\text{(B) The locus of } 1 + G(s)H(s) \text{ should encircle the origin twice in the clockwise direction}}$$

Key Points

- For the Nyquist criterion,

$$N = P - Z,$$

where N is the net number of encirclements of the origin by the locus of $1 + G(s)H(s)$, P is the number of poles of $G(s)H(s)$ inside the contour, and Z is the number of zeros of $1 + G(s)H(s)$ inside the contour.

- For closed-loop stability, $Z = 0$.

Q7.23.1 NAT | 2023 | 1M | Ans: 2 to 2 | ● ○ ○

Given,

$$G(s)H(s) = \frac{1}{2}\frac{(s-1)(s-2)}{(s+1)(s+2)}$$

Poles: $s = -1, s = -2$ and Zeros: $s = 1, s = 2$
Hence it has two zeros and 0 poles in the right half plane.
From Nyquist criterion, $N = P - Z = 0 - 2 = -2$
Therefore, the Nyquist plot encircles the origin twice (Direction is not asked).

$$\boxed{2}$$

Q7.22.1 MCQ | 2022 | 2M | Ans: A | ● ● ○

Since point $P = -0.77 - 0.64j$ lies on the unit circle, $|G(j\omega_{gc})| = 1$. Therefore, P corresponds to the gain crossover frequency.
The phase at P is

$$\angle G(j\omega_{gc}) = -180^\circ + \tan^{-1}\left(\frac{0.64}{0.77}\right) = -180^\circ + 39.7^\circ = -140.3^\circ$$

$$PM = 180^\circ + \angle G(j\omega_{gc}) = 180^\circ - 140.3^\circ = 39.7^\circ$$

At the phase crossover frequency, the Nyquist plot intersects the negative real axis at $R = -0.21 + 0j$

$$|G(j\omega_{pc})| = 0.21$$

$$GM = \frac{1}{|G(j\omega_{pc})|} = \frac{1}{0.21} = 4.76$$

$$\boxed{(A)\ PM = 39.7^\circ \text{ and } GM = 4.76}$$

Key Points

- If the Nyquist plot intersects the unit circle, then $|G(j\omega_{gc})| = 1$, and the corresponding point gives the gain crossover frequency, then the phase margin is: $PM = 180^\circ + \angle G(j\omega_{gc})$.
- If the Nyquist plot intersects the negative real axis at $G(j\omega_{pc}) = -a$, $a > 0$, then the gain margin is:

$GM = \frac{1}{a}.$

Q7.21.1 NAT | 2021 | 1M | Ans: -1 or 1 | ● ● ●

Given,

$$G(s) = \frac{3}{s-1}$$

The open-loop transfer function has one right-half plane pole at $s = 1$. Hence, $P = 1$
For the closed-loop system,

$$1 + G(s) = 0$$
$$1 + \frac{3}{s-1} = 0$$
$$s - 1 + 3 = 0$$
$$s = -2$$

Hence, the closed-loop system has no right-half plane poles. So, $Z = 0$
Using Nyquist stability criterion (assuming nyquist contour in clockwise direction),

$$N = P - Z = 1 - 0 = 1$$

Since it is given that the N is positive for clockwise encirclement. Hence the nyquist plot encirlces $(1, j0)$ point in clockwise direction.
But as N is positive, that is, clockwise encirclement and we assumed the nyquist contour is already in clockwise direction. This contradicts the fact that if $P > Z$, then the nyquist contour and nyquist plot should be in opposite directoin.
This creates an ambiguity of sign convention in nyquist stability criterion. And this can be solved by:

As positive N itself denotes a clockwise encirclement of the $(-1, j0)$ point. Using $N = P - Z$ would make the contour traversal and plot encirclement have the same direction, which is inconsistent with the Nyquist argument principle. Hence, for this question, we use

$$N = Z - P$$
$$N = 0 - 1 = -1$$

From question, encirclement of nyquist plot is anti-clockwise if N is negative. Hence the fact that the nyquist plot and nyquist contour should be in opposite direction is preserved. Therefore,

$$\boxed{-1}$$

Alternatively, since the question defines positive N for clockwise encirclements of the $(-1, j0)$ point, assuming nyquist plot in clockwise direction caused inconsistency so we may assume that the Nyquist contour is traversed in the anticlockwise direction. Under this convention,

$$N = P - Z$$
$$N = 1 - 0 = 1$$

From question, encirclement of nyquist plot is clockwise if N is positive. Hence the fact that the nyquist plot and nyquist contour should be in opposite direction is preserved. Therefore,

$$\boxed{1}$$

Mistakes to be avoided

- Do not blindly assume the formula as $N = P - Z$. Always check the sign convention adopted for N and the assumed direction of traversal of the Nyquist contour.
- The quantities P and Z always satisfy $P \geq 0$ and $Z \geq 0$.
- The quantity N can be positive, negative, or zero.
- Always compare the direction of traversal of the Nyquist contour and the direction of encirclement of the critical point. If $P > Z$, the two directions must be opposite, whereas if $Z > P$, the two directions must be the same.

Q7.20.1 | MCQ | 2020 | 2M | Ans: A | ● ● ○

The open-loop transfer function is

$$G(s)H(s) = \frac{1}{s(s-2)}$$

The open-loop poles are at $s = 0$, and $s = 2$
One pole lies in the right-half plane. Hence, $P = 1$

The characteristic equation is, $1 + G(s)H(s) = 0$

$$1 + \frac{1}{s(s-2)} = 0$$

$$s^2 - 2s + 1 = 0$$

$$s = 1, 1$$

The characteristic equation has 2 Right half zeros. Hence, $Z = 2$

Using the Nyquist stability criterion (assuming nyquist contour is in clockwise direction),

$$N = P - Z$$

$$N = 1 - 2$$

$$Z = -1$$

$\boxed{(A) \text{ encircles } (-1 + j0) \text{ point once in the clockwise direction}}$

Key Points

- By default, the nyquist contour is taken in clockwise direction.
- The Nyquist plot of $G(s)H(s) = 0$ encircles origin. The Nyquist plot of $G(s)H(s) = -1$ encircles $-1, j0$ point.
- In order to find the encirclements of $G(s)H(s) = -1$, we need to find encirclements of $1 + G(s)H(s) = 0$ of origin.
- Thus, the number of encirclements of $(-1, j0)$ by $G(s)H(s)$ is equal to the number of encirclements of the origin by $1 + G(s)H(s)$. Therefore, the Nyquist Stability Criterion is applied to the characteristic equation $1 + G(s)H(s) = 0$.
- Since adding unity to the transfer function only translates the Nyquist plot and does not change its poles, the poles of $1 + G(s)H(s)$ are the same as those of the open-loop transfer function $G(s)H(s)$. Hence, the number of right-half-plane poles P is obtained directly from the open-loop transfer function.

Q7.18.1 | NAT | 2018 | 2M | Ans: 0 to 0 | ● ● ○

The loop transfer function is

$$G(s)H(s) = \frac{s^2 + 0.2s + 100}{2(s^2 - 0.2s + 100)}$$

The characteristic equation is $1 + G(s)H(s) = 0$

$$1 + \frac{s^2 + 0.2s + 100}{2(s^2 - 0.2s + 100)} = 0$$

$$3s^2 - 0.2s + 300 = 0$$

Applying the Routh-Hurwitz criterion,

$$\begin{array}{c|cc} s^2 & 3 & 300 \\ s^1 & -0.2 & 0 \\ s^0 & 300 & \end{array}$$

There are two sign changes in the first column.
Hence, $Z = 2$

The poles of the loop transfer function are obtained from:

$$s^2 - 0.2s + 100 = 0$$

Applying the Routh-Hurwitz criterion,

$$\begin{array}{c|cc} s^2 & 1 & 100 \\ s^1 & -0.2 & 0 \\ s^0 & 100 & \end{array}$$

Again, there are two sign changes in the first column.
Hence, $P = 2$

Using the Nyquist Stability Criterion,

$$N = P - Z = 2 - 2 = 0$$

$$\boxed{0}$$

Q7.16.1 NAT 2016 1M Ans: 1 to 1

Given,

$$G(s) = \frac{s-1}{s+1}$$

Pole: $s = -1$ and Zero: $s = 1$
Hence, the given transfer function has one right half zero and no right half pole. Therefore, $P = 0$ and $Z = 1$
Using Nyquist Stability Criterion (assuming nyquist contour in clockwise direction),

$$N = P - Z = 0 - 1 = -1$$

Therefore, the nyquist plot of the given transfer function encircles origin once in clockwise direction.

$$\boxed{1}$$

Q7.15.1 MCQ 2015 2M Ans: D

From the given Nyquist plot, as $\omega \to 0$, the phase starts from $-270°$. As $\omega \to \infty$, the phase approaches $-180°$. Since the numerator degree is 0 and the denominator degree is 2, let us verify each option by considering a typical transfer function.

Option (A): Stable, Type-0

$$G(s) = \frac{K}{s^2 + as + b} \qquad a, b > 0$$

$$G(j\omega) = \frac{K}{-\omega^2 + ja\omega + b}$$

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{a\omega}{b - \omega^2}\right)$$

$$\omega \to 0$$

$$\angle G(j0) = 0°$$

Starting phase is not $-270°$.
Hence, Option (A) is eliminated.

Option (B): Stable, Type-1

$$G(s) = \frac{K}{s(s+a)} \qquad a > 0$$

$$G(j\omega) = \frac{K}{j\omega(j\omega + a)}$$

$$\angle G(j\omega) = -90° - \tan^{-1}\left(\frac{\omega}{a}\right)$$

$$\omega \to 0$$

$$\angle G(j0) = -90°$$

Starting phase is not $-270°$.
Hence, Option (A) is eliminated.

Option (C): Unstable, Type-0

$$G(s) = \frac{K}{(s-a)(s+b)} \qquad a, b > 0$$

$$G(j\omega) = \frac{K}{(j\omega - a)(j\omega + b)}$$

$$\angle G(j\omega) = -180° + \tan^{-1}\left(\frac{\omega}{a}\right) - \tan^{-1}\left(\frac{\omega}{b}\right)$$

$$\omega \to 0$$

$$\angle G(j0) = -180°$$

Starting phase is not $-270°$.
Hence, Option (C) is eliminated.

Option (D): Unstable, Type-1

$$G(s) = \frac{K}{s(s-a)} \qquad a > 0$$

$$G(j\omega) = \frac{K}{j\omega(j\omega - a)}$$

$$\angle G(j\omega) = -90° - 180° + \tan^{-1}\left(\frac{\omega}{a}\right)$$

$$\omega \to 0$$

$$\angle G(j0) = -270°$$

Starting phase matches the given plot.
Hence, Option (D) is correct.

$$\boxed{(D)\ \text{an unstable, type-1 system}}$$

Q7.05.1 MCQ 2005 2M Ans: A

The open-loop transfer function has two right-half-plane poles, $P = 2$.
Since the closed-loop system is stable, $Z = 0$.
Using the Nyquist stability criterion (clockwise Nyquist contour),

$$N = P - Z = 2 - 0 = 2$$

Therefore, the Nyquist plot must encircle the point $(-1, 0)$ two times in the clockwise direction.

$$\boxed{(A)\ 2}$$

Q7.02.1 MCQ 2002 1M Ans: A

Assume Nyquist contour is in clockwise direction.
The open-loop transfer function has one right-half-plane pole, $P = 1$
The Nyquist plot makes one clockwise encirclement of the point $(-1, 0)$, $N = -1$
Using the Nyquist stability criterion,

$$N = P - Z$$

$$-1 = 1 - Z$$

$$Z = 2$$

Hence, the closed-loop system has 2 right-half-plane poles. Hence unstable.

$$\boxed{(A)\ \text{Unstable}}$$

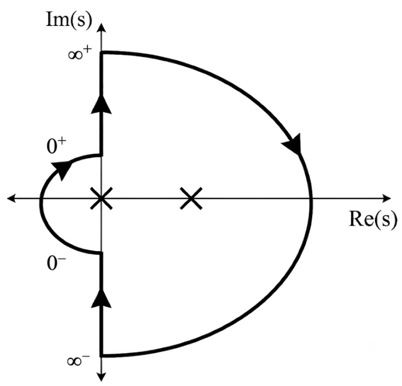
Q7.01.1 NAT 2001 Ans: *

The open-loop transfer function is

$$G(s) = \frac{s+1}{s(s-1)}$$

The poles of the open-loop transfer function are $s = 0, 1$
The given Nyquist contour contains an indentation around the pole at the origin towards the left.

Segment	Mapping
Segment–I	$j0^+ \rightarrow j\infty^+$
Segment–II	$j\infty^+ \rightarrow j\infty^-$
Segment–III	$j\infty^- \rightarrow j0^-$
Segment–IV	$j0^- \rightarrow j0^+$



Segment–I:
Put $s = j\omega$

$$G(j\omega) = \frac{1 + j\omega}{j\omega(j\omega - 1)} = \frac{\sqrt{1+\omega^2}}{\omega\sqrt{1+\omega^2}} \angle \tan^{-1}\omega - 90^\circ - \left(180^\circ - \tan^{-1}\omega\right) = \frac{1}{\omega} \angle -270^\circ + 2\tan^{1}(\omega)$$

$$G(j0^+) = \infty \angle -270^\circ \qquad \text{and} \qquad G(j\infty^+) = 0 \angle -90^\circ$$

On rationalizing the given transfer function, we get,

$$G(j\omega) = -\frac{2}{1+\omega^2} + j\,\frac{(1-\omega^2)}{\omega(1+\omega^2)}$$

$$\Re\{G(j\omega)\} = -\frac{2}{1+\omega^2} \qquad \Im\{G(j\omega)\} = \frac{1-\omega^2}{\omega(1+\omega^2)}$$

Here, $\Re\{G(j\omega)\} < 0$ for every $\omega > 0$
But, $\Im\{G(j\omega)\} > 0$ for $\omega < 1$ and $\Im\{G(j\omega)\} < 0$ for $\omega > 1$
The nyquist plot cuts the real axis when $\omega = 1$ and at:

$$G(j1)H(j1) = -\frac{2}{2} = -1$$

Therefore, the nyquist plot crosses the real axis at $(-1, j0)$

ω	$\Re\{G(j\omega)\}$	$\Im\{G(j\omega)\}$	Quadrant
$\omega < 1$	$-$	$+$	Quadrant - II
$\omega > 1$	$-$	$-$	Quadrant - III

Segment–II:
Put $s = Re^{j\theta}$ where, $R \to \infty$ and $\theta : +90^\circ$ to -90° in Clockwise Direction

$$G(Re^{j\theta}) = \frac{Re^{j\theta} + 1}{Re^{j\theta}(Re^{j\theta} - 1)}$$

As $R \to \infty$, then, $Re^{j\theta} + 1 \approx Re^{j\theta}$ and $Re^{j\theta} - 1 \approx Re^{j\theta}$

$$G(Re^{j\theta})H(Re^{j\theta}) \approx \frac{Re^{j\theta}}{R^2 e^{j2\theta}} = \frac{1}{R}e^{-j\theta}$$

As $R \to \infty$, then, $\frac{1}{R} \to 0$
As $\theta : 90^\circ \to -90^\circ$, then $-\theta : -90^\circ \to 90^\circ$
Therefore, Segment-II maps to a semicircle of approximately zero radius about the origin and is traced in the anticlockwise direction from -90° to 90°
Segment–III:
Segment–III is the mirror image of Segment–I.
Segment–IV:
Put $s = re^{j\theta}$ where, $r \to 0$ and $\theta : -90^\circ \to 90^\circ$ in Clockwise direction

$$G(re^{j\theta}) = \frac{re^{j\theta} + 1}{re^{j\theta}(re^{j\theta} - 1)}$$

As $r \to 0$, then, $re^{j\theta} + 1 \approx 1$ and $re^{j\theta} - 1 \approx -1$

$$G(re^{j\theta})H(re^{j\theta}) \approx -\frac{1}{re^{j\theta}} = \frac{1}{r}e^{j(\pi-\theta)}$$

As $r \to 0$, then, $\frac{1}{r} \to \infty$
As $\theta : -90^\circ \to 90^\circ$, then $\pi - \theta : 270^\circ \to 90^\circ$
Therefore, Segment-IV maps to an infinite semicircle in the anticlockwise direction from 270° to 90°
Hence the complete Nyquist Plot is:

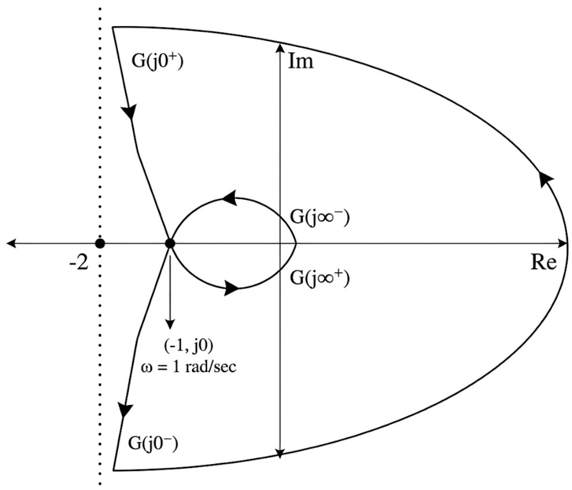


Fig. Solution Q7.01.1

Check:
The given contour encircles two poles in clockwise direction. Hence the mapped nyquist plot should encircle the origin two times in opposite direction of the contour, that is, in anticlockwise direction.
Hence, the obtained nyquist plot is correct.

Q7.01.2 NAT 2001 Ans: *

The given Nyquist contour is traversed in the clockwise direction. With this choice of contour, a clockwise encirclement of the critical point $(-1, j0)$ is taken as negative.
The Nyquist stability criterion is: $N = P - Z$, where N is the net clockwise encirclements of $(-1, 0)$, P is the number of right-half-plane poles of the open-loop transfer function and Z is the number of right-half-plane poles of the closed-loop system.
For closed-loop stability, $Z = 0$. Therefore,

$$N = 2 - 0 = 2$$

Thus, if the nyquist plot of open-loop transfer function encircles the point $(-1, j0)$ twice in anti clockwise direction, then the system would be stable.
Hence, the encirclement criterion for stability of this system with the type of contour chosen is:

$$\boxed{N = 2}$$

Q7.01.3 NAT 2001 Ans: K>1

With proportional gain K, the open-loop transfer function is:

$$G(s) = \frac{K(s+1)}{s(s-1)}$$

The characteristic equation of the closed-loop system is:

$$1 + G(s) = 0$$

$$1 + \frac{K(s+1)}{s(s-1)} = 0$$

$$s^2 + (K-1)s + K = 0$$

For stability, all the coefficients should be of same sign. Hence,

$$K - 1 > 0, \qquad K > 0$$

Hence, $K > 1$

$$\boxed{K > 1}$$

DEBUG INFO

Chapter VII

Nyquist Plot

Total Questions : 13

MCQ : 6

MSQ : 0

NAT : 7

SOLUTIONS

VIII

Bode Plot

Topics

1. Bode Gain and Phase Plots
2. Minimum and Non-Minimum Phase Systems
3. Bode Plots for Repeated Poles and Zeros
4. Recovery of Transfer Function from Bode Plot
5. Error in Asymptotic Bode Gain Plot
6. Bode Plots of 2nd Order Systems

Q8.23.1 MCQ 2023 2M Ans: B

The resonant peak occurs at approximately $\omega = 100$ rad/s.
Hence, the transfer function must have a pair of complex poles with natural frequency $\omega_n = 100$ rad/s.
Therefore, from the options, the denominator would be $s^2 + 2s + 10000$.
From the given phase plot, at $\omega = 100$ rad/s, we have phase as $-360°$.
Now we will check phase at $\omega = 100$ rad/sec of options A, B and C.
Option A:

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{2\omega}{10000-\omega^2}\right)$$

$$\angle G(j100) = -\tan^{-1}\left(\frac{2(100)}{10000-(100)^2}\right) = -90°$$

Option B:

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{2\omega}{10000-\omega^2}\right) - 0.05\omega$$

$$\angle G(j100) = -90° - 286.48° = -376.48°$$

Option C:

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{2\omega}{10000-\omega^2}\right) - 0.05\omega \times 10^{-12}$$

$$\angle G_C(j100) = \left(-90° - 2.86 \times 10^{-9}\right)^{\circ} \approx -90°$$

From the above phases, only option B is closer to $-360°$.

$$\boxed{(B)\ \frac{10000e^{-0.05s}}{s^2+2s+10000}}$$

Mistakes to be avoided

- Do not mix radians and degrees while computing the phase contribution of a time delay term.
- For $G(s) = e^{-sT}$, the phase is $\angle G(j\omega) = -\omega T$ which is always in radians. Convert to degrees only if required:

$$\angle G(j\omega) = -\omega T\left(\frac{180°}{\pi}\right).$$

Q8.19.1 MCQ 2019 2M Ans: A

Since the disturbance is added at the output,

$$\frac{Y(s)}{D(s)} = \frac{1}{1+L(s)}$$

From the Bode magnitude plot,

$$20\log|L(j5)| = 40\,\text{dB}$$

$$|L(j5)| = 10^{40/20} = 100$$

Hence,

$$\left|\frac{Y(j5)}{D(j5)}\right| = \frac{1}{|1+L(j5)|} \approx \frac{1}{|L(j5)|} = \frac{1}{100}, \qquad \text{since } |L(j5)| \gg 1$$

The disturbance amplitude is, $|D| = 0.1$.

Therefore, the steady-state output amplitude is, $|Y| = 0.1 \times \frac{1}{100} = 10^{-3}$

$$\boxed{(A)\ 1.00 \times 10^{-3}}$$

Key Points

- If the input to a transfer function $G(s)$ is $x(t) = A\sin(\omega_0 t + \phi)$, then the steady-state output is:

$$y(t) = A|G(j\omega_0)|\sin(\omega_0 t + \phi + \angle G(j\omega_0))$$

- The amplitude of the steady state output is, $A|G(j\omega_0)|$ and the phase is, $\phi + \angle G(j\omega_0)$

Q8.18.1 MCQ | 2018 | 1M | Ans: A

The given transfer function is

$$G(s) = \frac{100^2 e^{-0.01s}}{s^2 + 0.2s + 100^2}$$

The above transfer function is a second order low-pass filter cascaded by a transportation lag.
The phase of practical second order low-pass filters changes gradually from 0° to −180°.
But in Option (B), it shows an abrupt phase change. So it an ideal plot. Hence, it is eliminated.
Option (D) is also eliminated since it represents a second-order high-pass filter whose phase changes from 0° to +180°.
We are left with Options (A) and (C). Both plots starts at 0 degree phase but:
Options (A) has decreasing plot (negative phase) for frequency range 0 – 100 rad/sec.
Options (C) has increasing plot (positive phase) for frequency range 0 – 100 rad/sec.
Let us evaluate the phase at $\omega = 50$ rad/s,

$$G(j\omega) = \frac{100^2 e^{-j0.01\omega}}{100^2 - \omega^2 + j0.2\omega}$$

$$\angle G(j\omega) = \angle 100^2 + \angle e^{-j0.01\omega} - \angle\left(100^2 - \omega^2 + j0.2\omega\right) = 0^\circ - \omega(0.01)\left(\frac{180}{\pi}\right) - \tan^{-1}\left(\frac{0.2\omega}{100^2 - \omega^2}\right)$$

$$= -\tan^{-1}\left(\frac{0.2\omega}{100^2 - \omega^2}\right) - \omega(0.01)\left(\frac{180}{\pi}\right)$$

$$\angle G(j50) = -\tan^{-1}\left(\frac{10}{7500}\right) - \left(\frac{180}{2\pi}\right) = -28.72^\circ$$

Since the phase is negative between 0 and 100 rad/s, Option (C) is eliminated.
Therefore, the correct phase plot is Option (A).

$$\boxed{(A)}$$

Q8.16.1 MCQ | 2016 | 2M | Ans: B

From the given plot,

Frequency Range	Slope Calculation	Inference
$\omega < 1$	0 dB/dec	No Pole or Zero
$1 < \omega < 100$	$\frac{80 - 0}{\log_{10} 10^2 - \log_{10} 10^0} = 40$ dB/dec	2 Zeros
$\omega > 100$	0 dB/dec	2 Poles

Hence,

$$G(s) = K\left(\frac{1+\frac{s}{1}}{1+\frac{s}{100}}\right)^2$$

For $\omega < 1$ rad/s,

$$G(j\omega) \approx K$$

From the magnitude plot,

$$20\log K = 0$$

$$K = 1$$

Therefore,

$$G(s) = \left(\frac{1+s}{1+\frac{s}{100}}\right)^2 = 10^4\left(\frac{s+1}{s+100}\right)^2$$

$$\boxed{(B)\ 10^4\left(\frac{s+1}{s+100}\right)^2}$$

Q8.13.1 MCQ | 2013 | 1M | Ans: B

From the given Bode magnitude plot,

$$\text{Slope} = \frac{-8-32}{\log_{10}10 - \log_{10}1} = -40\,\text{dB/dec}$$

A slope of -40 dB/dec indicates two poles at the origin. Also, the phase is negative for all ω. Hence, the system contains only poles. Therefore, the transfer function can be written as,

$$G(s) = \frac{K}{s^2}$$

For $\omega < 10$ rad/sec, only the two poles at the origin are effective.

$$G(s) \approx \frac{K}{s^2}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{K}{(j\omega)^2}$$

$$G(j\omega) = \frac{-K}{\omega^2}$$

Therefore, magnitude is,

$$|G(j\omega)| = \frac{K}{\omega^2}$$

In dB,

$$20\log(|G(j\omega)|) = 20\log\left(\frac{K}{\omega^2}\right)$$

At $\omega = 1$ rad/sec, the magnitude from the plot is 32 dB. Hence,

$$32 = 20\log\left(\frac{K}{1^2}\right)$$

$$20\log K = 32$$

$$K = 10^{1.6} = 39.8$$

$$\therefore G(s) = \frac{39.8}{s^2}$$

$$\boxed{\text{(B) } \frac{39.8}{s^2}}$$

Q8.08.1 MCQ 2008 1M Ans: C ● ○ ○

The slope of the Bode magnitude plot indicates the number of poles and zeros encountered at each corner frequency.
Initially, the slope is 0 dB/decade, which indicates that no pole or zero has been encountered.
At $\omega = \omega_1$, the slope changes from 0 dB/decade to +20 dB/decade. An increase of 20 dB/decade corresponds to one zero.
At $\omega = \omega_2$, the slope changes from +20 dB/decade to −20 dB/decade. The change in slope is $-20 - (+20) = -40$ dB/decade, which corresponds to two poles.
Therefore, the transfer function contains 1 zero and 2 poles

$$\boxed{(C)\ \text{2 poles and 1 zero}}$$

Q8.05.1 MCQ 2005 2M Ans: C ● ● ○

From the Bode magnitude plot,

- At $\omega = 1$ rad/s, the slope changes from −40 to −20 dB/decade. Hence, there is a zero at $\omega = 1$ rad/s.
- At $\omega = 100$ rad/s, the slope changes from −20 to −40 dB/decade. Hence, there is a pole at $\omega = 100$ rad/s.

Therefore,

$$G(s) = \frac{K\left(1 + \frac{s}{1}\right)}{s^2\left(1 + \frac{s}{100}\right)}$$

The phase angle is:

$$\phi(\omega) = -180^\circ + \tan^{-1}\omega - \tan^{-1}\left(\frac{\omega}{100}\right)$$

Since the plot is symmetric about ω_c,

$$\omega_c = \sqrt{1 \times 100} = 10 \text{ rad/s}$$

The minimum absolute phase angle occurs at $\omega = \omega_c$.

$$\phi(10) = -180^\circ + \tan^{-1}(10) - \tan^{-1}(0.1) = -180^\circ + 84.3^\circ - 5.7^\circ = -101.4^\circ$$

Hence, the minimum absolute phase angle contribution is, $|\phi_{\min}| = 101.4^\circ$

$$\boxed{(C)\ 101.4^\circ}$$

Key Points

- A slope increase of 20 dB/decade indicates one zero, while a slope decrease of 20 dB/decade indicates one pole.
- If the Bode magnitude plot is symmetric, the frequency of minimum absolute phase angle is the geometric mean of the break frequencies.

Q8.04.1 MCQ | 2004 | 2M | Ans: B | ● ● ○

From the Bode magnitude plot,

- Initial slope is -20 dB/decade $\Rightarrow$ one pole at the origin.
- At $\omega = 2$ rad/s, the slope changes from -20 to 0 dB/decade $\Rightarrow$ one zero at $\omega_z = 2$ rad/s.
- At $\omega = 20$ rad/s, the slope changes from 0 to -20 dB/decade $\Rightarrow$ one pole at $\omega_p = 20$ rad/s.

Hence, the transfer function is of the form:

$$G(s) = K\frac{\left(1 + \frac{s}{2}\right)}{s\left(1 + \frac{s}{20}\right)}$$

Between $2 < \omega < 20$, the magnitude plot is flat at 20 dB. In this frequency range, $\left(1 + \frac{s}{2}\right) \approx \frac{s}{2}$, and $\left(1 + \frac{s}{20}\right) \approx 1$. Therefore,

$$|G(j\omega)| \approx K\frac{\omega/2}{\omega} = \frac{K}{2}$$

$$20\log\left(\frac{K}{2}\right) = 20$$

$$\frac{K}{2} = 10$$

$$K = 20$$

Hence,

$$G(s) = \frac{20\left(1 + \frac{s}{2}\right)}{s\left(1 + \frac{s}{20}\right)}$$

$$\boxed{(B)\ \frac{20\left(\frac{s}{2} + 1\right)}{s\left(\frac{s}{20} + 1\right)}}$$

Q8.02.1 NAT | 2002 | 5M | Ans: * | ● ● ○

From the Bode magnitude plot:

Frequency Range	Slope	Inference
$\omega < 1$	-20 dB/dec	Pole at origin
$1 < \omega < \omega_1$	-40 dB/dec	Pole at $\omega = 1$ rad/sec
$\omega > \omega_1$	-20 dB/dec	Zero at $\omega = \omega_1$ rad/sec

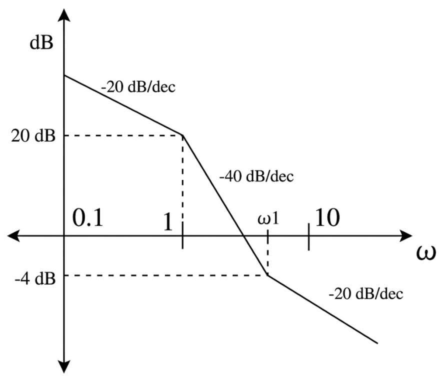


Fig. Solution Q8.02.1

Hence, the transfer function is of the form:

$$G(s) = K\frac{1 + \frac{s}{\omega_1}}{s(1 + \frac{s}{1})}$$

For $\omega < \omega_1$ rad/sec, pole at the origin and pole at $\omega = 1$ rad/sec are active. So,

$$G(s) \approx \frac{K}{s(s)}$$

$$G(j\omega) \approx -\frac{K}{\omega^2}$$

$$|G(j\omega)|_{dB} \approx 20\log\frac{K}{\omega^2}$$

At $\omega = 1$ rad/sec, magnitude is 20 dB.

$$|G(j1)|_{dB} \approx 20\log K = 20$$

$$K = 10$$

For ω_1:

$$\text{Slope} = \frac{20-(-4)}{\log_{10}1 - \log_{10}(\omega_1)} = -40\,\text{dB/dec}$$

$$\log_{10}(\omega_1) = 0.6$$

$$\omega_1 \approx 4\text{ rad/sec}$$

Hence, the transfer function is:

$$G(s) = \frac{10\left(1+\frac{s}{4}\right)}{s(1+s)} = \frac{2.5(s+4)}{s(s+1)}$$

$$\boxed{G(s) = \frac{2.5(s+4)}{s(s+1)}}$$

Q8.00.1 MCQ | 2000 | 1M | Ans: A | ● ○ ○

From the asymptotic Bode magnitude plot:

- Initial slope is 0 dB/decade $\Rightarrow$ no pole or zero at the origin.
- The slope changes from 0 to -20 dB/decade at $\omega = 10$ rad/s, indicating a pole at 10 rad/s.
- The slope changes from -20 to -40 dB/decade at $\omega = 250$ rad/s, indicating another pole at 250 rad/s.

Hence,

$$G(s) = \frac{K}{(1+s/10)(1+s/250)}$$

The low-frequency gain is 40 dB. Hence,

$$20\log_{10}K = 40$$

$$K = 10^{40/20} = 100$$

Therefore,

$$G(s) = \frac{100}{(1+s/10)(1+s/250)}$$

$$\boxed{(A)\ \frac{100}{(1+s/10)(1+s/250)}}$$

Q8.96.1 MCQ | 1996 | 1M | Ans: D | ● ○ ○

A minimum phase system has all its poles and zeros in the left half of the s-plane and does not contain a time delay.

(A) $\dfrac{1+s}{1-s}$ has a pole at $s = 1$ (RHP) $\Rightarrow$ Not minimum phase.

(B) $\dfrac{1-s}{1+s}$ has a zero at $s = 1$ (RHP) $\Rightarrow$ Not minimum phase.

(C) $\dfrac{e^{-3s}}{s}$ contains a time delay e^{-3s}, which introduces additional phase lag $\Rightarrow$ Not minimum phase.

(D) $\dfrac{s}{(s+1)(s+2)}$ has poles at $s = -1, -2$ and a zero at $s = 0$. There are no right-half-plane poles or zeros and no time delay. Hence, it is a minimum phase system.

$$\boxed{(D)\ \frac{s}{(s+1)(s+2)}}$$

Q8.92.1 MCQ 1992 Ans: C ● ● ○

From the Bode magnitude plot:

- Initial slope is 0 dB/dec.
- At ω_1, the slope changes to -6 dB/octave ($= -20$ dB/dec), indicating a pole at ω_1.
- At $\omega_2 = 2$ rad/s, the slope changes to -12 dB/octave ($= -40$ dB/dec), indicating another pole at $\omega_2 = 2$ rad/s.

Between ω_1 and ω_2, the magnitude decreases from 20 dB to 12 dB. Hence,

$$20\log_{10}\left(\frac{\omega_2}{\omega_1}\right) = 8$$

$$20\log_{10}\left(\frac{2}{\omega_1}\right) = 8$$

$$\omega_1 = \frac{2}{10^{0.4}} \approx 0.8\,\text{rad/s}$$

Beyond ω_2, the slope is -40 dB/dec. The gain crossover frequency is obtained from:

$$40\log_{10}\left(\frac{\omega_c}{\omega_2}\right) = 40\log_{10}\left(\frac{\omega_{gc}}{2}\right) = 12$$

$$\omega_{gc} = 2\times 10^{0.3} \approx 4\,\text{rad/s}$$

Since the system is minimum phase,

$$G(s) = \frac{K}{\left(1+\frac{s}{0.8}\right)\left(1+\frac{s}{2}\right)}$$

The phase at the gain crossover frequency is:

$$\angle G(j4) = -\tan^{-1}\left(\frac{4}{0.8}\right) - \tan^{-1}\left(\frac{4}{2}\right) = -78.69^\circ - 63.43^\circ = -142.12^\circ$$

$$\therefore\ \text{PM} = 180^\circ - 142.12^\circ \approx 38^\circ$$

Since the phase approaches -180° only asymptotically, there is no finite phase crossover frequency.

$$\therefore\quad \text{GM} = \infty$$

$$\boxed{(C)\ 4,\ \infty,\ 38^\circ}$$

Key Points

- 6 dB/octave = 20 dB/decade.
- Gain margin is infinite when the phase never reaches -180° at any finite frequency.

DEBUG INFO

Chapter VIII

Bode Plot

Total Questions : 12

MCQ : 11

MSQ : 0

NAT : 1

SOLUTIONS

IX

State Space Analysis

Topics

1. State Variables and State Diagram
2. State Space Representation
3. Eigen Values and Eigen Vectors in State Space
4. Solution of State Space Equations
5. State Transition Matrix and Its Properties
6. Recovery of Transfer Function from State Space
7. Similarity Transforms
8. Controllable and Observable Canonical Forms
9. Parallel and Cascade Decomposition
10. Controllability — Kalman's and Gilbert's Test
11. Observability — Kalman's and Gilbert's Test
12. Observable and Detectable Systems

Q9.26.1 MCQ 2026 2M Ans: A

The poles of the system are the eigenvalues of the system matrix

$$A = \begin{bmatrix} 0 & 1 \\ -10 & -7 \end{bmatrix}.$$

The characteristic equation is obtained from

$$|sI - A| = 0.$$

$$\begin{vmatrix} s & -1 \\ 10 & s+7 \end{vmatrix} = 0$$

$$s(s+7) - (-1)(10) = 0$$

$$s^2 + 7s + 10 = 0$$

$$(s+2)(s+5) = 0$$

$$s = -2, -5$$

$$\boxed{(A) \quad -2, -5}$$

Q9.24.1 MCQ 2024 1M Ans: B

A system has a pole at the origin if one of the eigenvalues of the system matrix A is zero.
Hence, the characteristic equation must satisfy $|sI - A| = 0$, with $s = 0$. Therefore, $|A| = 0$
Check the determinant of each option.

Option A: $|A| = (0)(-3) - (1)(-2) = 2 \neq 0$

Option B: $|A| = (1)(3) - (-1.5)(-2) = 3 - 3 = 0$

Option C: $|A| = (1)(-3) - (1.5)(2) = -6 \neq 0$

Option D: $|A| = (0)(3) - (1)(-2) = 2 \neq 0$

Only Option B has zero determinant. Hence, the system has a pole at the origin.

$$\boxed{(B) \begin{bmatrix} 1 & -1.5 \\ -2 & 3 \end{bmatrix}}$$

Key Points

- A system matrix has a pole at the origin if one of its eigenvalues is zero.
- An eigenvalue of zero implies that the determinant of the system matrix is zero.

Q9.22.1 NAT 2022 1M Ans: 3 to 3

The given transfer function is

$$G(s) = \frac{(s+1)(s+3)}{(s+5)(s+7)(s+9)}$$

The order of the system is equal to the degree of the denominator. Thus, Order = 3.
The minimum number of state variables required is same as the order of the transfer function. Hence, minimum 3 state variables are required.

$$\boxed{3}$$

Key Points

- The minimum number of state variables is equal to the order of the transfer function after removing any common pole-zero cancellations.
- Pole-zero cancellation reduces the system order and hence the minimum number of state variables.

Q9.19.1 MCQ 2019 2M Ans: D

The given state equation is

$$\dot{X}(t) = AX(t) + BU(t),$$

where

$$A = \begin{bmatrix} 1 & 2 \\ -3 & -4 \end{bmatrix}, \qquad B = \begin{bmatrix} 20 \\ 10 \end{bmatrix} \qquad \text{and} \qquad X(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The state response is $X(s) = \phi(s)X(0) + \phi(s)BU(s)$, where $\phi(s) = (sI - A)^{-1}$

Since $X(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

$$X(s) = \phi(s)BU(s)$$

Now,

$$sI - A = \begin{bmatrix} s-1 & -2 \\ 3 & s+4 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{(s-1)(s+4)+6} \begin{bmatrix} s+4 & 2 \\ -3 & s-1 \end{bmatrix}$$

For a unit-step input, $U(s) = \frac{1}{s}$

Therefore,

$$X(s) = \frac{1}{s\left[(s-1)(s+4)+6\right]} \begin{bmatrix} 20(s+5) \\ 10(s-7) \end{bmatrix}$$

Applying the Final Value Theorem,

$$X(\infty) = \lim_{s \to 0} sX(s) = \frac{1}{(-1)(4)+6} \begin{bmatrix} 20(5) \\ 10(-7) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 100 \\ -70 \end{bmatrix} = \begin{bmatrix} 50 \\ -35 \end{bmatrix}$$

$$\boxed{(D) \begin{bmatrix} 50 \\ -35 \end{bmatrix}}$$

Q9.18.1 MCQ 2018 2M Ans: D

The given system is

$$\dot{x} = Ax, \qquad \text{where} \quad A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

The solution of the state equation is, $x(t) = e^{At}x(0)$

Since A is a diagonal matrix,

$$e^{At} = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix}$$

Therefore,

$$x(t) = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\boxed{(D)\ x(t) = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

Q9.15.1 NAT | 2015 | 2M | Ans: -3

The controllability matrix is

$$C = \begin{bmatrix} B & AB \end{bmatrix}$$

First, compute

$$AB = \begin{bmatrix} 1 & 2 \\ \alpha & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ \alpha + 6 \end{bmatrix}$$

Hence,

$$C = \begin{bmatrix} 1 & 3 \\ 1 & \alpha + 6 \end{bmatrix}.$$

For the system to be uncontrollable, $|C| = 0$

$$\begin{vmatrix} 1 & 3 \\ 1 & \alpha + 6 \end{vmatrix} = 0$$

$$\alpha + 6 - 3 = 0$$

$$\alpha + 3 = 0$$

$$\boxed{\alpha = -3}$$

Q9.12.1 MCQ | 2012 | 2M | Ans: D

Given state space model,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The controllability matrix is

$$Q_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

First, compute AB:

$$AB = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix}$$

Next, compute A^2B:

$$A^2B = A(AB)$$

$$= \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1 a_2 \\ 0 \\ 0 \end{bmatrix}$$

Hence,

$$Q_c = \begin{bmatrix} 0 & 0 & a_1 a_2 \\ 0 & a_2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Its determinant is

$$|Q_c| = -a_1 a_2^2$$

For complete controllability,

$$|Q_c| \neq 0$$

$$-a_1 a_2^2 \neq 0$$

$$\therefore \quad a_1 \neq 0, \qquad a_2 \neq 0$$

Observe that a_3 does not appear in the controllability determinant. Therefore, controllability is independent of a_3.

$$\boxed{(D)\ a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0}$$

Q9.11.1 MCQ 2011 2M Ans: A

The transfer function is

$$G(s) = C(sI - A)^{-1} B$$

Here,

$$A = \begin{bmatrix} -4 & -1 \\ -3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

First,

$$sI - A = \begin{bmatrix} s+4 & 1 \\ 3 & s+1 \end{bmatrix}$$

$$|sI - A| = (s+4)(s+1) - 3 = s^2 + 5s + 1$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 5s + 1} \begin{bmatrix} s+1 & -1 \\ -3 & s+4 \end{bmatrix}$$

Now,

$$G(s) = C(sI - A)^{-1} B = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s^2 + 5s + 1} \begin{bmatrix} s+1 & -1 \\ -3 & s+4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{s}{s^2 + 5s + 1}$$

$$\boxed{(A)\ \frac{s}{s^2 + 5s + 1}}$$

Q9.09.1 MCQ 2009 2M Ans: B

The characteristic equation is obtained from,

$$|sI - F| = 0$$

$$\begin{vmatrix} s & -1 \\ 4 & s+2 \end{vmatrix} = 0$$

$$s(s+2) + 4 = 0$$

$$s^2 + 2s + 4 = 0$$

Comparing with the standard second-order characteristic equation, $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$, we get:

$$\omega_n^2 = 4, \qquad 2\zeta\omega_n = 2$$

$$\omega_n = 2 \qquad 2\zeta(2) = 2$$

$$\zeta = \frac{1}{2} = 0.5$$

$$\boxed{(B)\ 0.5}$$

Q9.08.1 MCQ 2008 2M Ans: A ● ○ ○

Method 1

The transfer function of a state-space model is given by,

$$G(s) = C(sI - A)^{-1}B + D$$

Here,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

Now,

$$sI - A = \begin{bmatrix} s & -1 \\ 0 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s(s+3)} \begin{bmatrix} s+3 & 1 \\ 0 & s \end{bmatrix}$$

Therefore,

$$C(sI-A)^{-1}B = \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s(s+3)} \begin{bmatrix} s+3 & 1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{s}$$

Method 2

The poles of the transfer function are the eigen values of the state matrix, A Here,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 0 & 1 \\ 0 & -3 \end{vmatrix} = 0$$

Option (A): Pole is: $s = 0$
Option (B): Poles are: $s = 0, -3$
Option (C): Pole is: $s = -3$
Option (D): Poles are: $s = 0, 0$

Only option (A) matches.

Therefore, the transfer function is $\frac{1}{s}$

$$\boxed{(A)\ \frac{1}{s}}$$

Q9.06.1 MCQ 2006 1M Ans: C ● ○ ○

For the state-space model,

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

the output under zero initial conditions is:

$$y(t) = C\int_0^t \exp[A(t-\tau)]\,Bu(\tau)\,d\tau$$

The impulse response is obtained by applying $u(t) = \delta(t)$.
Using the sifting property of the Dirac delta function,

$$\int_0^t \exp[A(t-\tau)]\,B\delta(\tau)\,d\tau = \exp(At)B$$

Therefore,

$$h(t) = C\exp(At)B$$

$$\boxed{(C)\ C\exp(At)B}$$

Q9.05.1 MCQ 2005 1M Ans: B ● ○ ○

The state-space model is:

$$\dot{x} = Ax + bu$$

$$y = cx + du$$

Taking the Laplace transform under zero initial conditions,

$$sX(s) = AX(s) + bU(s)$$

$$(sI - A)X(s) = bU(s)$$
$$X(s) = (sI - A)^{-1}bU(s)$$

The output is:

$$Y(s) = cX(s) + dU(s) = c(sI - A)^{-1}bU(s) + dU(s)$$

Therefore, the transfer function is:

$$\frac{Y(s)}{U(s)} = c(sI - A)^{-1}b + d$$

$$\boxed{(B)\ c(sI - A)^{-1}b + d}$$

Q9.04.1 MCQ 2004 2M Ans: B ● ● ○

The transfer function of a state-space model is:

$$G(s) = C(sI - A)^{-1}B$$

Given,

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

First,

$$sI - A = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 5s + 6}\begin{bmatrix} s+5 & 1 \\ -6 & s \end{bmatrix}$$

Therefore,

$$(sI - A)^{-1}B = \frac{1}{s^2 + 5s + 6}\begin{bmatrix} 1 \\ s \end{bmatrix}$$

Let $C = \begin{bmatrix} c_1 & c_2 \end{bmatrix}$, then:

$$G(s) = \frac{c_1 + c_2 s}{s^2 + 5s + 6}$$

Comparing with the given transfer function, $\frac{s+6}{s^2 + 5s + 6}$,

$$c_2 = 1, \qquad c_1 = 6.$$

Hence,

$$C = \begin{bmatrix} 6 & 1 \end{bmatrix}$$

$$\boxed{(B)\ \begin{bmatrix} 6 & 1 \end{bmatrix}}$$

Q9.03.1 MCQ 2003 2M Ans: A ● ○ ○

The poles of the system are the eigenvalues of the state matrix A.
The characteristic equation is:

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = 0$$

$$s^2 + 3s + 2 = 0$$

$$s = -1, -2$$

$$\boxed{(A)\ -2, -1}$$

Q9.94.1 MCQ 1994 1M Ans: D

The state-space matrices are

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad D = 0.$$

The transfer function is: $G(s) = C(sI - A)^{-1}B + D$.

$$sI - A = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$C(sI - A)^{-1}B = \begin{bmatrix} 1 & 1 \end{bmatrix} \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \frac{2+2s}{(s+1)(s+2)} = \frac{2(s+1)}{(s+1)(s+2)} = \frac{2}{s+2}$$

$$\boxed{(D)\ \frac{2}{s+2}}$$

Q9.94.2 MCQ 1994 1M Ans: C

The output is: $Y(s) = G(s)U(s) + C(sI - A)^{-1}x(0)$.

For a unit step input, $U(s) = \frac{1}{s}$, and given $x(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

From Q9.94.12,

$$G(s) = \frac{2}{s+2}$$

$$G(s)U(s) = \frac{2}{s(s+2)}$$

Also,

$$C(sI - A)^{-1}x(0) = \begin{bmatrix} 1 & 1 \end{bmatrix} \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{s+1}{(s+1)(s+2)} = \frac{1}{s+2}$$

Therefore,

$$Y(s) = \frac{2}{s(s+2)} + \frac{1}{s+2} = \frac{s+2}{s(s+2)} = \frac{1}{s}$$

Applying Inverse Laplace Transform,

$$y(t) = u(t)$$

$$\boxed{(C)\ u(t)}$$

Q9.94.3 MCQ 1994 1M Ans: B

The number of state variables is equal to the number of independent energy storage elements.
The given network contains one induction L and one capacitor C
There is no capacitor-only loop or inductor-only cut-set that introduces a dependency between the storage elements.
Hence, both storage elements contribute independent state variables.
Therefore, 2 state variables are required to represent the given network.

$$\boxed{(B)\ 2}$$

DEBUG INFO

Chapter IX

State Space Analysis

Total Questions : 17

MCQ : 15

MSQ : 0

NAT : 2

SOLUTIONS

X

Controllers and Compensators

Topics

1. Basics and Types of Controllers — P, PI, PD, PID
2. Lag Compensator
3. Lead Compensator
4. Lag-Lead Compensator
5. Design Problems on Controllers and Compensators

Q10.25.1 MCQ 2025 2M Ans: B ● ● ● CLICK FOR VIDEO SOLUTION

The open-loop transfer function is $G(s) = \dfrac{1}{s^2}$

The desired dominant closed-loop poles are at $s = -1.5 \pm j\dfrac{\sqrt{27}}{2}$.

Now, we were asked to find the phase contribution at the given dominant poles location by the compensator $C(s)$

As the poles of the closed loop system are given, these implies the zeros of the characteristic equation.

The characteristic equation is given by,

$$1 + G(s)C(s) = 0$$

The phase of $C(s)$ is to be calculated at $s = s_p = -1.5 \pm j\dfrac{\sqrt{27}}{2}$

$$(G(s) \times C(s))_{s=s_p} = -1$$

$$(\angle G(s) + \angle C(s))_{s=s_p} = -180^\circ$$

$$\left(\angle \frac{1}{s^2} + \angle C(s)\right)_{s=s_p} = -180^\circ$$

$$\left(\angle \frac{1}{s^2}\right)_{s=s_p} + (\angle C(s))_{s=s_p} = -180^\circ$$

$$s = s_{p1} = -1.5 + j\frac{\sqrt{27}}{2}$$

$$\angle \frac{1}{\left(-1.5 + j\frac{\sqrt{27}}{2}\right)^2} = -2 \times \left(180^\circ - \tan^{-1}\left(\frac{\sqrt{27}/2}{1.5}\right)\right)$$

$$= -240^\circ$$

$$-240^\circ + (\angle C(s))_{s=s_p} = -180^\circ$$

$$(\angle C(s))_{s=s_p} = 60^\circ$$

$$s = s_{p2} = -1.5 - j\frac{\sqrt{27}}{2}$$

$$\angle \frac{1}{\left(-1.5 - j\frac{\sqrt{27}}{2}\right)^2} = -2 \times \left(180^\circ + \tan^{-1}\left(\frac{\sqrt{27}/2}{1.5}\right)\right)$$

$$= -480^\circ$$

$$-480^\circ + (\angle C(s))_{s=s_p} = -180^\circ$$

$$(\angle C(s))_{s=s_p} = 300^\circ \quad (= -60^\circ)$$

In the options, only 60° is given. Therefore, the phase contributed by the compensator is 60°

$$\boxed{(B)\ 60^\circ}$$

Q10.25.2 MSQ 2025 2M Ans: A, B, C ● ● ○

The given plant is

$$P(s) = \frac{a}{s^2 - b^2}, \quad a > 0 \text{ and } b > 0$$

The plant has poles at $s = \pm b$

Thus, one pole lies in the right half-plane, making the plant unstable.

Let the controller be $C(s)$. The characteristic equation is: $1 + C(s)P(s) = 0$

We examine each controller separately.

Option A: Proportional (P) controller

$$C(s) = K$$

The characteristic equation becomes

$$1 + \frac{Ka}{s^2 - b^2} = 0.$$

$$s^2 - b^2 + Ka = 0.$$

$$s^2 + (Ka - b^2) = 0.$$

There is no s-term in the characteristic equation. Hence, the poles are either purely imaginary or one lies in the right half-plane. Therefore, a proportional controller cannot stabilize the plant.

Option B: Integral (I) controller

$$C(s) = \frac{K}{s}$$

The characteristic equation becomes

$$1 + \frac{Ka}{s(s^2 - b^2)} = 0$$

$$s^3 - b^2 s + Ka = 0$$

There is no s^2-term in the characteristic equation. Therefore, an integral controller cannot stabilize the plant.

Option C: Proportional-Integral (PI) controller

$$C(s) = K_p + \frac{K_i}{s}$$

The characteristic equation becomes

$$1 + \frac{a(K_p s + K_i)}{s(s^2 - b^2)} = 0$$

$$s^3 + (aK_p - b^2)s + aK_i = 0$$

There is no s^2-term in the characteristic equation. Therefore, a Proportional-Integral controller cannot stabilize the plant.

Option D: Proportional-Derivative (PD) controller

$$C(s) = K_p + K_d s$$

The characteristic equation becomes

$$1 + \frac{a(K_p + K_d s)}{s^2 - b^2} = 0$$

$$s^2 + aK_d s + (aK_p - b^2) = 0$$

For stability, $aK_d > 0$ and $aK_p - b^2 > 0$

Since $a > 0$, choose $K_d > 0$ and $K_p > \frac{b^2}{a}$

Both conditions can be satisfied. Hence, a PD controller can stabilize the plant.

$\boxed{(A)\ \text{Proportional (P) controller}}$

$\boxed{(B)\ \text{Integral (I) controller}}$

$\boxed{(C)\ \text{Proportional-Integral (PI) controller}}$

Q10.23.1 NAT | 2023 | 2M | Ans: 0.9 to 1.1 | ● ● ○

The open loop transfer function is,

$$G(s)C(s) = \left(K_P + \frac{K_I}{s} + K_D s\right)\left(\frac{1}{(s-2)(s-3)}\right) = \left(\frac{sK_P + sK_I + K_D s^2}{s(s-2)(s-3)}\right)$$

Therefore, the open-loop transfer function has three poles: $s = 0,\ 2,\ 3$
From the given root-locus plot:

- The pole at $s = 0$ terminates at the zero at $s = -1$.
- The poles at $s = 2$ and $s = 3$ meet at the breakaway point.
- After the breakaway, one branch terminates at $s = -1$ while the other goes to infinity.

Hence, there must be two coincident zeros at $s = -1$. Therefore,

$$K_D s^2 + K_P s + K_I = A(s+1)^2 \qquad \text{where A being any constant}$$

$$K_D s^2 + K_P s + K_I = As^2 + 2As + A$$

Comparing coefficients,

$$K_D = A, \qquad K_P = 2A, \qquad K_I = A$$

It is given that $|C(j\omega)| = 2$ for $\omega = 1$ rad/s
Substituting $s = j$ in $C(j\omega)$,

$$C(j1) = K_P + \frac{K_I}{j} + jK_D = K_P + j\,(K_D - K_I) = 2A$$

$$|C(j1| = 2A$$

$$2 = 2A$$

$$A = 1$$

Since $K_D = A$,

$$K_D = 1$$

$$\boxed{1}$$

Q10.22.1 MCQ 2022 2M Ans: C ● ● ○

The plant is $P(s) = \dfrac{1}{(s+1)(s+2)}$

The PID controller is $C(s) = K_P + \dfrac{K_I}{s} + K_D s$

Since the controller contains an integral term, the open-loop system is of Type-1. For a Type-1 system, the steady-state error for a unit ramp input is

$$e_{ss} = \frac{1}{K_v}$$

where $K_v = \lim\limits_{s\to 0} sC(s)P(s)$

$$K_v = \lim_{s\to 0} s\left(K_P + \frac{K_I}{s} + K_D s\right)\frac{1}{(s+1)(s+2)} = \lim_{s\to 0}\frac{K_P s + K_I + K_D s^2}{(s+1)(s+2)} = \frac{K_I}{2}$$

Thus, the velocity error constant depends only on the integral gain K_I
After changing the controller gains, $K_P' = 2K_P$, $K_I' = 3K_I$, $K_D' = K_D$.
Therefore,

$$K_v' = \frac{3K_I}{2} = 3K_v$$

Hence, the new steady-state error becomes

$$e_{ss}' = \frac{1}{K_v'} = \frac{1}{3K_v} = \frac{e_{ss}}{3}$$

Therefore, the steady-state error decreases by a factor of 3

$$\boxed{(C)\ \text{Decreases by a factor of 3}}$$

Q10.22.2 NAT 2022 2M Ans: -60 ● ● ○

The open-loop transfer function is

$$G(s) = C(s)P(s) = \frac{s+10}{s+0.1} \times \frac{0.001}{s(2s+1)(0.01s+1)}$$

The corner frequencies are $0.1,\ 0.5,\ 10,\ 100$ rad/s.
At $\omega = 3$ rad/s, the contributing factors are:

Factor	Slope (dB/decade)
Pole at origin	−20
Pole at 0.5 rad/s	−20
Pole at 0.1 rad/s	−20
Total	**−60**

Hence, the slope of the asymptotic Bode magnitude plot at $\omega = 3$ rad/s is −60 dB/dec.

$$\boxed{-60}$$

Key Points

- While finding the slope at a particular frequency, consider only the corner frequencies below that frequency.
- Each pole contributes −20 dB/decade and each zero contributes +20 dB/decade after its corner frequency.
- A pole at the origin contributes −20 dB/decade over the entire frequency range.

Q10.21.1 MCQ 2021 2M Ans: A

The open-loop transfer function is

$$G(s)C(s) = \frac{K(s+3-j)(s+3+j)}{s(s+1)(s+3)} = \frac{K(s^2+6s+10)}{s(s+1)(s+3)}$$

The characteristic equation is

$$1 + G(s)C(s) = 0$$
$$s(s+1)(s+3) + K(s^2+6s+10) = 0$$
$$s^3 + 4s^2 + 3s + Ks^2 + 6Ks + 10K = 0$$
$$s^3 + (4+K)s^2 + (3+6K)s + 10K = 0$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 3+6K \\ s^2 & 4+K & 10K \\ s^1 & \dfrac{(4+K)(3+6K)-10K}{4+K} & 0 \\ s^0 & 10K & \end{array}$$

For stability, all the elements in the first column must be positive.

Condition	Simplification	Range of K
$4+K>0$	$K>-4$	$K>-4$
$10K>0$	$K>0$	$K>0$
$(4+K)(3+6K)-10K>0$	$(3K+4)(2K+3)>0$	$K<-\frac{3}{2}$ or $K>-\frac{4}{3}$

For $K>0$, all the above conditions are satisfied. Hence, the closed-loop system is stable only for $K>0$.

$$\boxed{(A)\text{ only stable for }K>0}$$

Q10.20.1 MCQ 2020 1M Ans: A

The transfer function is

$$G(s) = \frac{1 + 0.2s}{1 + 0.05s}$$

Method 1

Compare with the standard phase lead network

$$G(s) = \frac{1 + sT}{1 + s\alpha T}, \qquad \alpha < 1.$$

Here, $T = 0.2$ and $\alpha = \frac{0.05}{0.2} = 0.25$

The frequency at maximum phase lead is

$$\omega_m = \frac{1}{T\sqrt{\alpha}} = \frac{1}{0.2 \times 0.5} = 10 \text{ rad/s}$$

Method 2

The zero frequency is $\omega_z = \frac{1}{0.2} = 5$ rad/s

The pole frequency is $\omega_p = \frac{1}{0.05} = 20$ rad/s

The maximum phase shift occurs at the geometric mean of the zero and pole frequencies.

$$\omega_m = \sqrt{\omega_z \omega_p} = \sqrt{5 \times 20} = 10 \text{ rad/s}$$

$$\boxed{(A)\ 10 \text{ rad/s}}$$

Key Points

- For a phase lead network, the maximum phase lead occurs at the geometric mean of the zero and pole frequencies.
- The formula is $\omega_m = \frac{1}{T\sqrt{\alpha}}, \quad \alpha < 1.$

Q10.19.1 MCQ 2019 2M Ans: B

The open-loop transfer function is

$$G(s) = C(s)P(s) = \left(10 + \frac{1}{s} + 2s\right)\frac{3}{s} = \frac{30}{s} + \frac{3}{s^2} + 6 = \frac{30s + 3 + 6s^2}{s^2}$$

The system is a Type-2 system and the reference input is $r(t) = t^2$. Therefore,

$$R(s) = \frac{2}{s^3}$$

For a Type-2 system, the steady-state error for a parabolic input is

$$e_{ss} = \frac{A}{K_a},$$

where A = Magnitude of the parabolic input and $K_a = \lim_{s\to 0} s^2 G(s)$

Hence,

$$e_{ss} = \frac{2}{\lim_{s\to 0} s^2\left(\frac{30s + 3 + 6s^2}{s^2}\right)} = \frac{2}{3}$$

$$\boxed{(B)\ \frac{2}{3}}$$

Q10.13.1 MCQ 2013 2M Ans: C ● ● ○

Given

$$G(s) = \frac{1}{(s+1)^3}$$

$$G(j\omega) = \frac{1}{(1+j\omega)^3}$$

The phase is,

$$\angle G(j\omega) = -3\tan^{-1}\omega$$

At the ultimate frequency, $\angle G(j\omega_u) = -180°$

$$-3\tan^{-1}\omega_u = -180°$$

$$\omega_u = \sqrt{3}\text{ rad/s}$$

Hence, $T_u = \frac{2\pi}{\omega_u} = \frac{2\pi}{\sqrt{3}}$

The magnitude at ω_u is,

$$|G(j\omega_u)| = \frac{1}{(1+\omega_u^2)^{3/2}} = \frac{1}{(1+3)^{3/2}} = \frac{1}{8}$$

Therefore,

$$K_u = \frac{1}{|G(j\omega_u)|} = 8$$

$$\boxed{(C)\ 8 \text{ and } \frac{2\pi}{\sqrt{3}}}$$

Q10.13.2 MCQ 2013 2M Ans: D ● ● ○

From Q10.13.1, $K_u = 8$.
According to the Ziegler-Nichols tuning rule, $K = 0.5K_u = 4$.
The transfer function from the disturbance input to the output is,

$$\frac{Y(s)}{D(s)} = \frac{1}{1+KG(s)}$$

At $\omega = \frac{2\pi}{T_u} = \sqrt{3}$,

$$KG(j\omega) = 4\left(\frac{1}{8}\angle - 180°\right) = -\frac{1}{2}$$

$$\therefore \frac{Y(j\omega)}{D(j\omega)} = \frac{1}{1-\frac{1}{2}} = 2$$

$$\boxed{(D)\ 2.0}$$

Q10.12.1 MCQ 2012 2M Ans: A ● ○ ○

Given,

$$G_c(s) = \frac{s+a}{s+b}$$

For a lead compensator, the zero must be dominant. That means the zero should be nearer to the origin than the pole.
Zero location: $s = -a$ and Pole location: $s = -b$
For lead compensation,

$$-a > -b$$

$$a < b$$

Checking the options:

Option A:

$$a = 1, \quad b = 2$$

$$1 < 2$$

It satisfies the lead compensator condition.

$$\boxed{\text{(A) a=1, b=2}}$$

Q10.12.2 MCQ 2012 2M Ans: A ● ○ ○

From Q10.12.1, the lead compensator is

$$G_c(s) = \frac{s+1}{s+2}$$

Maximum phase occurs at the geometric mean of pole and zero locations.

$$\omega_m = \sqrt{z \times p}$$

Here, $z = -1$ and $p = -2$

$$\omega_m = \sqrt{1 \times 2}$$

$$\omega_m = \sqrt{2}\,\text{rad/s}$$

$$\boxed{(A)\ \sqrt{2}\,\text{rad/s}}$$

Q10.10.1 MCQ 2010 2M Ans: B ● ● ○

From the asymptotic Bode magnitude plot, the slope changes from 0 to +20 dB/decade at $\omega_z = 0.1$ rad/s and changes back to 0 at $\omega_p = 0.5$ rad/s.
For a lead network,

$$G(s) = \frac{1 + s/\omega_z}{1 + s/\omega_p}, \qquad \omega_p > \omega_z.$$

The maximum phase lead occurs at the geometric mean of the zero and pole frequencies.

$$\omega_m = \sqrt{\omega_z \omega_p} = \sqrt{0.1 \times 0.5} = \sqrt{0.05} = \sqrt{\frac{1}{20}} = \frac{1}{\sqrt{20}}\,\text{rad/s}$$

$$\boxed{(B)\ \frac{1}{\sqrt{20}}}$$

Q10.09.1 MCQ 2009 1M Ans: A ● ● ○

The plant transfer function is,

$$G_p(s) = \frac{2}{s(s+3)}$$

The PI controller with $K_p = 1$ is,

$$G_c(s) = K_p + \frac{K_i}{s} = 1 + \frac{K_i}{s} = \frac{s + K_i}{s}$$

Therefore, the open-loop transfer function is,

$$G(s) = G_c(s)G_p(s) = \left(\frac{s + K_i}{s}\right)\left(\frac{2}{s(s+3)}\right) = \frac{2(s + K_i)}{s^2(s+3)}$$

For a unit step input, $e_{ss} = \frac{1}{1+K}$, where the position error constant is $K = \lim_{s\to 0} G(s)$.

If $K_i = 0$,

$$G(s) = \frac{2}{s(s+3)},$$

which is a Type-1 system.

$$K = \lim_{s\to 0} \frac{2}{s(s+3)} = \infty$$

and therefore $e_{ss} = \frac{1}{1+\infty} = 0$

If $K_i > 0$,

$$K = \lim_{s\to 0} \frac{2(s+K_i)}{s^2(s+3)} = \infty$$

and hence $e_{ss} = \frac{1}{1+\infty} = 0$

Thus, the steady-state error is zero for both $K_i > 0$ and $K_i = 0$. We now verify the stability at the boundary value $K_i = 0$.

The characteristic equation becomes,

$$1 + \frac{2}{s(s+3)} = 0$$

$$s(s+3) + 2 = 0$$

$$s^2 + 3s + 2 = 0$$

$$(s+1)(s+2) = 0$$

The closed-loop poles are $s = -1, -2$, which lie entirely in the left-half of the s-plane. Hence, the closed-loop system is stable even when $K_i = 0$.

$$\boxed{(A)\ 0}$$

Q10.09.2 MCQ 2009 2M Ans: D ● ● ○

To determine the feedforward controller, consider only the disturbance input. Hence, $r(t) = 0$.

Let $P(s) = \frac{1}{s(s+1)}$

The output due to the disturbance is,

$$Y(s) = \frac{P(s)G_{ff}(s) + 1}{1 + P(s)} D(s)$$

To eliminate the effect of the disturbance at $\omega = 1$ rad/s, the numerator must be zero.

$$P(j1)G_{ff}(j1) + 1 = 0$$

$$G_{ff}(j1) = -\frac{1}{P(j1)}$$

Now,

$$P(j1) = \frac{1}{j(1+j)} = \frac{1}{-1+j} = \frac{1}{\sqrt{2}}\angle -\frac{3\pi}{4}$$

Therefore,

$$G_{ff}(j1) = -\sqrt{2}\,\angle\frac{3\pi}{4} = \sqrt{2}\,\angle -\frac{\pi}{4}$$

$$\boxed{(D)\ \sqrt{2}\angle -\frac{\pi}{4}}$$

Q10.09.3 MCQ 2009 2M Ans: B

Let $P(s) = \dfrac{1}{s(s+1)}$

Let the PD controller be,

$$G_{ff}(s) = K_P + sK_D$$

$$G_{ff}(j\omega) = K_P + j\omega K_D$$

From Q10.09.2,

$$G_{ff}(j1) = \sqrt{2}\angle -\frac{\pi}{4}$$

Hence,

$$K_P = \sqrt{2}\times\cos\left(-\frac{\pi}{4}\right) = 1 \qquad \text{and} \qquad \omega K_D = K_D = \sqrt{2}\times\sin\left(-\frac{\pi}{4}\right) = -1$$

Therefore,

$$G_{ff}(j\omega) = K_P + j\omega K_D = 1 - j\omega$$

The output due to the disturbance is,

$$Y(s) = \frac{P(s)G_{ff}(s)+1}{1+P(s)}D(s)$$

$$\frac{Y(s)}{D(s)} = \frac{P(s)G_{ff}(s)+1}{1+P(s)} = \frac{s^2+(1+K_D)s+K_P}{s^2+s+1} = \frac{s^2+1}{s^2+s+1}$$

$$\frac{Y(j\omega)}{D(j\omega)} = \frac{1-\omega^2}{1-\omega^2+j\omega}$$

For the disturbance input $d(t) = sin(2t)$, the output amplitude will be:

$$|y(t)| = 1\times\left|\frac{1-(2)^2}{1-(2)^2+j(2)}\right| = \left|\frac{-3}{-3+j2}\right| = \frac{3}{\sqrt{13}} = \sqrt{\frac{9}{13}}$$

$$\boxed{(B)\ \sqrt{\frac{9}{13}}}$$

Key Points

- If the input to a transfer function $G(s)$ is $x(t) = A\sin(\omega_0 t+\phi)$, then the steady-state output is: $y(t) = A|G(j\omega_0)|\sin(\omega_0 t+\phi+\angle G(j\omega_0))$
- Here, the amplitude of the output is: $A|G(j\omega_0)|$ and phase of the output is: $\phi + \angle G(j\omega_0)$

Q10.07.1 MCQ 2007 2M Ans: -

The transfer function of the inner loop is

$$G_i(s) = \frac{K_2}{2s^2+3s+1+K_2}$$

The outer loop forward path is,

$$G_{eq}(s) = \frac{K_1}{3s+1}G_i(s) = \frac{K_1}{3s+1}\times\frac{K_2}{2s^2+3s+1+K_2} = \frac{K_1K_2}{(3s+1)\left(2s^2+3s+1+K_2\right)}$$

The equivalent characteristic equation of the overall closed-loop system is,

$$1+\frac{K_1K_2}{(3s+1)\left(2s^2+3s+1+K_2\right)} = 0$$

$$(3s+1)\left(2s^2+3s+1+K_2\right)+K_1K_2 = 0$$

$$6s^3 + 11s^2 + (6 + 3K_2)s + (1 + K_2 + K_1K_2) = 0$$

The Routh array is:

$$\begin{array}{c|cc} s^3 & 6 & 6 + 3K_2 \\ s^2 & 11 & 1 + K_2 + K_1K_2 \\ s^1 & \dfrac{66 + 27K_2 - 6K_1K_2}{11} & 0 \\ s^0 & 1 + K_2 + K_1K_2 & \end{array}$$

For stability, $66 + 27K_2 - 6K_1K_2 > 0$ and $1 + K_1K_2 + K_2 > 0$

Checking each option:

Option	$\mathbf{66 + 27K_2 - 6K_1K_2}$	$\mathbf{1 + K_1K_2 + K_2}$	$\mathbf{66 + 27K_2 - 6K_1K_2 > 0}$ and $\mathbf{1 + K_1K_2 + K_2 > 0}$
$(A)\ K_1 = 10,\ K_2 = 100$	-3234	1101	$\times$
$(B)\ K_1 = 100,\ K_2 = 10$	-5664	1011	$\times$
$(C)\ K_1 = 10,\ K_2 = 10$	-264	111	$\times$
$(D)\ K_1 \to \infty,\ K_2 \to \infty$	$-\infty$	∞	$\times$

No option matches

Q10.07.2 MCQ 2007 2M Ans: D

The open-loop transfer function is,

$$G(s) = \frac{K_c e^{-s}}{s + 1}$$

$$\angle G(j\omega) = -\omega - \tan^{-1}\omega$$

At the phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180^\circ = -\pi$$

$$\omega + \tan^{-1}\omega = \pi$$

Solving the above equation by trail of options, we find $\omega = 2.03$ rad/s satisfies the above equation.

$\boxed{(D)\ 2.03}$

Key Points

- A transport delay e^{-sT} contributes a phase lag of $-\omega T$ radians.
- We can also solve the equation by using $\tan^{-1}\omega \approx \omega - \dfrac{\omega^3}{3} + \ldots..$

Q10.07.3 MCQ 2007 2M Ans: B

Given, $K_c = 1$.

The open-loop transfer function is,

$$G(s) = \frac{e^{-s}}{s + 1}$$

The steady-state error for a unit step input is,

$$e_{ss} = \frac{1}{1 + K_p}, \qquad \text{where} \quad K_p = \lim_{s \to 0} G(s)$$

$$K_p = \lim_{s\to 0} \frac{e^{-s}}{s+1} = 1$$

Therefore,

$$e_{ss} = \frac{1}{1+1} = \frac{1}{2}$$

$$\boxed{(B)\ \frac{1}{2}}$$

Key Points

- For steady-state error analysis, the transport delay e^{-sT} is evaluated at $s = 0$, giving $e^{-sT} = 1$. Hence the transport delay does not affect the steady-state error.

Q10.06.1 MCQ 2006 2M Ans: C

Let the compensator be

$$G_c(s) = K\frac{s+z}{s+p}$$

The open-loop transfer function is

$$G_{OL}(s) = \frac{K(s+z)}{s(s+1)(s+p)}$$

The desired closed-loop poles are $-1 \pm j$ and -4
Hence, the desired characteristic equation is:

$$(s+4)(s+1-j)(s+1+j) = 0$$

$$(s+4)(s^2+2s+2) = 0$$

$$s^3 + 6s^2 + 10s + 8 = 0$$

The characteristic equation of the compensated system is given by,

$$1 + G_{OL}(s) = 0$$

$$1 + \frac{K(s+z)}{s(s+1)(s+p)} = 0$$

$$s(s+1)(s+p) + K(s+z) = 0$$

$$s^3 + (1+p)s^2 + (p+K)s + Kz = 0$$

Comparing with $s^3 + 6s^2 + 10s + 8 = 0$, gives,

$$p + 1 = 6$$
$$p = 5$$

$$p + K = 10$$
$$5 + K = 10$$
$$K = 5$$

$$Kz = 8$$
$$(5)z = 8$$
$$z = \frac{8}{5} = 1.6$$

Hence, the compensator is

$$G_c(s) = 5\frac{s+1.6}{s+5}$$

$$\boxed{(C)\ \frac{5(s+1.6)}{s+5}}$$

Q10.06.2 MCQ 2006 2M Ans: D

The open-loop transfer function is

$$G(s) = \left(K_p + \frac{K_I}{s}\right)\frac{1}{(s+2)(s+10)}$$

The characteristic equation of the closed-loop system is,

$$1 + G(s) = 0$$

$$1 + \frac{K_p s + K_I}{s(s+2)(s+10)} = 0$$

$$s(s+2)(s+10) + K_p s + K_I = 0$$

$$s^3 + 12s^2 + (20 + K_p)s + K_I = 0$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 20 + K_p \\ s^2 & 12 & K_I \\ s^1 & \dfrac{240 + 12K_p - K_I}{12} & 0 \\ s^0 & K_I & \end{array}$$

For stability, all the elements of the first column must be positive. So,

$$240 + 12K_p - K_I > 0 \quad \text{and} \quad K_I > 0$$

$$12K_p > K_I - 240$$

$$K_p > \frac{K_I}{12} - 20$$

$$\boxed{(D)\ K_I > 0,\ K_p > \frac{K_I}{12} - 20}$$

Q10.06.3 MCQ 2006 2M Ans: B

To determine the effect of disturbance alone, $r(s) = 0$.
The controller and plant transfer functions are:

$$G_c(s) = 0.5\left(1 + \frac{1}{2s}\right) = \frac{2s+1}{4s} \quad \text{and} \quad G_p(s) = \frac{2e^{-s}}{1+2s}$$

The transfer function from disturbance to output is:

$$\frac{C(s)}{D(s)} = \frac{1}{1 + G_c(s)G_p(s)}$$

$$|G_c(j0.2)| = \frac{\sqrt{1^2 + 0.4^2}}{4(0.2)} = 1.346 \quad \text{and} \quad |G_p(j0.2)| = \frac{2 \times 1}{\sqrt{1 + (0.4)^2}} = 1.857$$

$$|G_c(j0.2)G_p(j0.2)| = 2.50$$

Hence,

$$\left|\frac{C(j0.2)}{D(j0.2)}\right| = \frac{1}{2.50} = 0.40$$

$$\boxed{(B)\ 0.40\ \text{unit}}$$

Q10.06.4 MCQ 2006 2M Ans: D ● ○ ○

Since the disturbance is measurable, its effect can be compensated before it propagates through the plant by using a feedforward path in addition to the feedback controller.
Therefore, the recommended control action is:

$$\boxed{(D)\ \text{feedback-feedforward control}}$$

Q10.05.1 MCQ 2005 2M Ans: A ● ○ ○

The distance from the roll gap where the thickness is measured is, $L = 0.5\,\text{m}$, and the speed of the sheet is $v = 0.2\,\text{m/s}$.
Therefore, the time taken by the sheet to travel this distance is:

$$T = \frac{L}{v} = \frac{0.5}{0.2} = 2.5\,\text{s}$$

Hence, the measured thickness is delayed by 2.5 s with respect to the roll gap, i.e.,

$$m(t) = d(t - 2.5)$$

$$M(s) = e^{-2.5s} D(s)$$

Therefore,

$$G(s) = \frac{M(s)}{D(s)} = e^{-2.5s}$$

$$\boxed{(A)\ e^{-2.5s}}$$

Q10.05.2 MCQ 2005 2M Ans: C,D ● ● ○

From Q10.05.1,

$$G(s) = e^{-2.5s}$$

The open-loop transfer function is, $KG(s) = Ke^{-2.5s}$
At the stability limit,

$$\angle G(j\omega) = -180^\circ$$

$$-2.5\omega = -\pi$$

$$\omega = \frac{\pi}{2.5} = 0.4\pi \text{ rad/s}$$

$$\boxed{(C)\ \omega = 0.4\pi}$$

$$\boxed{(D)\ \omega = \frac{\pi}{2.5}}$$

Q10.04.1 MCQ 2004 1M Ans: C ● ○ ○

The transfer function of a PI controller is,

$$G_c(s) = K_p\left(1 + \frac{1}{T_i s}\right), \qquad \text{where } T_i = \text{reset time.}$$

Given, Proportional Band (PB) = 50% and Reset time, $T_i = 0.2\,\text{s}$.
The proportional gain is related to the proportional band by,

$$K_p = \frac{100}{\text{PB}} = \frac{100}{50} = 2$$

Also,

$$\frac{1}{T_i s} = \frac{1}{0.2s} = \frac{5}{s} = \frac{1}{s/5}$$

Hence,

$$G_c(s) = 2\left(1 + \frac{5}{s}\right) = 2\left(1 + \frac{1}{0.2s}\right)$$

$$\boxed{(C)\ 2\left(1 + \frac{1}{0.2s}\right)}$$

Key Points

- Proportional gain and proportional band are related by, $K_p = \dfrac{100}{\text{PB}(\%)}$
- The standard PI controller when reset time is given is, $G_c(s) = K_p\left(1 + \dfrac{1}{T_i s}\right)$

Q10.03.1 | MCQ | 2003 | 2M | Ans: D | ● ● ○

The transfer function of a PI controller is:

$$G_c(s) = K_p\left(1 + \frac{1}{T_i s}\right)$$

Given,

$$G_c(s) = 2 + \frac{0.4}{s}$$

Comparing with the standard form,

$$K_p = 2, \qquad \text{and} \qquad \frac{K_p}{T_i} = 0.4$$

$$\frac{2}{T_i} = 0.4$$

$$T_i = 5 \text{ min}$$

The proportional band is: $PB = \dfrac{100}{K_p} = \dfrac{100}{2} = 50\%$.

$$\boxed{(D)\ 50\% \text{ and } 5 \text{ min.}}$$

Q10.03.2 | MCQ | 2003 | 2M | Ans: A | ● ● ○

The transfer function of the inner loop is

$$G_i(s) = \frac{K_1}{s + 1 + K_1}$$

The open loop transfer function of the outer loop is

$$G_{eq}(s) = \frac{8K_2}{(s+2)(s+4)} G_i(s) = \frac{8K_1K_2}{(s+2)(s+4)(s+1+K_1)}$$

The characteristic equation of the overall closed-loop system is

$$1 + \frac{8K_1K_2}{(s+2)(s+4)(s+1+K_1)} = 0$$

$$(s+2)(s+4)(s+1+K_1) + 8K_1K_2 = 0$$

$$(s+2)(s+4)\left(\frac{s+1}{K_1} + 1\right) + 8K_2 = 0$$

As $K_1 \to \infty$, $\dfrac{s+1}{K_1} \to 0$.
Hence, the characteristic equation reduces to,

$$(s+2)(s+4) + 8K_2 = 0$$

$$s^2 + 6s + 8 + 8K_2 = 0$$

The Routh array is:

$$\begin{array}{c|cc} s^2 & 1 & 8+8K_2 \\ s^1 & 6 & 0 \\ s^0 & 8+8K_2 & \end{array}$$

For stability,

$$8 + 8K_2 > 0$$

$$K_2 > -1$$

Therefore, there is no finite upper limit on either controller gain.

$$\boxed{(A)\ \infty,\ \infty}$$

Q10.03.3 | MCQ | 2003 | 2M | Ans: B | ● ● ○

From the given root locus, the system reaches the stability limit when the roots lie on the imaginary axis, that is, $s = \pm j\sqrt{2}$.
The open-loop transfer function is:

$$G(s) = \frac{2K}{s(s+1)(s+2)}$$

Using the magnitude condition,

$$\left|\frac{2K}{s(s+1)(s+2)}\right| = 1$$

$$K = \left.\frac{|s(s+1)(s+2)|}{2}\right|_{s=j\sqrt{2}}$$

$$K = \left|\frac{\sqrt{2}\,j(1+\sqrt{2}\,j)(2+\sqrt{2}\,j)}{2}\right| = 3$$

$$\boxed{(B)\ 3.0}$$

Q10.03.4 | MCQ | 2003 | 2M | Ans: D | ● ● ○

From the given root locus, the system exhibits the maximum non-oscillatory response at the breakaway point $s = -0.423$. Beyond this point, the closed-loop poles become complex and the response becomes oscillatory.
Using the magnitude condition,

$$\left|\frac{2K}{s(s+1)(s+2)}\right| = 1$$

$$K = \left.\frac{|s(s+1)(s+2)|}{2}\right|_{s=-0.423}$$

$$K = \frac{|(-0.423)(0.577)(1.577)|}{2} \approx 0.2$$

$$\boxed{(D)\ 0.2}$$

Q10.02.1 MCQ 2002 1M Ans: C ● ○ ○

The integral action continuously integrates the error signal. Therefore, even a small steady-state error accumulates with time, forcing the controller output to change until the error becomes zero. Hence, integral control eliminates the steady-state error (offset).

$$\boxed{(C)\text{ Eliminate the offset}}$$

Q10.02.2 MCQ 2002 2M Ans: C ● ● ○

The given PID controller is:

$$G(s) = 4\left(1 + \frac{1}{2s} + 0.5s\right)$$

$$G(j\omega) = 4\left(1 + \frac{1}{2j\omega} + 0.5j\omega\right) = 4\left[1 + j\left(\frac{\omega}{2} - \frac{1}{2\omega}\right)\right]$$

As $\omega \to \infty \implies \dfrac{1}{2\omega} \to 0$. Hence,

$$G(j\omega) \approx 4\left(1 + j\frac{\omega}{2}\right)$$

Therefore,

$$|G(j\omega)| = 4\sqrt{1 + \left(\frac{\omega}{2}\right)^2} \to \infty \qquad \text{and} \qquad \angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{2}\right) \to 90^\circ$$

$$\boxed{(C)\text{ Magnitude of } G(j\omega) \text{ tends to infinity and phase angle tends to } +90^\circ}$$

Key Points

- At high frequencies, the derivative term dominates the PID controller. Hence, $G(j\omega) \approx 2j\omega$, so the magnitude increases without bound and the phase approaches $+90^\circ$.

Q10.01.1 MCQ 2001 2M Ans: B ● ○ ○

According to the Ziegler-Nichols closed-loop tuning rules, the proportional gain for a P controller is:

$$K_p = 0.5K_u$$

Given, $K_u = 10$,

$$K_p = 0.5 \times 10 = 5$$

$$\boxed{(B)\ 5}$$

Key Points

- Ziegler-Nichols closed-loop tuning rules:

Controller	K_p	T_i	T_d
P	$0.5K_u$	–	–
PI	$0.45K_u$	$\dfrac{P_u}{1.2}$	–
PID	$0.6K_u$	$\dfrac{P_u}{2}$	$\dfrac{P_u}{8}$

- The oscillation period (or frequency) is not required for tuning a P controller.

Q10.00.1 MCQ 2000 2M Ans: B

The proportional band (PB) and proportional gain (K_p) are related by

$$PB = \frac{100}{K_p}$$

$$K_p = \frac{100}{PB}$$

Therefore, the proportional gain is:

$$\boxed{(B)\ \frac{100}{PB}}$$

Key Points

- Proportional band and proportional gain are inversely proportional.
- Increasing proportional gain decreases the proportional band.

Q10.00.2 MCQ 2000 2M Ans: C

The plant consists of two first-order systems connected in parallel. Hence,

$$G_p(s) = \frac{2}{1+0.5s} - \frac{1}{1+0.1s}$$

The overall open loop transfer function will be,

$$\frac{Y(s)}{R(s)} = KG_p(s) = K\left[\frac{-6s+20}{(s+2)(s+10)}\right]$$

For a unit step input, $R(s) = \frac{1}{s}$

$$Y(s) = \frac{1}{s} \times K\left[\frac{-6s+20}{(s+2)(s+10)}\right]$$

$$Y(s) = K\left[\frac{-6s+20}{s(s+2)(s+10)}\right]$$

$$Y(s) = K\left[\frac{1}{s} - \frac{2}{s+2} + \frac{1}{s+10}\right]$$

Taking Inverse Laplace transform,

$$y(t) = K\left[1 - 2e^{-2t} + e^{-10t}\right]$$

For:

$$t = 0 \qquad y(t) = 0$$

$$t = 1 \qquad y(t) = 0.26K$$

$$t = \infty \qquad y(t) = K$$

Thus, the open-loop step response starts from zero and asymptotically approaches a constant value (K) without overshoot or oscillation. Hence, the correct option is:

$$\boxed{(C)}$$

Q10.00.3 NAT 2000 2M Ans: 0.67

From Q10.00.2, the overall open loop transfer function is:

$$G(s) = K\left[\frac{-6s+20}{(s+2)(s+10)}\right]$$

The characteristic equation is:

$$1 + G(s) = 0$$

$$1 + K\left[\frac{-6s+20}{(s+2)(s+10)}\right] = 0$$

$$1 + K\left[\frac{-6s+20}{(s+2)(s+10)}\right] = 0$$

$$s^2 + s(12-6K) + 20(1+K) = 0$$

For stability:

$$12 - 6K > 0 \quad \text{and} \quad 1 + K > 0$$

$$K < 2 \quad \text{and} \quad K > -1$$

The closed-loop transfer function is

$$T(s) = \frac{\dfrac{K(20-6s)}{(s+2)(s+10)}}{1 + \dfrac{K(20-6s)}{(s+2)(s+10)}}$$

For steady state, substitute $s = 0$:

$$T(0) = \frac{\dfrac{K(20)}{(2)(10)}}{1 + \dfrac{K(20)}{(2)(10)}} = \frac{\dfrac{20K}{20}}{1 + \dfrac{20K}{20}} = \frac{K}{1+K}$$

Evaluating at the stability limits:

For $K = -1$

$$T(0) = \frac{-1}{1-1} \Rightarrow \infty$$

The steady-state gain is unbounded.

For $K = 2$

$$T(0) = \frac{2}{1+2} = \frac{2}{3} = 0.67$$

Hence, the maximum finite steady-state gain is:

$$\boxed{0.67}$$

Q10.99.1 MCQ 1999 Ans: B

A proportional controller alone cannot eliminate the steady-state error (offset) for a step input in a Type-0 system.
Introducing an integral action adds a pole at the origin, thereby increasing the system type by one.
As a result, the steady-state error for a step input becomes zero, $e_{ss} = 0$.
Hence, the offset is eliminated by adding an integral mode to the controller.

$$\boxed{(B)\ \text{Adding an integral mode to the controller.}}$$

Key Points

- Integral action eliminates the steady-state error for a step input.
- Derivative action improves the transient response but does not reduce the steady-state error.

- Changing the proportional gain changes the magnitude of the offset but cannot eliminate it completely.

Q10.98.1 MCQ 1998 Ans: B

The transfer function of an ideal PD controller is:

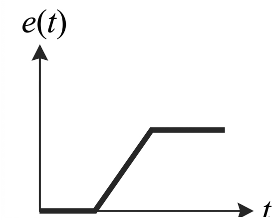
$$G_c(s) = K(1 + T_d s)$$

Hence, the controller output is:

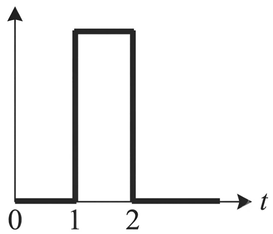
$$e_o(t) = K\left[e(t) + T_d \frac{de(t)}{dt}\right]$$

From the given input signal,

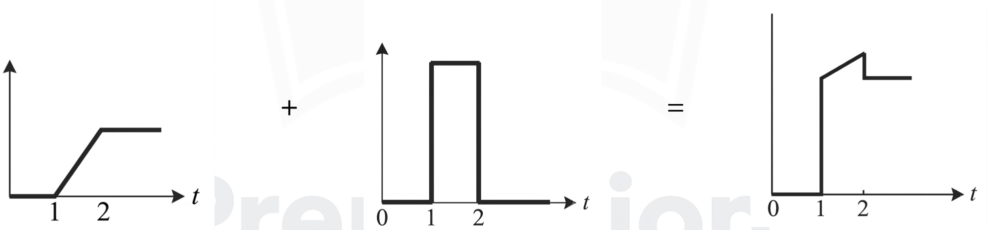
- For $0 < t < 1$, $e(t) = 0$.
- For $1 < t < 2$, $e(t)$ is a ramp.
- For $t > 2$, $e(t)$ is constant.



$$\frac{de(t)}{dt} = \begin{cases} 0, & 0 < t < 1, \\ \text{constant}, & 1 < t < 2, \\ 0, & t > 2. \end{cases}$$



Hence, the controller output is obtained by adding the proportional and derivative components:



The obtained waveform is similar to option (B).

$$\boxed{\text{Option (B)}}$$

Q10.97.1 MCQ 1997 1M Ans: B

A basic proportional-derivative (PD) controller consists of proportional and derivative actions only. Hence, its transfer function is:

$$G_c(s) = K_p + K_d s$$

Comparing with the given options, $K_p = K_0$ and $K_d = K_2$,

$$G_c(s) = K_0 + K_2 s$$

$$\boxed{(B)\ K_0 + K_2 s}$$

Key Points

- A P controller: K_p

- A PI controller: $K_p + \frac{K_i}{s}$
- A PD controller: $K_p + K_d s$
- A PID controller: $K_p + \frac{K_i}{s} + K_d s$

Q10.94.1 MCQ 1994 Ans: C

Cascade control employs two feedback loops: an inner (secondary) loop and an outer (primary) loop.
It is particularly effective for processes having widely different dynamic characteristics, where the inner loop controls the faster dynamics before they affect the primary controlled variable.
Therefore, cascade control is preferred for processes with widely different time constants.

$$\boxed{(C) \text{ Has widely different time constants}}$$

Q10.93.1 MCQ 1993 2M Ans: D

A pure time delay has the transfer function: $G(s) = e^{-sT}$.
Its magnitude is $|G(j\omega)| = 1$, while its phase is $\angle G(j\omega) = -\omega T$.
Thus, a time delay does not affect the magnitude response but introduces additional phase lag. Consequently, the phase margin decreases, whereas the gain crossover frequency remains unchanged.

$$\boxed{(D) \text{ phase margin decreases}}$$

Q10.92.1 MCQ 1992 1M Ans: D

Temperature control systems are generally slow due to their large thermal time constants.
A proportional-derivative (PD) controller improves the transient response by providing anticipatory action, thereby increasing the speed of response without introducing integral-induced sluggishness.
Hence, a PD controller is preferred for improving the dynamics of a sluggish temperature control system.

$$\boxed{(D) \text{ a PD controller can be used}}$$

DEBUG INFO

Chapter X

Controllers and Compensators

Total Questions : 42

MCQ : 38

MSQ : 1

NAT : 3